

# MAS2001 Probability and Statistics

## Illustrated reference deck

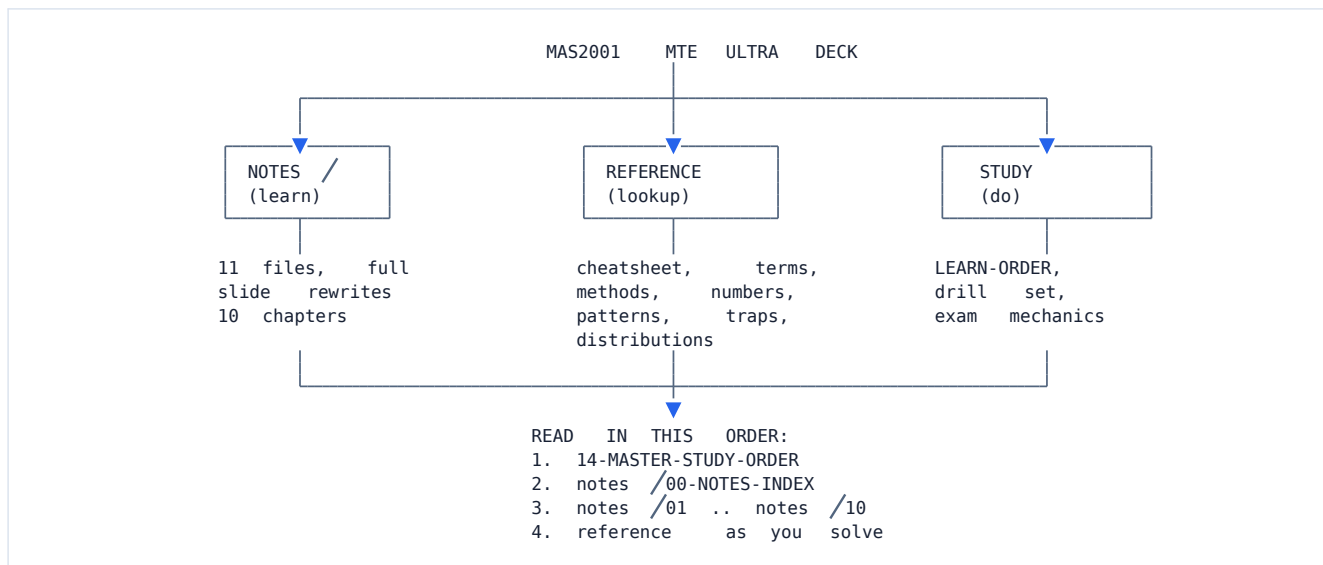
Audited Markdown content with topic diagrams, semantic tables, vectorized source drawings, exact formulas, and worked examples.

### Contents

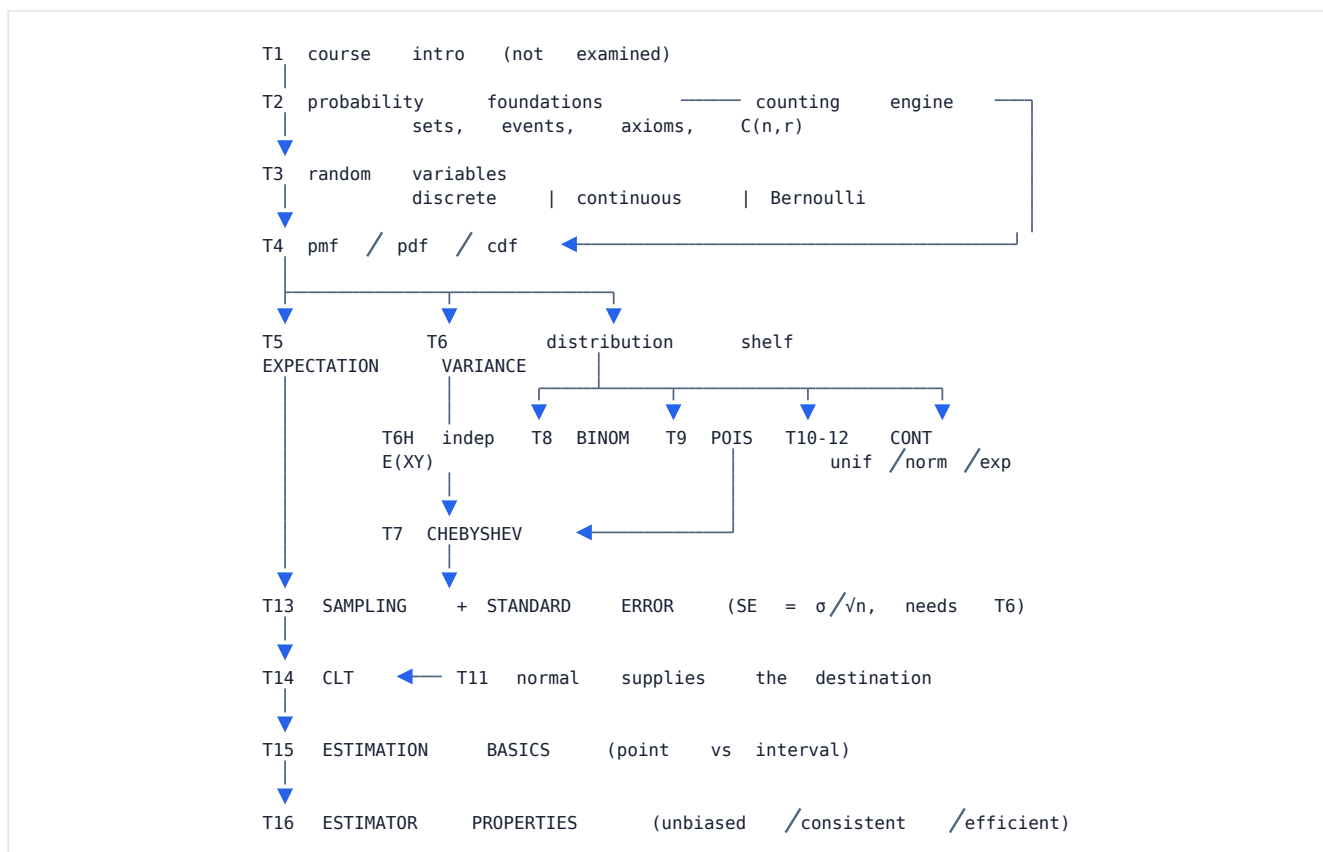
- 00. 00 INDEX: the map
- 01. 01 CHEATSHEET: everything, one pass
- 02. 02 TERMS: every term, exact meaning, the trap in each
- 03. 03 LINGUISTICS: the question wording decoder
- 04. 04 METHODS: solving path per question type
- 05. 05 DISTRIBUTIONS: all five, all slots, all values
- 06. 06 NUMBERS: constants, table values, landmarks, identities
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- 15. MAS2001 MTE ultra deck: notes + reference

# 00 INDEX: the map

## THE WHOLE FOLDER AS A MAP



## THE COURSE SPINE, TOP TO BOTTOM



## THE HIDDEN LAYER, ON THE SPINE

the nodes the slides UNDER-TAUGHT but the papers ASK:

T6 —[H1 independence rules] → T7, T14, T16 <- gates 3 topics  
 T7 —[H2 Chebyshev] → graded 9+ times  
 T12 —[H3 memoryless] → A2 MCQ5, A2 C1  
 T11 —[H4 68 /95 /99.7] → A2 MCQ11  
 T5 —[H5 N x P count] → A1 app, A2 B4, A2 D1  
 T3 —[H6 geometric 1/p] → A1 short4  
 T3 —[H7 hypergeometric nK /N] → A1 long3  
 T11 —[H8 two table conventions] → every normal /CLT numeric

= the cheapest marks on the paper. See notes /10 + 11-BEYOND-SLIDES.

### 1. What the MTE is

|                                                     |                                                            |
|-----------------------------------------------------|------------------------------------------------------------|
| paper                                               | MAS2001 Statistics and Probability, mid term               |
| marks                                               | 30                                                         |
| scope                                               | lectures 1 to 21 (the 16 syllabus lines below)             |
| shape                                               | Section A MCQ (a few marks each), Section B short answers, |
| Section C longer. All compulsory.                   |                                                            |
| closed book so the sheet below must be in your head |                                                            |
| calculator is allowed on both supplied MTE papers   |                                                            |

## 2. The 16 syllabus lines and where each is taught

| #                             | syllabus line                                 | lecture                               | file                                     |
|-------------------------------|-----------------------------------------------|---------------------------------------|------------------------------------------|
| 1                             | Introduction of the Course                    | 1                                     | md/notes-lecture-series-01-09 p001-011   |
| 2                             | Basic terminology and concepts of probability | 2                                     | notes p012-050                           |
| 3                             | Random variables (discrete, continuous)       | 3-4                                   | notes p051-062, p112                     |
| 4                             | PMF, pdf, CDF                                 | 5-6                                   | notes p063-087, p112-139                 |
| 5                             | Expectation of random variables               | 7                                     | notes p088-094, p140-147                 |
| 6                             | Expectation and Independent random variables  | 8-9                                   | notes p095-111 (indep rules: hidden, 11) |
| 7                             | Chebyshev's inequality                        | 10-11                                 | deck L10-11 (9 pages)                    |
| 8                             | Binomial distribution                         | 12                                    | md/ppt3 p001-018                         |
| 9                             | Poisson distribution                          | 13                                    | md/ppt3 p019-028                         |
| 10                            | Uniform distribution (continuous)             | 14                                    | md/ppt4 p001-006                         |
| 11                            | Normal distribution                           | 15                                    | md/ppt4 p007-037                         |
| 12                            | Exponential distribution                      | 16                                    | md/ppt4 p038-043                         |
| 13                            | Sampling concepts, standard error             | 17                                    | md/lms-standard-error-clt p001-019       |
| 14                            | Central limit theorem                         | 18                                    | md/lms-standard-error-clt p006-018       |
| 15                            | Theory of estimation: parameter, statistic,   |                                       |                                          |
| point and interval estimation | 19                                            | md/ppt5 p001-007, lms-theory p001-015 |                                          |
| 16                            | Characteristics of a good estimator           | 20-21                                 | md/ppt5 p008-021, lms-theory p016-030    |

### 3. The 16-topic tree (full detail in 05 and 10)

|                                 |                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------|
| T1                              | course intro (not examined)                                                         |
| T2                              | probability foundations: sets, events, axioms, counting, conditional, independence  |
| T3                              | random variables: definition, discrete, continuous, functions of an rv              |
| T4                              | pmf, pdf, cdf: conditions, build, read, invert                                      |
| T5                              | expectation: definition, $E[h(X)]$ , linearity, expected count                      |
| T6                              | variance: definition, shortcut, transforms; independent rvs ( $E(XY)$ , Var of sum) |
| T7                              | Chebyshev: tail form, complement form, k extraction, apply, inverse                 |
| T8                              | binomial                                                                            |
| T9                              | Poisson                                                                             |
| T10 uniform                     |                                                                                     |
| T11 normal                      |                                                                                     |
| T12 exponential                 |                                                                                     |
| T13 sampling and standard error |                                                                                     |
| T14 central limit theorem       |                                                                                     |
| T15 estimation basics           |                                                                                     |
| T16 estimator properties        |                                                                                     |

### 4. The learning order (why, and the evidence)

The batch-2 S&P decks are re-teaches, proved by shared-phrase counts:

|                      |                                                                   |                                                  |
|----------------------|-------------------------------------------------------------------|--------------------------------------------------|
| L1-7                 | vs notes deck                                                     | 1,129 shared 5-grams (54 percent of the smaller) |
| L8-9                 | vs notes deck                                                     | 456 shared (55 percent)                          |
| L12-13 vs ppt3       | 874 shared (53 percent), same pens numbers 0.2301, 0.2824, 0.3766 |                                                  |
| L14-15 vs ppt4       | 630 shared (40 percent)                                           |                                                  |
| L10-11 vs notes deck | 12 shared (2 percent)                                             | <- the only genuinely new deck                   |

So the order is: Chebyshev and the hidden layer first (nothing else teaches them, they gate four downstream topics), then the main theory, with each S&P re-teach used as the second pass on its topic, then the papers, then the assignments.

|        |                                                                         |              |
|--------|-------------------------------------------------------------------------|--------------|
| PASS 0 | Chebyshev deck + independence rules + the hidden set + formula sheet    | 1h40         |
| PASS 1 | notes p012-087 (foundations, rvs, pmf / cdf) + its questions            | 3h           |
| PASS 2 | notes p088-147 (expectation, variance) + S&P L1-7 and L8-9 questions    | 3h15         |
| PASS 3 | binomial, Poisson, uniform, normal, exponential + their re-teach blocks | 4h           |
| PASS 4 | sampling and CLT + questions                                            | 2h           |
| PASS 5 | estimation (ppt5 full, lms-theory 2 items only) + questions             | 2h30         |
| PASS 6 | mock paper, closed book, timed, then mark it                            | 3h           |
| PASS 7 | MTE papers, then assignments, then ETE IN blocks                        | several days |
| PASS 8 | sheet, landmarks, hidden set, traps, from memory                        | last day     |

Full per-question sequencing: [../reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md](https://reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md).

## 5. The counts you can trust (as of 16 Sep 2026)

|                            |                                         |                                        |
|----------------------------|-----------------------------------------|----------------------------------------|
| base branch ledger         | 383 instances                           |                                        |
| assignment-bundle addition | 119 instances on PR 14                  |                                        |
| combined enumerated total  | 502 instances after PR 14 is merged     |                                        |
| MTE source/family leads    | 10 of 16 blocks                         |                                        |
| MTE generic concept checks | 5 of 16 blocks                          |                                        |
| MTE open composite         | 1 of 16 blocks                          |                                        |
| content family ids         | 0 of 502 rows, dedup remains unfinished |                                        |
| exam papers on disk        | 9                                       | (2 MTE + 4 ETE + 2 re-sess + 1 summer) |
| marking schemes on disk    | 2                                       | (kept separate from exam-paper count)  |

The 16 MTE blocks have all been searched. “Searched” is not the same as “traced”. A generic definition match is not evidence that a question was copied from a particular source.

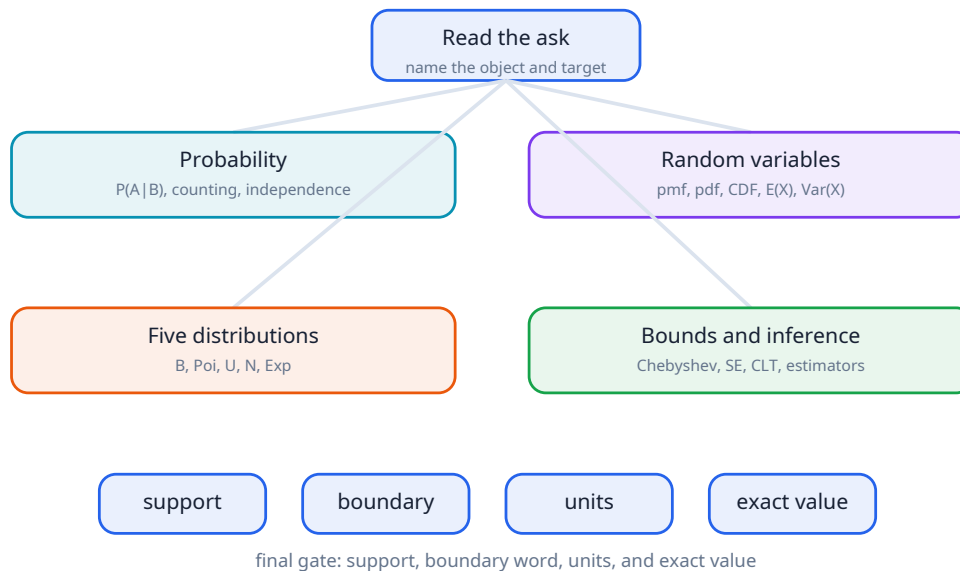
## 6. The hidden layer, one line each (full detail in 11)

|    |                                                                            |                                |
|----|----------------------------------------------------------------------------|--------------------------------|
| H1 | independence rules of rvs: $E(XY)$ , $\text{Var}(X+Y)$ , $\text{Var}(X-Y)$ | no teaching slide              |
| H2 | Chebyshev entire topic                                                     | taught only in the L10-11 deck |
| H3 | memoryless property of the exponential                                     | no slide                       |
| H4 | the 68.27 / 95.45 / 99.73 landmarks                                        | no slide                       |
| H5 | expected count, $N \times P$                                               | no slide                       |
| H6 | geometric mean $1/p$                                                       | no slide                       |
| H7 | hypergeometric mean $nK/N$                                                 | no slide                       |
| H8 | two normal table conventions ( $\phi$ vs $F$ )                             | no slide                       |

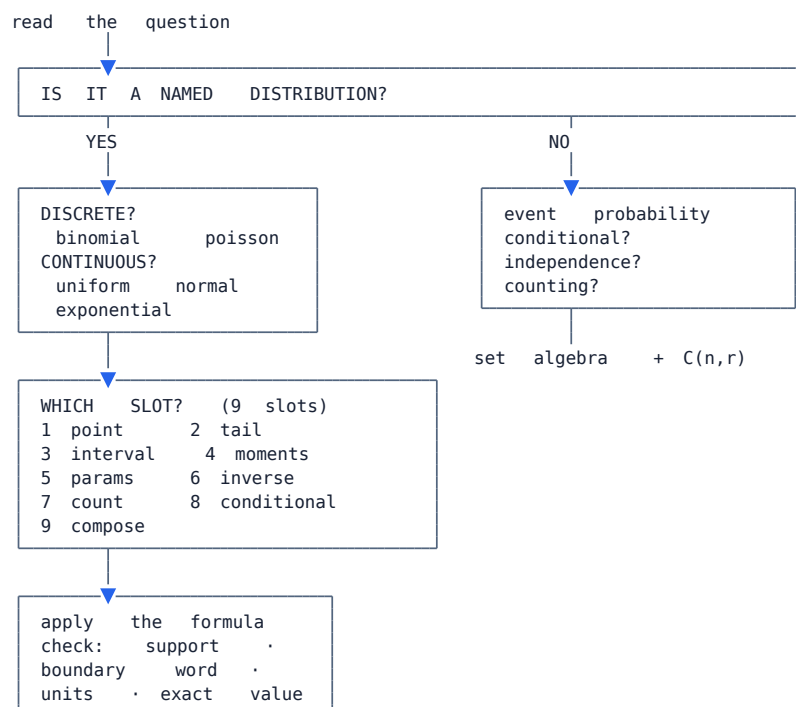
# 01 CHEATSHEET: everything, one pass

## THE WHOLE PAPER AS ONE FLOWCHART

Formula landscape: classify first, then compute

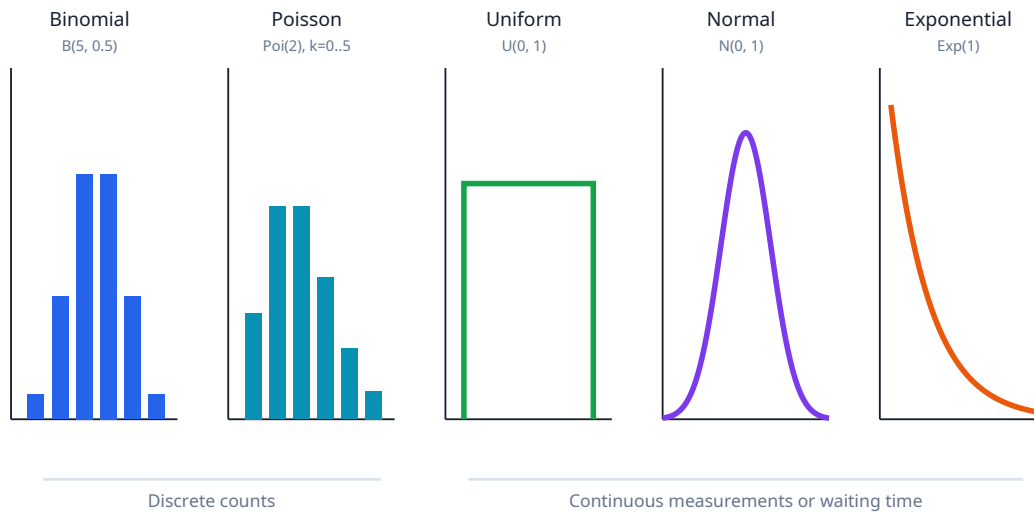


Formula landscape and final checks



## THE FIVE DISTRIBUTIONS AT A GLANCE

## Five distribution shapes



## Five parameterized distribution shapes

### DISCRETE

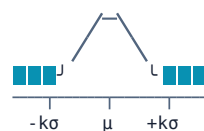
|                                          |
|------------------------------------------|
| <b>BINOMIAL</b><br>$np, npq$<br>         |
| <b>POISSON</b><br>$\lambda, \lambda$<br> |
| (mean=var!)                              |

### CONTINUOUS

|                                                    |
|----------------------------------------------------|
| <b>UNIFORM</b><br>$(a+b)/2, (b-a)^2/12$<br>        |
| <b>NORMAL</b><br>$\mu, \sigma^2$<br>               |
| <b>EXPONENTIAL</b><br>$1/\lambda, 1/\lambda^2$<br> |

## CHEBYSHEV, ONE PICTURE

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2$$



■ = the tail event (bounded)

| k | tail  | centre |
|---|-------|--------|
| 2 | .25   | .75    |
| 3 | .111  | .889   |
| 4 | .0625 | .9375  |

Closed book paper. This page is the whole exam in formulas. Read it daily, last thing at night.



## A. Probability

$P(S)=1$   $P(A) \geq 0$  disjoint additivity on S  
 $P(A') = 1 - P(A)$   
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$   
 disjoint:  $P(A \cup B) = P(A) + P(B)$   
 $P(A|B) = P(A \cap B)/P(B)$ ,  $P(B) > 0$   
 $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$   
 independent:  $P(A \cap B) = P(A)P(B)$   
 $(A \cup B)' = A' \cap B'$   $(A \cap B)' = A' \cup B'$   
 $n! / 0! = 1$   $C(n,r) = n! / (r!(n-r)!)$   $P(n,r) = n! / (n-r)!$   $C(n,r) = C(n,n-r)$

## B. Random variables, pmf, pdf, cdf

discrete  $p(x) \geq 0$ ,  $\sum p(x) = 1$ ,  $P(X \in A) = \sum$   
 continuous  $f(x) \geq 0$ ,  $\int f dx = 1$ ,  $P(a \leq X \leq b) = \int_a^b f dx$ ,  $P(X=x)=0$   
 cdf  $F(x) = P(X \leq x)$ , non decreasing,  $F(-\infty)=0$ ,  $F(\infty)=1$   
 $P(a < X \leq b) = F(b) - F(a)$   
 discrete  $p(x) = F(x) - F(x-)$  (the jump)  
 continuous  $F(x) = \int_{-\infty}^x f$ ,  $f(x) = F'(x)$   
 integer rv  $P(a \leq X \leq b) = F(b) - F(a-1) \leftarrow$  the a-1, the commonest slip

## C. Expectation

discrete  $E(X) = \sum x p(x)$  continuous  $E(X) = \int x f dx$   
 $E[h(X)] = \sum h(x)p(x)$  or  $\int h(x) f dx$   
 $E(aX+b) = aE(X)+b$   $E(c) = c$   $E(X+Y) = E(X)+E(Y)$  ALWAYS  
 $E(XY) = E(X)E(Y)$  only if INDEPENDENT  
 expected count: out of N items, expected =  $N \cdot P$

## D. Variance

$\text{Var}(X) = E[(X-\mu)^2] = E(X^2) - [E(X)]^2$   
 $sd = \sqrt{\text{Var}}$   
 $\text{Var}(aX+b) = a^2 \text{Var}(X)$   $sd(aX+b) = |a| sd(X)$   $\text{Var}(c)=0$   
 $\text{Var}(X+Y) = \text{Var}(X)+\text{Var}(Y)$  independent only  
 $\text{Var}(X-Y) = \text{Var}(X)+\text{Var}(Y)$  independent only, the MINUS still ADDS  
 $\text{Var}(\bar{X}) = \sigma^2 / n$

## E. Chebyshev

$P(|X-\mu| \geq k \sigma) \leq 1/k^2$   $k > 0$   
 $P(|X-\mu| \geq \epsilon) \leq \sigma^2/\epsilon^2$   
 $P(|X-\mu| < k \sigma) \geq 1 - 1/k^2$  complement form  
 $k=2 \rightarrow 3/4$   $k=3 \rightarrow 8/9$   $k=4 \rightarrow 15/16$   $k=1 \rightarrow 1$ , useless  
 $k$  is in standard deviations:  $k = \epsilon/\sigma$   
 no distribution assumed, only finite  $\mu$  and variance  
 always an UPPER bound on the tail, so actual  $\leq$  bound, say so when asked

## F. The five distributions

BINOMIAL  $X \sim B(n, p)$   $q = 1 - p$   
 $P(X=x) = C(n, x) p^x q^{(n-x)}$   $E = np$   $\text{Var} = npq$   
 needs: 2 outcomes, finite  $n$ , independent trials,  $p$  constant  
 POISSON  $X \sim \text{Poi}(l)$   
 $P(X=x) = e^{-l} l^x / x!$   $E = l$   $\text{Var} = l$  (mean = variance)  
 $l = np$  for the rare-event approximation,  $n$  large  $p$  small  
 UNIFORM  $X \sim U(a, b)$   
 $f = 1/(b-a)$   $E = (a+b)/2$   $\text{Var} = (b-a)^2/12$   $F = (x-a)/(b-a)$   
 probability = length ratio  
 NORMAL  $X \sim N(\mu, \sigma^2)$  (second slot is VARIANCE)  
 $Z = (X - \mu)/\sigma$   $X = \mu + Z \sigma$   
 $P(a < X < b) = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$   
 $\Phi(-z) = 1 - \Phi(z)$   $P(Z > z) = 1 - \Phi(z)$   
 landmarks: 68.27 within 1s, 95.45 within 2s, 99.73 within 3s  
 EXPONENTIAL  $T \sim \text{Exp}(l)$  ( $l$  is a RATE)  
 $f = l e^{-l t}$   $F = 1 - e^{-l t}$   $P(T > t) = e^{-l t}$   
 $E = 1/l$   $\text{Var} = 1/l^2$   
 memoryless:  $P(T > s + t \mid T > s) = P(T > t)$   
 if the question gives the MEAN  $\mu$  instead,  $l = 1/\mu$  and  $f = (1/\mu) e^{-t/\mu}$

## G. Sampling, SE, CLT

SE of the mean  $\sigma_{\bar{X}} = \sigma / \sqrt{n}$   
 SE estimate  $S / \sqrt{n}$   
 SE of a proportion  $\sqrt{p(1-p)/n}$   
 CLT as  $n$  grows:  $\bar{X} \sim N(\mu, \sigma^2/n)$  approximately, under the theorem's conditions  
 $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$   
 exact, not approx, when the population itself is normal (any  $n$ )  
 $n \geq 30$  is the course rule of thumb, not a universal theorem cutoff  
 for small  $n$  and a non-normal population, normal approximation needs justification  
 sum of  $n$  draws  $\sim N(n\mu, n\sigma^2)$   
 quadruple  $n \rightarrow$  halve the SE (sqrt rule)

## H. Estimation

|                                                                         |                                                                                               |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| estimator = a rule (a random variable), estimate = the number           |                                                                                               |
| unbiased                                                                | $E(\hat{\theta}) = \theta$                                                                    |
| consistent                                                              | converges in probability; sufficient pair $E \rightarrow \theta$ , $\text{Var} \rightarrow 0$ |
| efficient                                                               | smallest variance AMONG UNBIASED                                                              |
| relative efficiency of $T_1$ vs $T_2 = \text{Var}(T_2)/\text{Var}(T_1)$ |                                                                                               |
| sufficient                                                              | conditional sample distribution given the statistic is parameter-free;                        |
| Neyman-Fisher factorisation is the usual test in this course            |                                                                                               |
| $\text{MSE} = \text{Var} + \text{Bias}^2$                               |                                                                                               |
| $\bar{X}$ : unbiased, $\text{Var} = \sigma^2/n$ , consistent            |                                                                                               |
| $S^2 = \sum (X_i - \bar{X})^2 / (n-1)$ : unbiased for $\sigma^2$        | (the $n-1$ matters)                                                                           |
| $\hat{p} = X/n$ : unbiased, $\text{Var} = p(1-p)/n$                     |                                                                                               |

## I. The four sentence patterns that carry most marks

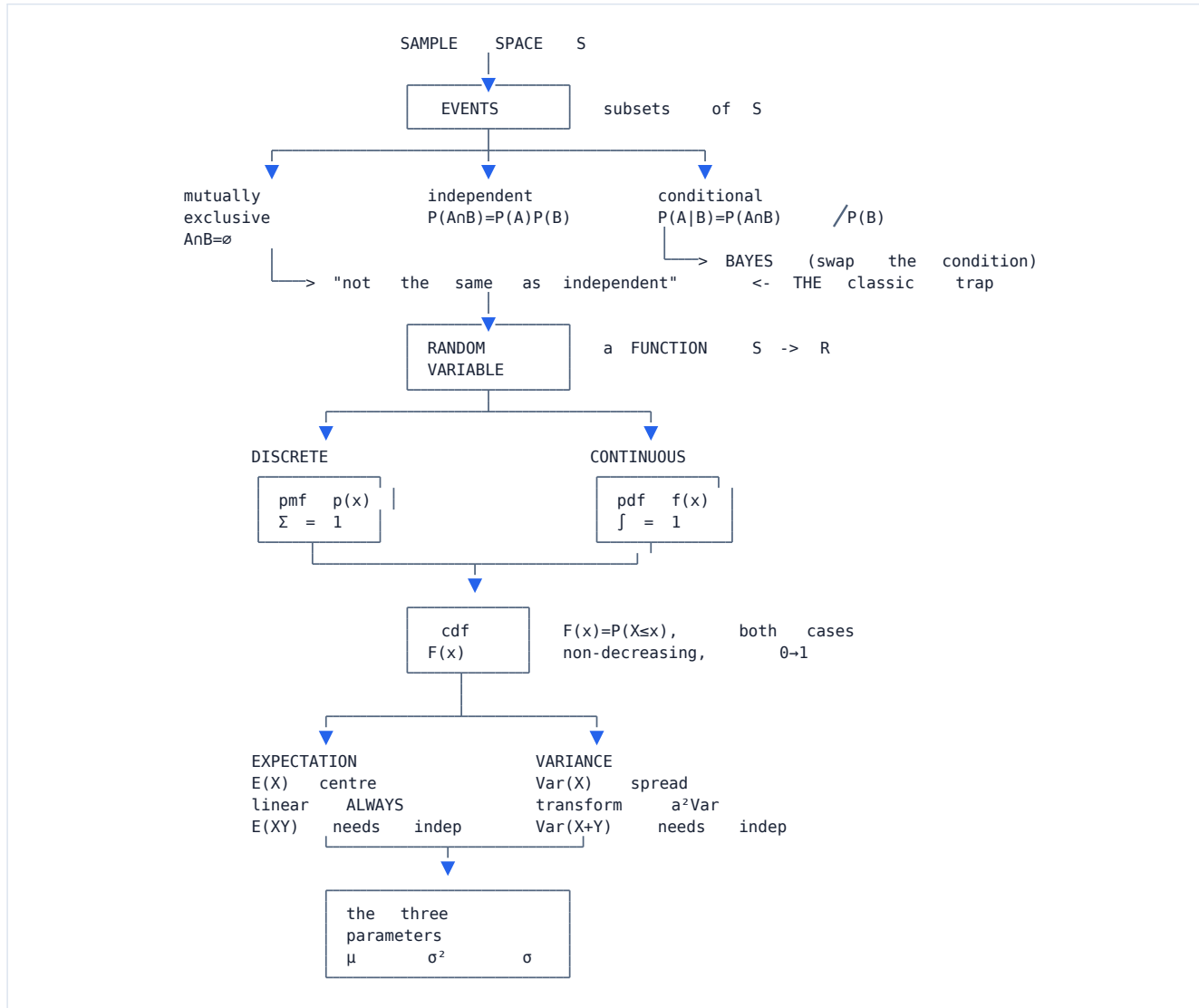
|   |                                                                                |                                                                 |
|---|--------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1 | "state which estimator is unbiased"                                            | check E of each, sum the coefficients                           |
| 2 | "find the constant that makes it unbiased" solve sum of coefficients = divisor |                                                                 |
| 3 | "which is most efficient"                                                      | variances, smallest among the unbiased                          |
| 4 | "show $\bar{X}$ is consistent"                                                 | $E = \mu$ for all $n$ , $\text{Var} = \sigma^2/n \rightarrow 0$ |

## J. Boundaries

|                                                                             |                                                                                 |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| in scope                                                                    | lectures 1 to 21, the four MTE papers plus their schemes, the in-scope parts of |
| the 2024-25 and 2025-26 assignments, all four S&P decks, the Chebyshev deck |                                                                                 |
| boundary                                                                    | the confidence interval shape (named in lecture 19, mechanics in lecture 25)    |
| out                                                                         | MLE, method of moments, hypothesis testing, t/F/chi-square, ANOVA,              |
| confidence interval mechanics, lms-theory p031-p040                         |                                                                                 |

## 02 TERMS: every term, exact meaning, the trap in each

### THE TERM MAP (how the words connect)



### THE "WORDS THAT DECIDE MARKS" CHEAT

|               |                                                                       |                              |
|---------------|-----------------------------------------------------------------------|------------------------------|
| exactly $k$   | $\rightarrow P(X = k)$                                                | point                        |
| at least $k$  | $\rightarrow P(X \geq k) = 1 - P(X \leq k-1)$ tail, boundary INCLUDED |                              |
| more than $k$ | $\rightarrow P(X > k) = 1 - P(X \leq k)$                              | tail, boundary EXCLUDED      |
| at most $k$   | $\rightarrow P(X \leq k) = F(k)$                                      | cdf                          |
| less than $k$ | $\rightarrow P(X < k) = F(k-1)$                                       | cdf, step back one           |
| none          | $\rightarrow P(X = 0)$                                                |                              |
| at least one  | $\rightarrow 1 - P(X=0)$                                              |                              |
| expected no.  | $\rightarrow N \times P$                                              | (a COUNT, not a probability) |

Read with the theory. Each entry is: the term, what it exactly means, and the trap or the reason the term matters. Terms are grouped by topic.

## Probability

|                                                                                            |                                                                        |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| sample space S                                                                             | the set of all possible outcomes. TRAP: for two dice write 36 ORDERED  |
| outcomes, not 21 unordered. The counting must match the space.                             |                                                                        |
| event                                                                                      | a subset of S. TRAP: "at least one" is the complement of "none".       |
| mutually exclusive                                                                         | no two share an outcome (disjoint). TRAP: NOT the same as independent. |
| A die roll even vs prime SHARE the outcome 2, so they are not exclusive.                   |                                                                        |
| independent                                                                                | $P(A \cap B) = P(A)P(B)$ . TRAP: exclusive events with $P > 0$ are NOT |
| independent (if one happens the other cannot, so knowing A changes B).                     |                                                                        |
| conditional $P(A B)$                                                                       | the chance of A given B already happened. TRAP: the denominator is     |
| $P(B)$ , not $P(A)$ . $P(A B) \neq P(B A)$ .                                               |                                                                        |
| Bayes                                                                                      | swapping the condition: $P(A B) = P(B A)P(A)/P(B)$ . Used in the rare  |
| disease and county-forms questions.                                                        |                                                                        |
| complement discipline "at least one", "none", "rejected", "more than" are all complements. |                                                                        |
| This single move solves more probability marks than any formula.                           |                                                                        |

## Counting

|                                                                         |                                                                     |
|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| permutation $P(n,r)$                                                    | ordered selections, "arrangements", "lineups", "how many ways to    |
| order". Order matters.                                                  |                                                                     |
| combination $C(n,r)$                                                    | unordered selections, "choose", "committees", "sets of". Order does |
| NOT matter. TRAP: "how many ways to form a committee of 3" is C, not P. |                                                                     |
| factorial, $0! = 1$                                                     | the seed for every count.                                           |
| repeated items                                                          | if items repeat, divide by the repeats' factorials.                 |
| complement counting                                                     | count the easy case and subtract from the total.                    |

## Random variables

|                                                                 |                                                                                         |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| random variable                                                 | a FUNCTION from the sample space to the real line. TRAP: it is not the                  |
| outcome, it is the number you attach to the outcome.            |                                                                                         |
| discrete rv                                                     | countable support (a list). Uses sums.                                                  |
| continuous rv                                                   | an interval of support. Uses integrals. $P(X = \text{one point}) = 0$ .                 |
| support                                                         | the set where the rv lives. TRAP: ALWAYS check the support before                       |
| integrating or summing; a density is zero outside it.           |                                                                                         |
| function of an rv                                               | $Y = g(X)$ . TRAP: changing $g$ changes the DISTRIBUTION of $Y$ , so recompute          |
| the support of $Y$ before anything else.                        |                                                                                         |
| Bernoulli                                                       | one trial, $X = 1$ or $0$ , $P(1)=p$ . Mean $p$ , variance $pq$ . The atom of binomial. |
| geometric                                                       | counts trials until the first success. $E = 1/p$ . Memoryless like the                  |
| exponential, in the discrete world. (no teaching slide, see 11) |                                                                                         |
| hypergeometric                                                  | draws WITHOUT replacement from a finite lot. Mean $nK/N$ , the same as the              |
| binomial mean at the same $p$ . (no teaching slide, see 11)     |                                                                                         |

## pmf, pdf, cdf

|                                                                       |                                                                         |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| pmf $p(x)$                                                            | discrete, $P(X=x)$ . Must be $\geq 0$ and sum to 1.                     |
| pdf $f(x)$                                                            | continuous, a DENSITY, not a probability. $f(x)$ can exceed 1 (example: |
| uniform on width 0.5 has $f=2$ ). Only its INTEGRAL is a probability. |                                                                         |
| cdf $F(x)$                                                            | $P(X \leq x)$ for both. Non decreasing, 0 to 1, right continuous.       |
| pmf from cdf                                                          | $p(x) = F(x) - F(x-)$ , the jump. For integer rv, $F(b)-F(a-1)$ .       |
| pdf from cdf                                                          | $f = F'$ . Integrate to go up, differentiate to come down.              |
| "valid pdf" check                                                     | integrate over its support and it must equal 1. Finding $k$ is exactly  |
| this check rearranged.                                                |                                                                         |

## Expectation

|                                                 |                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------|
| expectation $E(X)$                              | the weighted average, the long run mean. Symmetry trick: for symmetric                 |
| densities $E =$ the centre, no integral needed. |                                                                                        |
| $E[h(X)]$                                       | the mean of a TRANSFORMED rv. TRAP: $E[g(X)] \neq g(E(X))$ in general.                 |
| linearity                                       | $E(aX+b)=aE(X)+b$ , $E(X+Y)=E(X)+E(Y)$ with NO independence needed.                    |
| product rule                                    | $E(XY)=E(X)E(Y)$ needs INDEPENDENCE. TRAP: this is the classic true/false.             |
| expected count                                  | out of $N$ items, expected = $N \times P(\text{success})$ . No teaching slide, see 11. |

## Variance

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| variance              | $E[(X-\mu)^2]$ , the spread squared. Never negative.                                  |
| shortcut              | $E(X^2) - [E(X)]^2$ . Always easier than the definition.                              |
| standard deviation    | $\sqrt{\text{Var}}$ , in the ORIGINAL units. Use sd when Chebyshev k or a z-score     |
| is involved.          |                                                                                       |
| $\text{Var}(aX+b)$    | $a^2 \text{Var}(X)$ . TRAP: the shift b does NOTHING, the scale a squares.            |
| $\text{Var}(X-Y)$     | $\text{Var}(X)+\text{Var}(Y)$ . TRAP: the MINUS stays PLUS because Var is on squares. |
| $\text{Var}(\bar{X})$ | $\sigma^2/n$ . The root of every CLT standard error.                                  |

## Chebyshev

|                                               |                                                                                |
|-----------------------------------------------|--------------------------------------------------------------------------------|
| Chebyshev inequality                          | bounds a tail using only $\mu$ and $\sigma$ . Any distribution.                |
| k                                             | distance in STANDARD DEVIATIONS, $k = \text{eps}/\sigma$ . TRAP: raw units in, |
| must convert to sigma units first.            |                                                                                |
| vacuous bound                                 | $k \leq 1$ gives a bound of 1 or more, which says nothing.                     |
| distribution-free                             | no normal table, no assumption. That is why "which is valid without            |
| knowing the distribution" always points here. |                                                                                |

## Distributions

|                                                                                 |                                                                                            |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| binomial                                                                        | n independent 2-outcome trials, p constant, X counts successes. $E=np$ .                   |
| Poisson                                                                         | counts of rare events in a fixed window. The signature is $E = \text{Var} = \lambda$ .     |
| uniform                                                                         | constant density. Probability is proportional to interval LENGTH.                          |
| normal                                                                          | bell, symmetric, mean = median = mode = $\mu$ . Second parameter is VARIANCE.              |
| standard normal Z                                                               | $N(0,1)$ , the table world. Every normal question goes through Z.                          |
| phi vs F                                                                        | TWO table conventions. $\phi(z)$ is usually the area from 0 to z, $F(z)$ is                |
| the cumulative from $-\infty$ . MIXING THEM is the single biggest normal error. |                                                                                            |
| Decide which the question gives BEFORE substituting. (see 11)                   |                                                                                            |
| exponential                                                                     | waiting time, $\lambda = \text{rate}$ , mean = $1/\lambda$ . TRAP: rate vs mean confusion. |
| memoryless                                                                      | surviving s more given you survived to now = surviving s. No ageing.                       |
| Poisson as a limit                                                              | binomial with n big and p small, $\lambda = np$ . "Use the approximation" means            |
| Poisson with that $\lambda$ .                                                   |                                                                                            |

## Sampling, CLT

|                                                                              |                                                                                |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| population vs sample                                                         | N with parameters ( $\mu$ , $\sigma$ ) vs n with statistics ( $\bar{X}$ , s).  |
| parameter                                                                    | a number describing the population ( $\mu$ , $\sigma$ , p).                    |
| statistic                                                                    | a number computed from the sample ( $\bar{X}$ , S, $\hat{p}$ ).                |
| sampling distribution the distribution of a statistic over repeated samples. |                                                                                |
| standard error                                                               | the standard deviation of that sampling distribution. $SE = \sigma/\sqrt{n}$ . |
| CLT                                                                          | as n grows, the SAMPLE MEAN approaches normality under the theorem's           |
| conditions. $n \geq 30$ is a course heuristic, not a universal cutoff.       |                                                                                |
| exact vs approx                                                              | a normal population gives an exactly normal $\bar{X}$ at ANY n. This is the    |
| fallback when a question uses a small n.                                     |                                                                                |
| sqrt-n rule                                                                  | quadruple n, halve SE. The answer to "how large a sample".                     |

## Estimation

|                                                                            |                                                                             |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| estimator                                                                  | a RULE, a random variable ( $\bar{X}$ is a rule).                           |
| estimate                                                                   | the NUMBER once data is plugged in ( $\bar{X} = 205$ ms).                   |
| unbiased                                                                   | $E(\hat{\theta}) = \theta$ . Correct ON AVERAGE, not per sample.            |
| bias                                                                       | $E(\hat{\theta}) - \theta$ .                                                |
| consistent                                                                 | converges to $\theta$ as n grows. A large-sample property.                  |
| consistency test                                                           | sufficient pair: $E(T_n) \rightarrow \theta$ and $Var(T_n) \rightarrow 0$ . |
| efficient                                                                  | smallest variance AMONG THE UNBIASED. TRAP: a biased estimator can have     |
| a smaller variance and still lose, it is disqualified first.               |                                                                             |
| relative efficiency                                                        | $Var(T_2)/Var(T_1)$ . Bigger than 1 means $T_1$ is more efficient.          |
| sufficient                                                                 | given the statistic, the conditional sample distribution does not           |
| depend on $\theta$ . Neyman-Fisher factorisation is the usual test here.   |                                                                             |
| MSE                                                                        | $Var + Bias^2$ . The honest score when bias is allowed to compete.          |
| n-1                                                                        | the sample variance divides by n-1, not n, to be unbiased. The reason is    |
| that deviations are measured from $\bar{X}$ , which already used the data. |                                                                             |

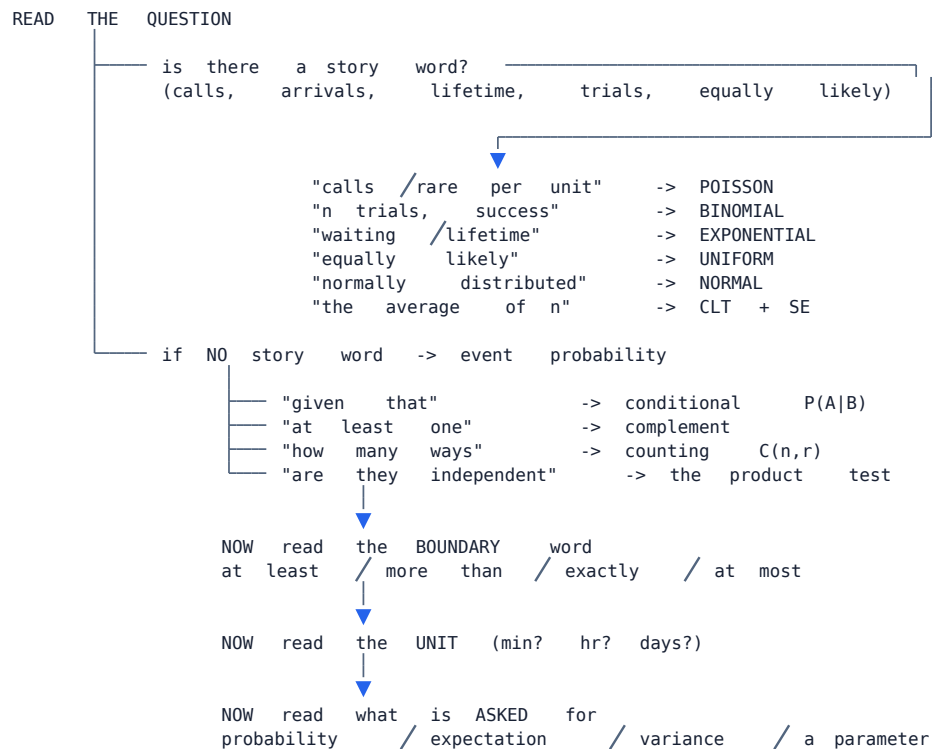


## Words that are not statistics but decide marks

|                                                                                   |                                   |                         |
|-----------------------------------------------------------------------------------|-----------------------------------|-------------------------|
| "exactly"                                                                         | $P(X=x)$ , point probability      |                         |
| "at least k"                                                                      | $P(X \geq k) = 1 - P(X \leq k-1)$ |                         |
| "at most k"                                                                       | $P(X \leq k)$                     |                         |
| "more than k"                                                                     | $P(X > k) = 1 - P(X \leq k)$      | NOTE the boundary moves |
| "no" / "none"                                                                     | $P(X=0)$                          |                         |
| "between a and b" depends on $\leq$ vs $<$ , for continuous rv it does not matter |                                   |                         |
| "expected number" $N \times P$ , expected COUNT not probability                   |                                   |                         |
| "on average"                                                                      | mean                              |                         |
| "spread"/"variation" variance or sd, read which unit is wanted                    |                                   |                         |
| "consistent with" a distributional claim, an applicability test                   |                                   |                         |
| "using the approximation" Poisson with $I = np$ , not the exact binomial          |                                   |                         |

# 03 LINGUISTICS: the question wording decoder

## THE DECODER FLOWCHART



## THE BOUNDARY-WORD MACHINE

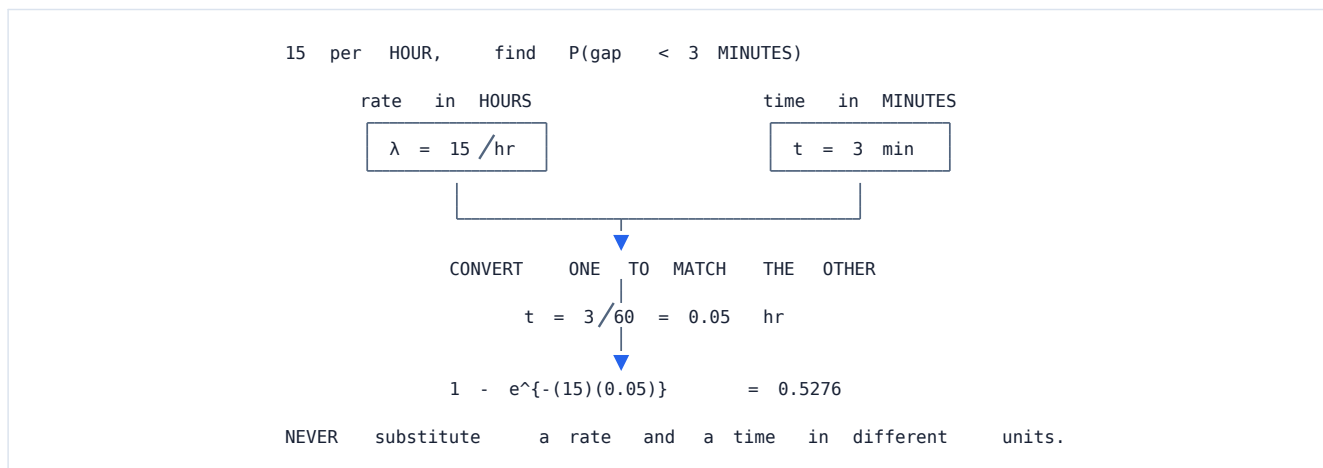
number line with the boundary:

... k-1      k      k+1 ...  
 $P(X \leq k-1)$      $P(X=k)$      $P(X \geq k+1)$

|               |            |                         |
|---------------|------------|-------------------------|
| "at least k"  | = includes | $k = 1 - P(X \leq k-1)$ |
| "more than k" | = excludes | $k = 1 - P(X \leq k)$   |
| "at most k"   | = includes | $k = F(k)$              |
| "less than k" | = excludes | $k = F(k-1)$            |

for a CONTINUOUS rv these boundaries do NOT matter ( $P(X=k)=0$ )  
 for a DISCRETE rv they are worth marks

## THE UNIT TRAP, DRAWN



The setters reuse a small vocabulary. Almost every wording maps to a fixed ask and a fixed method. This file is the map from phrase to ask. Read it before a drill pass.

### 1. The trigger words, mapped to the ask

| wording in the question              | the ask                         | the move                                              |
|--------------------------------------|---------------------------------|-------------------------------------------------------|
| "exactly k"                          | $P(X = k)$                      | direct pmf / pdf point value                          |
| "at least k" / "not less than k"     | $P(X \geq k)$                   | complement, $1 - P(X \leq k-1)$                       |
| "at most k" / "no more than k"       | $P(X \leq k)$                   | cdf directly                                          |
| "more than k" / "exceeds k"          | $P(X > k)$                      | complement, $1 - P(X \leq k)$                         |
| "fewer than k"                       | $P(X < k)$                      | cdf at k-1 (discrete)                                 |
| "none" / "no" / "zero of them"       | $P(X = 0)$                      | single pmf term                                       |
| "at least one"                       | $P(X \geq 1)$                   | complement, $1 - P(X=0)$                              |
| "between a and b"                    | $P(a < X < b)$                  | continuous: $F(b)-F(a)$                               |
| integer: $F(b-1)-F(a)$               |                                 |                                                       |
| "not more than ... apart"            | $ X - c  \leq d$                | interval then probability                             |
| "within k of the mean"               | $ X - \mu  \leq k$              | interval or Chebyshev                                 |
| "deviates by at least k"             | $ X - \mu  \geq k$              | Chebyshev tail form                                   |
| "is at least ..."                    | lower bound                     | Chebyshev COMPLEMENT form                             |
| "expected number" / "expected value" | expectation                     | $N \times P$ if a count of items                      |
| "on the average"                     | mean                            | $1/l$ for exponential                                 |
| "find k" / "find the constant"       | normalisation                   | integral or sum = 1                                   |
| "is it a valid pdf"                  | check                           | integral over support = 1                             |
| "how large a sample"                 | n from SE                       | $SE = \sigma / \sqrt{n}$ rearrange                    |
| "the probability that the mean ..."  | sampling distribution           | use SE in the denominator                             |
| "the average of n ..."               | $\bar{X}$                       | SE denominator, NOT sigma                             |
| "is the estimator unbiased"          | $E(\theta\text{-hat}) = \theta$ | compute E                                             |
| "most efficient"                     | min variance                    | among the unbiased only                               |
| "is it consistent"                   | large-sample claim              | $E \rightarrow \theta$ and $\text{Var} \rightarrow 0$ |

## 2. The boundary words, exact

|                                                                            |             |                                   |                   |
|----------------------------------------------------------------------------|-------------|-----------------------------------|-------------------|
| "at least k"                                                               | includes k. | $P(X \geq k) = 1 - P(X \leq k-1)$ |                   |
| "more than k" EXCLUDES k.                                                  | $P(X > k)$  | $= 1 - P(X \leq k)$               |                   |
| "at most k"                                                                | includes k. | $P(X \leq k) = F(k)$              |                   |
| "less than k" excludes k.                                                  | $P(X < k)$  | $= F(k-1)$                        | for an integer rv |
| for a CONTINUOUS rv these boundaries do not matter, $P(X=k)=0$ either way. |             |                                   |                   |
| for a DISCRETE rv they are worth marks. Read the word twice.               |             |                                   |                   |

## 3. The story templates (a story is a distribution in disguise)

| story words                                                                          | distribution           |
|--------------------------------------------------------------------------------------|------------------------|
| "n independent trials", "each trial", "success" binomial                             |                        |
| "number of arrivals", "calls", "defects per unit", "arrivals in an interval" Poisson |                        |
| "time between arrivals", "waiting time", "lifetime", "until failure" exponential     |                        |
| "equally likely", "random point in an interval", "uniformly" uniform                 |                        |
| "heights/weights/marks are normally distributed", "approximately normal" normal      |                        |
| "no memory", "as good as new", "does not age"                                        | exponential memoryless |
| "without replacement", "from the lot", "sample of k from N items"                    | hypergeometric         |
| "how large a sample", "how many observations"                                        | SE/CLT inverse         |
| "the whole population of ... ", "draws of n"                                         | sampling distribution  |

## 4. The instruction verbs (what the marker wants)

|                                  |                                                     |
|----------------------------------|-----------------------------------------------------|
| "find" / "compute" / "calculate" | a number, show the working, two lines of setup      |
| "show that" / "prove"            | a derivation, every algebra step written            |
| "state"                          | the formula or the theorem, no derivation needed    |
| "verify" / "check that"          | substitute and confirm, one line                    |
| "hence" / "using the above"      | reuse the earlier part, do not restart              |
| "compare" / "comment"            | state which is bigger/smaller and why, one sentence |
| "state the assumptions"          | list the conditions (n, independence, p constant)   |
| "explain whether"                | a true/false verdict WITH one reason                |

## 5. The words that decide which formula wins

|                                                                                            |                                                                                            |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| "independent"                                                                              | opens the product rule $E(XY)=E(X)E(Y)$ ,<br>$\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$ |
| "not independent"                                                                          | do not factor $E(XY)$ or drop covariance without more information.                         |
| A joint distribution or a stated covariance can still make it solvable.                    |                                                                                            |
| "population is normal" opens the exact-Xbar fallback at any n                              |                                                                                            |
| "large sample" / "n=50, n=40"                                                              | opens CLT with an approximate answer                                                       |
| "small sample, non-normal"                                                                 | do not assume a normal approximation without justification                                 |
| "known sigma"                                                                              | use $\sigma/\sqrt{n}$ directly                                                             |
| "sample standard deviation"                                                                | use $S/\sqrt{n}$                                                                           |
| "rate"                                                                                     | Poisson or exponential, never a mean                                                       |
| "mean time"                                                                                | reciprocal of the rate, check unit                                                         |
| "variance is 25"                                                                           | $\sigma = 5$ , square root it before z                                                     |
| THE UNIT TRAP: exponential and CLT questions mix minutes and hours, or hours and days.     |                                                                                            |
| Convert so the rate and the time share a unit BEFORE computing. ("15 per hour, 3 minutes") |                                                                                            |

## 6. The three question shapes the paper reuses

|                                                            |               |                                                              |
|------------------------------------------------------------|---------------|--------------------------------------------------------------|
| shape 1                                                    | single slot   | "find $P(X=k)$ " or "find $E(X)$ ". One formula, one answer. |
| shape 2                                                    | chain         | "find k, then the mean, then the probability". Each part     |
| uses the previous part's number. Do NOT restart.           |               |                                                              |
| shape 3                                                    | applicability | "which of these estimators is unbiased, and which is most    |
| efficient". A protocol, not a computation: unbiased first, |               |                                                              |
| then compare variances of the survivors.                   |               |                                                              |

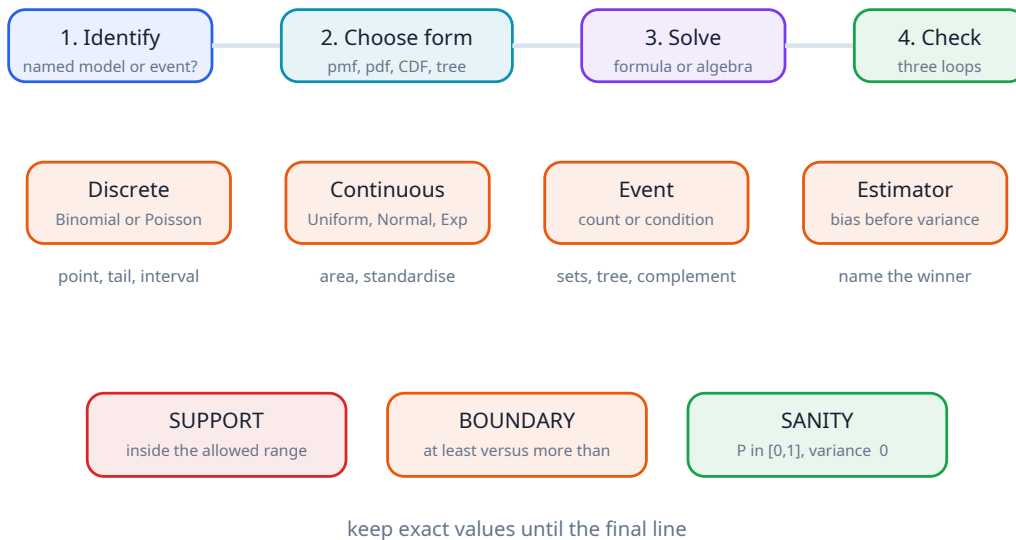
## 7. Words that appear in our corpus questions and what they meant (real examples)

|                                                                                              |                                                 |
|----------------------------------------------------------------------------------------------|-------------------------------------------------|
| "the expected number of sets with no even number"                                            | $N \times P$ , expected COUNT not a probability |
| "find the value of the constant $c$ such that $P(\dots) \leq 0.04$ "                         | Chebyshev, solve $1/k^2 = 0.04$                 |
| "using Chebyshev's inequality find the lower bound"                                          | complement form $1 - 1/k^2$                     |
| "determine a number $b$ such that $P(X < b) = P(X > b)$ "                                    | median via symmetry, $b = \text{midpoint}$      |
| "check that $f(x)$ is a pdf, and determine $k$ "                                             | normalise first, THEN answer the second ask     |
| "find the number of failures before the first success" geometric count, check the convention |                                                 |
| "probability that exactly two arrive in each minute"                                         | nest: Poisson per minute, then binomial         |
| over the minutes                                                                             |                                                 |
| "find the minimum value of $n$ "                                                             | solve an inequality in $n$ , not an equality    |
| "find the mean and variance of $X$ "                                                         | $E(X)$ , then $E(X^2) - [E(X)]^2$ , show both   |
| "is $T$ an unbiased estimator of $\theta$ "                                                  | compute $E(T)$ , compare to $\theta$ , verdict  |

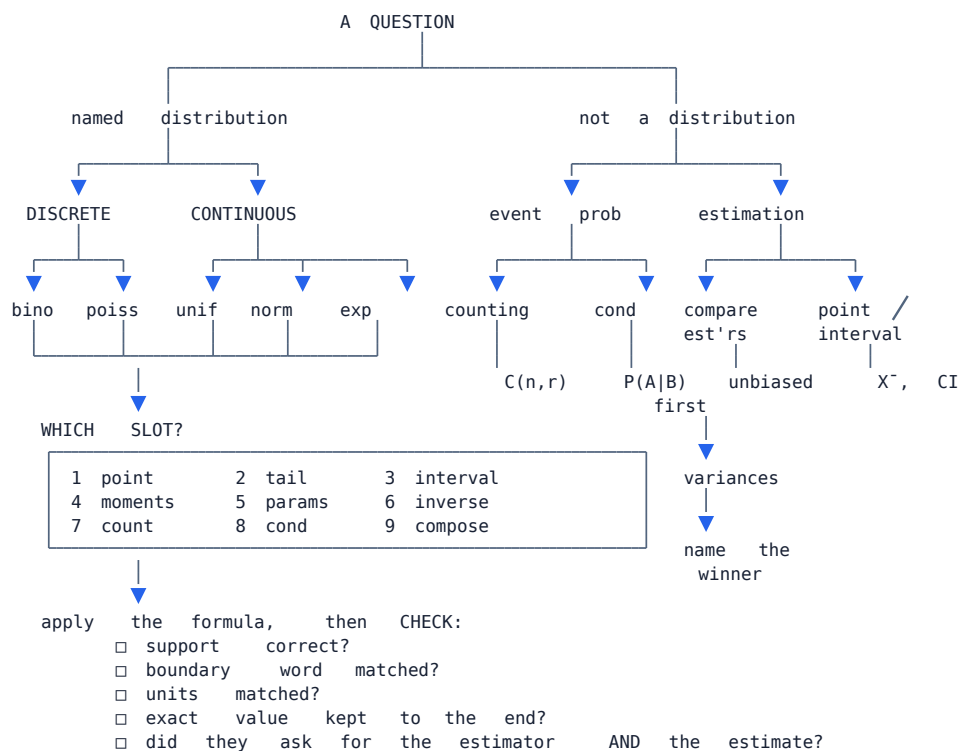
# 04 METHODS: solving path per question type

## THE MASTER DECISION TREE

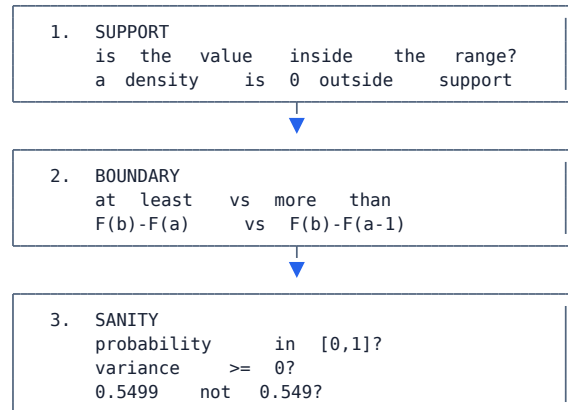
### Question-solving decision path



### Question-solving decision path



## THE THREE CHECK LOOPS AFTER ANY ANSWER



Every method here is the one the course teaches, plus the fastest valid alternative. Each entry: when to use it, the steps, and the alternate with its tradeoff.

### 1. Probability of an event

|                                                                                  |                                                                            |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| when                                                                             | any "find the probability that ..." that is not a named distribution       |
| steps                                                                            | 1 name the sample space and count its size (ordered if the count needs it) |
| 2 write the event as a set of outcomes, or as a complement                       |                                                                            |
| 3 if complement is easier, use $P(A) = 1 - P(A')$                                |                                                                            |
| 4 for "at least one" always go through the complement                            |                                                                            |
| alternate                                                                        | direct counting vs complement. Complement wins whenever the event says     |
| "at least", "none", "all of them". Direct counting wins for small explicit sets. |                                                                            |

### 2. Conditional and independence

|                                                                                     |                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| when                                                                                | "given that", "if ... what is the probability", or a two-stage draw        |
| steps                                                                               | 1 identify the condition B                                                 |
| 2 write $P(A B) = P(A \cap B)/P(B)$                                                 |                                                                            |
| 3 get the intersection from the multiplication rule if the story gives conditionals |                                                                            |
| alternate                                                                           | a tree diagram. Slower to draw but it stops the $P(A B)$ vs $P(B A)$ swap. |
| independence test                                                                   | $P(A \cap B) = P(A)P(B)$ , or $P(A B) = P(A)$ . Either form, one line.     |



### 3. Build a distribution (pmf or pdf) from a story

|                                                                                  |                                                                             |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| when                                                                             | "find the probability distribution of X", or "X has pdf $f(x)=\dots$ "      |
| steps                                                                            | 1 write the SUPPORT of X first                                              |
| 2 for discrete, tabulate x and $p(x)$ , then verify $\text{sum} = 1$             |                                                                             |
| 3 for continuous, write $f(x)$ with its range, then verify $\text{integral} = 1$ |                                                                             |
| alternate                                                                        | if the story is a named distribution, skip the build and name it (binomial, |
| Poisson, etc), then quote pmf and moments.                                       |                                                                             |

### 4. Find the constant k

|                                                                                       |                                                                                                |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| when                                                                                  | "find k such that $f(x)$ is a density" or "determine k"                                        |
| steps                                                                                 | 1 write the normalisation, integral (or sum) over the SUPPORT = 1                              |
| 2 integrate the polynomial, collect k                                                 |                                                                                                |
| 3 solve for k                                                                         |                                                                                                |
| traps                                                                                 | the support is finite here, not $(-\infty, \infty)$ . A scheme that prints infinite bounds and |
| then evaluates on a finite support is printing a slip (see errata 20).                |                                                                                                |
| the sum version: watch for a $k^2$ term, the equation can be quadratic with one valid |                                                                                                |
| positive root (the batch-1 assignment gives $k = 1/10$ from $10k^2 + 9k = 1$ )        |                                                                                                |

### 5. cdf from pmf or pdf, and back

|                                                                                    |                                                                                             |                                                                                           |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| when                                                                               | "find the cdf", "find $F(x)$ ", "given $F(x)$ find $f(x)$ ", "find $P(X < 6)$ from the cdf" |                                                                                           |
| steps                                                                              | pmf $\rightarrow$ cdf                                                                       | accumulate: $F(x) = \text{sum of } p \text{ up to } x$ . Piecewise on the INTEGER values. |
| pdf $\rightarrow$ cdf                                                              | integrate $f$ from the lower support edge to $x$                                            |                                                                                           |
| cdf $\rightarrow$ pdf                                                              | differentiate $F$ on each interval                                                          |                                                                                           |
| cdf $\rightarrow$ pmf                                                              | the jump $F(x) - F(x-)$                                                                     |                                                                                           |
| traps                                                                              | a cdf accumulates ALL earlier mass. For a piecewise cdf on $1 \leq x \leq 2$ you must add   |                                                                                           |
| the mass from 0 to 1 (errata 19: the 2024 MTE CDF is option D, not the scheme's B) |                                                                                             |                                                                                           |
| the minimum $c$ with $F(c) > 1/2$ is a cdf query, read it off the accumulation     |                                                                                             |                                                                                           |

## 6. Expectation

|                                                                                |                                                                                   |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| when                                                                           | "find $E(X)$ ", "find the expected value", "find $E[h(X)]$ "                      |
| steps                                                                          | 1 write the sum (discrete) or integral (continuous), over the support             |
| 2 plug, integrate or sum                                                       |                                                                                   |
| 3 for $E(X^2)$ repeat with $x^2$                                               |                                                                                   |
| alternate                                                                      | symmetry: a density symmetric about $c$ has $E(X) = c$ with no integration.       |
| traps                                                                          | $E[g(X)] \neq g(E(X))$ . Compute the integral of $g$ , do not plug $E$ into $g$ . |
| for a count of items: expected count = $N \times P$ , do not report $P$ alone. |                                                                                   |

## 7. Variance

|                                                                             |                                                                                           |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| when                                                                        | "find the variance", "find the sd", "find $E(X)$ and $\text{Var}(X)$ "                    |
| steps                                                                       | 1 $E(X)$                                                                                  |
| 2 $E(X^2)$ from its own sum/integral                                        |                                                                                           |
| 3 $\text{Var} = E(X^2) - [E(X)]^2$                                          |                                                                                           |
| 4 sd = $\sqrt{\text{Var}}$ when asked for spread in original units          |                                                                                           |
| alternate                                                                   | for a named distribution quote $np$ , $npq$ , $l$ , $(b-a)^2/12$ , $1/l^2$ directly.      |
| traps                                                                       | keep EXACT values until the final line (errata 12: rounding $\text{Var}$ then rooting it) |
| gives 0.975, the exact sd is 0.9734 $\rightarrow$ 0.97)                     |                                                                                           |
| the variance is never negative, if you get a negative check your arithmetic |                                                                                           |

## 8. Chebyshev

|                                                                                    |                                                                                |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| when                                                                               | "using Chebyshev's inequality", "find the lower bound", "without assuming a    |
| distribution", "is this distribution-free"                                         |                                                                                |
| steps                                                                              | 1 get $\mu$ and $\sigma$ (compute if not given, $\sigma = \sqrt{\text{Var}}$ ) |
| 2 convert the event to $ X - \mu  \geq \epsilon$ or $< \epsilon$                   |                                                                                |
| 3 $k = \epsilon / \sigma$                                                          | (raw units to sigma units)                                                     |
| 4 tail form $1/k^2$ , or complement form $1 - 1/k^2$ , whichever the wording wants |                                                                                |
| alternate                                                                          | if the exact distribution is known, compute the exact value TOO and report     |
| "bound 35/54, exact 1/3, the bound holds and is wider"                             |                                                                                |
| traps                                                                              | $k$ must be in sigma units. The complement form is for "at least"/"within".    |
| The bound can exceed 1 ( $k \leq 1$ ), then it is vacuous.                         |                                                                                |
| errata 15: the L10-11 deck's Q3 slide mis-states its own row, use Q1 and Q2 only.  |                                                                                |

## 9. Binomial

|                                                                                   |                                                                      |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| when                                                                              | n trials, two outcomes, independent, p constant, count the successes |
| steps                                                                             | 1 state $X \sim B(n, p)$ , write $q = 1-p$                           |
| 2 the ask: point, tail (complement), or interval                                  |                                                                      |
| 3 substitute the pmf, use $C(n, x)$                                               |                                                                      |
| alternate                                                                         | for "at least 1" use $1 - q^n$ (faster than summing)                 |
| traps                                                                             | "at least 2" = $1 - P(0) - P(1)$ , the classic two-term complement   |
| read "p is the success probability", if the story gives failures, invert it first |                                                                      |

## 10. Poisson

|                                                                                   |                                                                                           |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| when                                                                              | rare events, arrivals, in a fixed window, or "use the approximation" with n big p small   |
| steps                                                                             | 1 get l. If given a rate, scale it to the window. If given np in an approximation,        |
| $l = np$ .                                                                        |                                                                                           |
| 2 the ask: point $e^{-l} l^x / x!$ , tail (complement), or the zero case $e^{-l}$ |                                                                                           |
| 3 state $\text{Var} = l$ if asked, it equals the mean                             |                                                                                           |
| alternate                                                                         | for the ratio question $P(X=1) = c P(X=2)$ , the $e^{-l}$ cancels, leaving $l = 2/c$      |
| traps                                                                             | rate vs window scaling (2 per minute, then 5 minutes $\rightarrow l =$ the window's rate) |
| the nesting: Poisson per minute THEN binomial over the minutes (0.00145 question) |                                                                                           |
| a Poisson variable has equal mean and variance, but equality alone does not prove |                                                                                           |
| an otherwise unspecified distribution is Poisson                                  |                                                                                           |

## 11. Uniform

|                                                                                |                                                                                     |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| when                                                                           | "equally likely in an interval", "random point"                                     |
| steps                                                                          | 1 identify a and b                                                                  |
| 2 probability = length ratio, $P(a \leq X \leq b) = (\text{interval}) / (b-a)$ |                                                                                     |
| 3 moments: mean $(a+b)/2$ , variance $(b-a)^2/12$                              |                                                                                     |
| traps                                                                          | clip to the support, a value outside $[a, b]$ has probability 0 or 1, not the ratio |
| absolute-value events become intervals first: $ X-5  < 1$ is $4 < X < 6$       |                                                                                     |

## 12. Normal

|                                                                                         |                                                                         |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| when                                                                                    | normal, approximately normal, large-sample mean, or the CLT destination |
| steps                                                                                   | 1 identify $\mu$ and $\sigma$ ( $\sigma$ from variance, sqrt it)        |
| 2 DECIDE which table the question gives ( $\phi$ area-from-0, or F cumulative)          |                                                                         |
| 3 standardise $Z = (X - \mu) / \sigma$                                                  |                                                                         |
| 4 read the table, apply symmetry $\Phi(-z) = 1 - \Phi(z)$                               |                                                                         |
| 5 for the inverse: get $z$ for the probability, then $X = \mu + z \sigma$               |                                                                         |
| alternate                                                                               | landmarks: within 1s 68.27, 2s 95.45, 3s 99.73, for a fast sanity check |
| traps                                                                                   | the second parameter is VARIANCE, sqrt it                               |
| the inverse's forgotten last step: convert $z$ back to the $X$ scale                    |                                                                         |
| two-unknown questions: two given probabilities give two equations, solve simultaneously |                                                                         |
| (errata 10: use $\mu = 37.2$ , $\sigma = 28.2$ , cutoff = 30.4)                         |                                                                         |

## 13. Exponential

|                                                                                                |                                                                                   |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| when                                                                                           | waiting time, time between arrivals, lifetime, "no memory"                        |
| steps                                                                                          | 1 get $\lambda$ (a RATE). If given a mean, $\lambda = 1/\text{mean}$ .            |
| 2 match the units of $\lambda$ and $t$                                                         |                                                                                   |
| 3 point/interval: $F = 1 - e^{-\lambda t}$ .                                                   | tail: $e^{-\lambda t}$ .                                                          |
| 4 memoryless if the condition is on surviving                                                  |                                                                                   |
| alternate                                                                                      | mean-first form $f = (1/\mu) e^{-t/\mu}$ when the question gives $\mu$ directly   |
| traps                                                                                          | rate vs mean reciprocal (errata 4: the slide p040 states it backwards, read p041) |
| conditional in the exponential: $P(X < 1 \mid X < 2) = (1 - e^{-0.5}) / (1 - e^{-1}) = 0.6225$ |                                                                                   |
| (errata 9: the assignment prints 0.5679, that is wrong)                                        |                                                                                   |

## 14. Sampling and standard error

|                                                              |                                                                                                |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| when                                                         | "standard error", "sampling distribution", "the mean of a sample of $n$ "                      |
| steps                                                        | 1 SE = $\sigma / \sqrt{n}$ (or $S / \sqrt{n}$ if the sample sd is given)                       |
| 2 the "average of $n$ " template divides by SE, not $\sigma$ |                                                                                                |
| alternate                                                    | the sqrt- $n$ rule for sample-size questions: quadruple $n$ halves SE                          |
| traps                                                        | a single observation uses $\sigma$ ; an average of $n$ uses $\sigma / \sqrt{n}$ . Mixing these |
| is the commonest CLT slip.                                   |                                                                                                |

## 15. CLT

|                                                                                         |                                                                                               |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| when                                                                                    | "the sample mean is approximately normal", $n$ listed, population shape unknown               |
| steps                                                                                   | 1 if the population is normal, $\bar{X}$ is exactly normal; otherwise assess whether $n$ and  |
| the population shape make the CLT approximation reasonable                              |                                                                                               |
| 2 $\bar{X} \sim N(\mu, \sigma^2/n)$ , so $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$        |                                                                                               |
| 3 read the table as a normal                                                            |                                                                                               |
| alternate                                                                               | for a SUM not a mean, the sum is $N(n\mu, n\sigma^2)$ , $\sigma$ scales by $\sqrt{n}$ not $n$ |
| traps                                                                                   | $n < 30$ with a non-normal population: do not use the normal approximation automatically      |
| $n \geq 30$ is the course heuristic, not a theorem boundary                             |                                                                                               |
| a normal population is exact at any $n$ , this is the fallback for small- $n$ questions |                                                                                               |
| (errata 6: impurity $Z_2 = -0.94$ , the slide prints $-0.4$ )                           |                                                                                               |

## 16. Estimator comparison (the 6 to 8 mark question)

|                                                                                                            |                                                                                    |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| when                                                                                                       | "which estimator is unbiased", "most efficient", "find the constant"               |
| steps                                                                                                      | 1 write $E$ of each estimator, expand using linearity                              |
| 2 unbiased: $E(T) = \theta$ . Solve for the constant if one exists.                                        |                                                                                    |
| 3 for each UNBIASED estimator, compute $\text{Var}(T)$ using $\text{Var}(X_i) = \sigma^2$ and independence |                                                                                    |
| 4 the smallest variance wins                                                                               |                                                                                    |
| alternate                                                                                                  | relative efficiency $\text{Var}(T_2)/\text{Var}(T_1)$ when asked to compare a pair |
| traps                                                                                                      | a biased estimator is DISQUALIFIED before the variance comparison, even if its     |
| variance is smaller (errata 18, mock B4: $T_4$ is biased and third of four, the note                       |                                                                                    |
| calling it "smallest" is wrong)                                                                            |                                                                                    |
| consistency is a large- $n$ property, a fixed- $n$ small variance does not make a                          |                                                                                    |
| consistent estimator                                                                                       |                                                                                    |

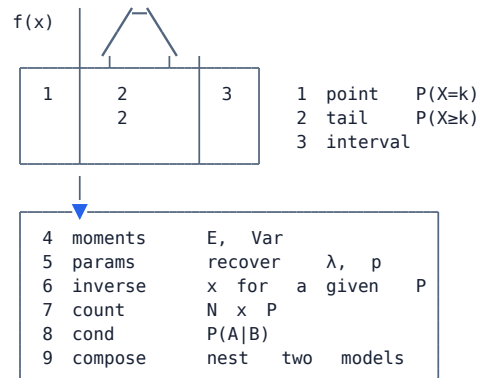
## 17. The protocol when a question chains parts

|                                                                                   |                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|
| when                                                                              | "find $k \dots$ hence find the mean ... hence find $P(X > \dots)$ " |
| steps                                                                             | 1 treat each part as independent given the previous answer          |
| 2 carry the EXACT value forward, round only at the final line                     |                                                                     |
| 3 if a part is impossible, say so, do not silently continue (the asgn-variant has |                                                                     |
| a row and a key that cannot both be right, errata 5.1)                            |                                                                     |

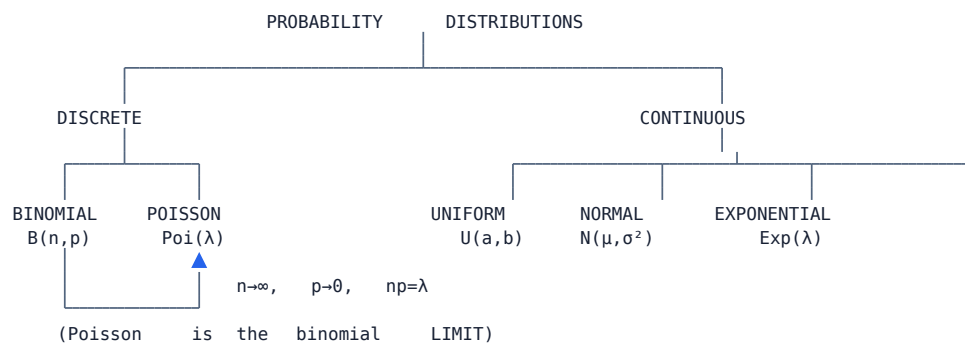
# 05 DISTRIBUTIONS: all five, all slots, all values

## THE NINE SLOTS, VISUALISED (every distribution)

a distribution is a SHAPE, and the exam asks 9 questions of it:



## THE DISTRIBUTION FAMILY TREE



## MEAN vs VARIANCE, EVERY DISTRIBUTION

| MEAN                                              | VARIANCE      |                                         |                           |
|---------------------------------------------------|---------------|-----------------------------------------|---------------------------|
| $B(n,p)$                                          | $  np$        | $  npq$                                 |                           |
| $Poi(\lambda)$                                    | $  \lambda$   | $  \lambda$                             | ← SAME!                   |
| $U(a,b)$                                          | $  (a+b)/2$   | $  (b-a)^2/12$                          |                           |
| $N(\mu,\sigma^2)$                                 | $  \mu$       | $  \sigma^2$                            | ← square                  |
| $Exp(\lambda)$                                    | $  1/\lambda$ | $  1/\lambda^2$                         | ← recip                   |
| the memory hooks:                                 |               |                                         |                           |
| binomial                                          | ->            | np and npq                              | (the q is the only extra) |
| Poisson                                           | ->            | ONE parameter does both jobs            |                           |
| uniform                                           | ->            | span <sup>2</sup> / 12                  |                           |
| normal                                            | ->            | second slot is VARIANCE, square-root it |                           |
| exponential -> reciprocal pair: rate in, time out |               |                                         |                           |

For each distribution: what it models, the setup, pmf/pdf, cdf, mean, variance, sd, the standardisation if any, the table if any, and every question slot the corpus has used. The slot names are the same nine in every table so you can see the pattern (full pattern logic in 07-PATTERNS.md).

### 1. Binomial, $X \sim B(n, p)$

|                                                                                                                            |                                                   |                                                                            |                                  |            |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------|----------------------------------|------------|
| models                                                                                                                     |                                                   | number of successes in n independent two-outcome trials                    |                                  |            |
| setup check                                                                                                                |                                                   | two outcomes, finite n, trials independent, p constant (ppt3 p011)         |                                  |            |
| pmf                                                                                                                        |                                                   | $P(X=x) = C(n,x) p^x q^{(n-x)},$                                           | $q = 1-p,$                       | $x = 0..n$ |
| cdf                                                                                                                        |                                                   | $P(X \leq x) = \sum_{i=0..x} C(n,i) p^i q^{(n-i)}$                         |                                  |            |
| mean                                                                                                                       |                                                   | np                                                                         |                                  |            |
| variance                                                                                                                   |                                                   | npq                                                                        |                                  |            |
| sd                                                                                                                         |                                                   | $\sqrt{npq}$                                                               |                                  |            |
| mode                                                                                                                       |                                                   | $\text{floor}((n+1)p)$ , with two adjacent modes if $(n+1)p$ is an integer |                                  |            |
| approximation -> Poisson in the limit $n \rightarrow \infty$ , $p \rightarrow 0$ , $np \rightarrow \text{finite } \lambda$ |                                                   |                                                                            |                                  |            |
| slot                                                                                                                       | ask                                               |                                                                            | used in                          |            |
| 1 point                                                                                                                    | $P(X=2)$ out of 12, $p=0.1$                       |                                                                            | ppt3 p013-016 pens               |            |
| 2 tail                                                                                                                     | $P(X \geq 2) = 1 - P_0 - P_1$                     |                                                                            | ppt3 p013-016 pens               |            |
| 3 interval                                                                                                                 | $P(2 \leq X \leq 4)$                              |                                                                            | NOT asked, high risk (see 07)    |            |
| 4 moments                                                                                                                  | $E=np, \text{Var}=npq$                            |                                                                            | A2 A2, A9, D3                    |            |
| 5 params                                                                                                                   | p from $P(X=5)=2P(X=4) \rightarrow 5/8$           |                                                                            | ppt3 p017-018                    |            |
| 6 inverse                                                                                                                  | smallest n for a guarantee                        |                                                                            | NOT asked, high risk             |            |
| 7 count                                                                                                                    | $N \times P$ , 10000 sets $x (3/8)^{10} = 0.5499$ |                                                                            | ppt3 p018                        |            |
| 8 conditional                                                                                                              | $P(X=k \mid X \geq j)$                            |                                                                            | sibling drill, not an A2 B5 item |            |

| slot      | ask                                     | used in   |
|-----------|-----------------------------------------|-----------|
| 9 compose | Poisson per minute then binomial over 5 | ppt3 p026 |

## 2. Poisson, $X \sim \text{Poi}(I)$

|                        |                                                        |                                       |
|------------------------|--------------------------------------------------------|---------------------------------------|
| models                 | counts of rare events in a fixed interval              |                                       |
| setup check            | rare, independent in time/space, rate constant         |                                       |
| pmf                    | $P(X=x) = e^{-I} I^x / x!$                             | $x = 0, 1, 2, \dots$                  |
| cdf                    | $P(X \leq x) = e^{-I} \sum_{i=0..x} I^i / i!$          |                                       |
| mean                   | $I$                                                    |                                       |
| variance               | $I$                                                    | (mean EQUALS variance, the signature) |
| sd                     | $\sqrt{I}$                                             |                                       |
| additivity             | sum of independent Poissons is Poisson with $I$ summed |                                       |
| from binomial $I = np$ |                                                        |                                       |
| slot                   | ask                                                    | used in                               |
| 1 point                | $P(X=4), I=5 \rightarrow 0.1755$                       | ppt3 p027 (slide prints 0.1745)       |
| 2 tail                 | $P(X>2) = 1 - P_0 - P_1 - P_2 = 0.8753$                | A2 C3                                 |
| 2 tail                 | "rejected" = $P(X \geq 4) = 0.2424$ when $I=2.5$       | A2 B1                                 |
| 3 interval             | $P(2 \leq X \leq 4)$                                   | NOT asked, medium risk                |
| 4 moments              | $E = \text{Var} = I$ , the signature test              | MCQ (R25S4-A2)                        |
| 5 params               | $I$ from $P(X=1)=0.2, P(X=2) \rightarrow I=10$         | ppt3 p025                             |
| 6 inverse              | not used                                               | low evidence                          |
| 7 count                | expected defective bottles 0.1 percent                 | E25SUM-Q17                            |
| 9 compose              | nest inside binomial, $32 e^{-10} = 0.00145$           | ppt3 p026                             |

## 3. Uniform, $X \sim U(a, b)$

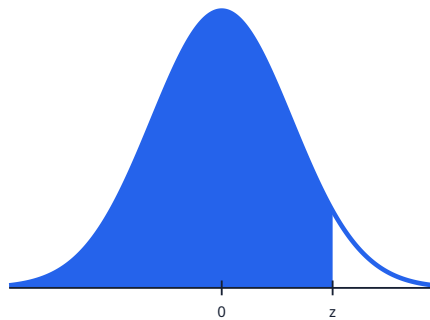
|            |                                                           |                                   |
|------------|-----------------------------------------------------------|-----------------------------------|
| models     | a random point in an interval, all lengths equally likely |                                   |
| pdf        | $f(x) = 1/(b-a)$ ,                                        | $a \leq x \leq b$                 |
| cdf        | $F(x) = 0$ below $a$ ,                                    | $(x-a)/(b-a)$ inside, 1 above $b$ |
| mean       | $(a+b)/2$                                                 |                                   |
| variance   | $(b-a)^2 / 12$                                            |                                   |
| sd         | $(b-a)/\sqrt{12}$                                         |                                   |
| shortcut   | probability = length ratio, no integration needed         |                                   |
| slot       | ask                                                       | used in                           |
| 1 point    | $f(x)$ value, and mean/variance                           | ppt4 p003-006                     |
| 2 tail     | $P(X>k)$ as a length ratio                                | ppt4 p006                         |
| 3 interval | $P(3<X<5) = 2/4 = 0.50$                                   | ppt4 p006                         |
| 4 moments  | $E=(a+b)/2, \text{Var}=(b-a)^2/12$                        | A2 A4                             |
| 5 params   | solve $a, b$ from mean and variance                       | A2 A10 variant                    |
| 6 inverse  | $K$ such that $P(X<K)=p$                                  | NOT asked, medium risk            |
| 9 compose  | sum of two stated uniforms                                | notes p126 example                |



## 4. Normal, $X \sim N(\mu, \sigma^2)$

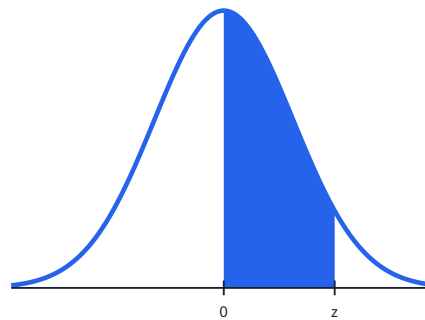
### Two normal-table conventions

p022: cumulative  $P(Z < z)$



$z=2: 0.9772$

p023: centre area  $P(0 < Z < z)$



$z=2: 0.4772$

### Two normal-table conventions

|               |                                                                                |                          |
|---------------|--------------------------------------------------------------------------------|--------------------------|
| models        | heights, weights, marks, measurement error, and the CLT destination            |                          |
| pdf           | $f(x) = 1/(\sigma \sqrt{2\pi}) \exp(-(x-\mu)^2/(2\sigma^2))$                   |                          |
| notation      | $X \sim N(\mu, \sigma^2)$ , the SECOND slot is VARIANCE                        |                          |
| standardise   | $Z = (X-\mu)/\sigma$ ,                                                         | $Z \sim N(0,1)$          |
| probability   | $P(a < X < b) = \Phi((b-\mu)/\sigma) - \Phi((a-\mu)/\sigma)$                   |                          |
| inverse       | $X = \mu + Z \sigma$                                                           |                          |
| symmetry      | $\Phi(-z) = 1 - \Phi(z)$ ,                                                     | $P(Z > z) = 1 - \Phi(z)$ |
| shape         | mean = median = mode = $\mu$ , tails never touch, $\sigma$ widens the curve    |                          |
| landmarks     | 68.27 within 1s, 95.45 within 2s, 99.73 within 3s                              |                          |
| slot          | ask                                                                            | used in                  |
| 1 point       | density value, symmetry facts                                                  | ppt4 p010-011            |
| 2 tail        | $P(X > 8.6) = 0.4522$                                                          | ppt4 p032-033            |
| 3 interval    | $P(45 < X < 62)$ , impurity=0.1645 by rounded table A2 B3, clt p013 (errata 6) |                          |
| 4 moments     | $E=\mu$ , $\text{Var}=\sigma^2$ , identify both                                | E25S4-A4, E25S3-A5       |
| 5 params      | two unknowns from two probabilities                                            |                          |
| 6 inverse     | $x$ for a 20 percent lower tail = 3.792                                        | ppt4 p034-037            |
| 7 count       | $N \times P$ , 5000 batteries, 10000 bulbs                                     |                          |
| 8 conditional | $P(X > a \mid X > \dots)$ normal ratio                                         | not used, low evidence   |
| 9 compose     | CLT $\rightarrow$ normal, then a table read                                    |                          |
|               |                                                                                | clt p013, p015           |

## 5. Exponential, $T \sim \text{Exp}(\lambda)$

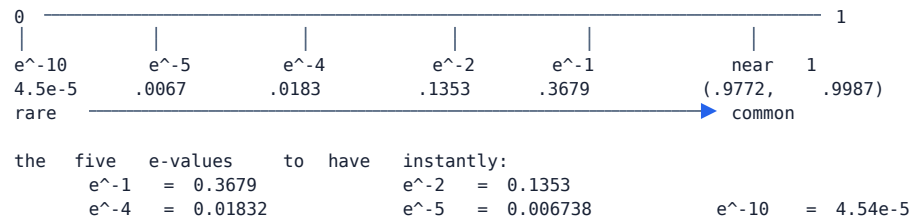
|                                                                                                 |                                                                                              |                                               |                                 |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------|
| models                                                                                          | waiting time between events, lifetimes, "until failure"                                      |                                               |                                 |
| pdf                                                                                             | $f(t) = \lambda e^{-\lambda t},$                                                             | $t > 0$                                       | ( $\lambda$ is a RATE)          |
| cdf                                                                                             | $F(t) = 0$ for $t \leq 0$ ; $1 - e^{-\lambda t}$ for $t > 0$                                 |                                               |                                 |
| tail                                                                                            | $P(T > t) = 1$ for $t < 0$ ; $e^{-\lambda t}$ for $t \geq 0$                                 |                                               |                                 |
| mean                                                                                            | $1/\lambda$                                                                                  |                                               |                                 |
| variance                                                                                        | $1/\lambda^2$                                                                                |                                               |                                 |
| sd                                                                                              | $1/\lambda$                                                                                  |                                               |                                 |
| memoryless                                                                                      | $P(T > s+t \mid T > s) = P(T > t)$                                                           |                                               |                                 |
| mean-first form                                                                                 | $f(t) = (\lambda) e^{-\lambda t}$ when the question gives $\lambda$ , then $\lambda = 1/\mu$ |                                               |                                 |
| slot                                                                                            |                                                                                              | ask                                           | used in                         |
| 1 point                                                                                         |                                                                                              | f value, pdf identification                   | E24S4-A4                        |
| 2 tail                                                                                          |                                                                                              | $P(T > 12) = 5e^{-4} = 0.0916$                | A1 short Q2 (errata 3)          |
| 3 interval                                                                                      |                                                                                              | $P(T < 3\text{min}) = 0.5276$ at 15/hour      | ppt4 p042-043                   |
| 4 moments                                                                                       |                                                                                              | $E = 1/\lambda, \text{Var} = 1/\lambda^2$     | ppt4 p043                       |
| 5 params                                                                                        |                                                                                              | $\lambda$ from a mean, or mean from $\lambda$ | A2 D2                           |
| 6 inverse                                                                                       |                                                                                              | not used, medium risk                         |                                 |
| 7 count                                                                                         |                                                                                              | not used                                      |                                 |
| 8 conditional                                                                                   |                                                                                              | $P(X < 1 \mid X < 2) = 0.6225$                | A2 B5 (errata 9: prints 0.5679) |
| 9 compose                                                                                       |                                                                                              | repair-time, memoryless, unit conversion      | A2 MCQ5, C1                     |
| ERRATUM 4: the ppt4 p040 slide says " $1/\lambda$ is the mean number per unit time", backwards. |                                                                                              |                                               |                                 |
| lambda is the rate, $1/\lambda$ is the mean TIME. p041 states it correctly.                     |                                                                                              |                                               |                                 |

## The one-table summary (memorise this block)

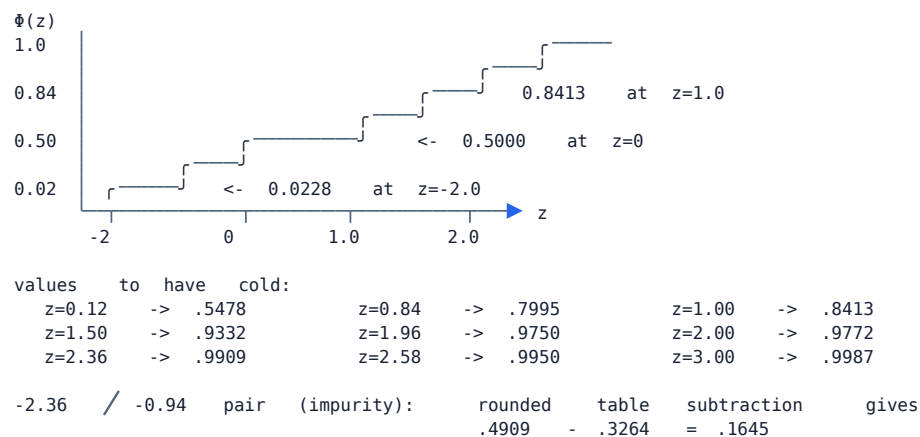
| pmf / pdf                                                                  | mean                                    | variance  |              |
|----------------------------------------------------------------------------|-----------------------------------------|-----------|--------------|
| $B(n,p)$                                                                   | $C(n,x) p^x q^{(n-x)}$                  | $np$      | $npq$        |
| $Poi(l)$                                                                   | $e^{-l} l^x / x!$                       | $l$       | $l$          |
| $U(a,b)$                                                                   | $1/(b-a)$                               | $(a+b)/2$ | $(b-a)^2/12$ |
| $N(\mu, s^2)$                                                              | $1/(s \sqrt{2\pi}) e^{-(x-\mu)^2/2s^2}$ | $\mu$     | $s^2$        |
| $Exp(l)$                                                                   | $l e^{-l t}$                            | $1/l$     | $1/l^2$      |
| memory hooks:                                                              |                                         |           |              |
| binomial: $np$ and $npq$ , the $q$ is the only extra                       |                                         |           |              |
| Poisson: mean IS variance, necessary but not enough to identify the family |                                         |           |              |
| uniform: centre and span squared over 12                                   |                                         |           |              |
| normal: $\mu$ and $\sigma^2$ , the second slot is variance                 |                                         |           |              |
| exponential: reciprocal pair, rate in, time out                            |                                         |           |              |

# 06 NUMBERS: constants, table values, landmarks, identities

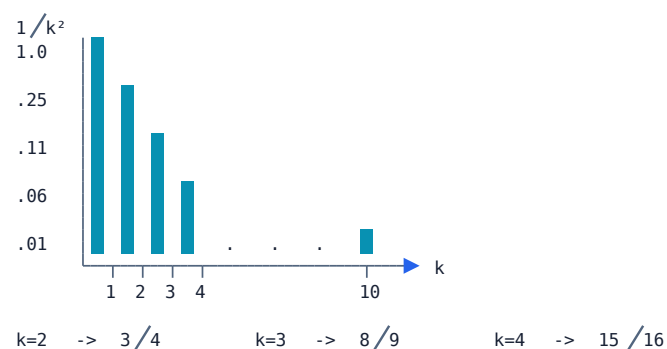
## THE NUMBER LINE OF EVERY PROBABILITY YOU WILL WRITE



## THE STANDARD NORMAL TABLE AS A PICTURE



## THE THREE CHEBYSHEV THRESHOLDS, DRAWN



Every number that can appear, with what it is and where it comes from. Grouped so you can find one fast.

## 1. Chebyshev thresholds (exact fractions)

| k                      | bound $1/k^2$                 | complement $1 - 1/k^2$ |           |
|------------------------|-------------------------------|------------------------|-----------|
| 1                      | 1                             | 0                      | (vacuous) |
| 2                      | $1/4 = 0.25$                  | $3/4 = 0.75$           |           |
| 3                      | $1/9 = 0.1111$                | $8/9 = 0.8889$         |           |
| 4                      | $1/16 = 0.0625$               | $15/16 = 0.9375$       |           |
| 5                      | $1/25 = 0.04$                 | $24/25 = 0.96$         |           |
| 10                     | $1/100 = 0.01$                | $99/100 = 0.99$        |           |
| sqrt(10) approx 3.1623 | -> $k^2 = 10$ from a 0.1 tail |                        |           |

## 2. Distribution moments worth having instantly

|                                                                                        |                                                |                                               |             |
|----------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------|-------------|
| one fair die                                                                           | $E = 7/2 = 3.5,$                               | $E(X^2) = 91/6, \text{ Var} = 35/12 = 2.9167$ |             |
| sum of two dice                                                                        | $E = 7,$                                       | $\text{Var} = 35/6 = 5.8333,$                 | sd = 2.4152 |
| B(1,p) Bernoulli                                                                       | $E = p,$                                       | $\text{Var} = pq$                             |             |
| B(12, 0.1)                                                                             | $E = 1.2,$                                     | $\text{Var} = 1.08$                           |             |
| B(600, 1/6) $E = 100,$                                                                 | $\text{Var} = 250/3 = 83.3333,$<br>sd = 9.1287 |                                               |             |
| Poi(2)                                                                                 | $E = \text{Var} = 2,$                          | sd = 1.4142                                   |             |
| Poi(10) $E = \text{Var} = 10$                                                          |                                                |                                               |             |
| U(2,6)                                                                                 | $E = 4,$                                       | $\text{Var} = 4/3 = 1.3333$                   |             |
| source example: mean 8, sd 5. The source writes N(8,5), but its arithmetic treats 5 as |                                                |                                               |             |
| sigma, not $\sigma^2$ . To avoid the ambiguity, write $\mu=8,$ $\sigma=5,$ or N(8,25). |                                                |                                               |             |
| Exp(l)                                                                                 | $E = 1/l,$                                     | $\text{Var} = 1/l^2$                          |             |
| Exp mean 2                                                                             | -> $l = 1/2$                                   |                                               |             |

### 3. Standard normal table values (cumulative, the common ones)

| z      | Phi(z) | use                              |
|--------|--------|----------------------------------|
| 0.00   | 0.5000 | centre                           |
| 0.10   | 0.5398 |                                  |
| 0.12   | 0.5478 | the 8.6 question, $P(X < 8.6)$   |
| 0.24   | 0.5948 |                                  |
| 0.50   | 0.6915 |                                  |
| 0.60   | 0.7257 |                                  |
| 0.84   | 0.7995 |                                  |
| 0.8416 | 0.8000 | inverse: a 20 percent lower tail |
| 1.00   | 0.8413 |                                  |
| 1.28   | 0.8997 | the 90 percent landmark          |
| 1.34   | 0.9099 |                                  |
| 1.50   | 0.9332 |                                  |
| 1.645  | 0.9500 | 95 percent one tail              |
| 1.96   | 0.9750 | 95 percent two tails             |
| 2.00   | 0.9772 |                                  |
| 2.36   | 0.9909 |                                  |
| 2.576  | 0.9950 | 99 percent two tails             |
| 3.00   | 0.9987 |                                  |

negative z, use  $\Phi(-z) = 1 - \Phi(z)$ , or the table's left tail directly.  
the -2.36 / -0.94 pair (impurity): rounded table subtraction gives  
 $0.4909 - 0.3264 = 0.1645$  (errata 6)

### 4. The 68 / 95 / 99.7 landmarks

|                                                                     |               |
|---------------------------------------------------------------------|---------------|
| for a normal distribution:                                          |               |
| within 1 sigma                                                      | 68.27 percent |
| within 2 sigma                                                      | 95.45 percent |
| within 3 sigma                                                      | 99.73 percent |
| (the ppt4 slides round these to 68.3, 95.4, 99.7)                   |               |
| used by: A2 MCQ Q11, and any "approximately what fraction" question |               |

## 5. Worked corpus values (the ones that reappear)

|                                      |                                                                                            |                                   |                          |
|--------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------|--------------------------|
| pens B(12,0.1)                       | $P(X=2) 0.2301$                                                                            | $P(X \geq 2) 0.3410$              | $P(X=0) 0.2824$          |
| irregular die                        | $p = 5/8$ , $P(X=0)$ over 10 = $5.499e-5$ , count over 10000 = 0.5499                      |                                   |                          |
| insurance Poi(5)                     | $P(X=4) 0.1755$                                                                            | (slide 0.1745 is wrong, errata 1) |                          |
| calls nesting                        | $32 e^{-10} = 0.0014528$                                                                   |                                   |                          |
| Poisson ratio                        | $l = 10$ , $P(X=0) = 4.54e-5$                                                              |                                   |                          |
| uniform U(2,6)                       | $P(3 < X < 5) 0.50$                                                                        |                                   |                          |
| normal $\mu=8$ , $sd=5$              | $P(X < 8.6) 0.5478$                                                                        | $P(X > 8.6) 0.4522$               | inverse 20 percent 3.792 |
| two-unknown normal                   | $\sigma = 28.2$ , $\mu = 37.2$ , cutoff 30.4                                               | (errata 10)                       |                          |
| exponential 15/hr                    | $P(T < 3 \text{ min}) 0.5276$                                                              |                                   |                          |
| exponential mean 2                   | $P(X < 1   X < 2) 0.6225$                                                                  | (slide 0.5679 wrong, errata 9)    |                          |
| exponential lifetime                 | $P(X > 12)$ with $f = x e^{-x/3} / 9 \rightarrow 5e^{-4} = 0.0916$ (errata 3)              |                                   |                          |
| clt lightbulbs                       | SE 20 at $n=25$ , SE 10 at $n=100$                                                         |                                   |                          |
| clt ATM                              | $Z = 1.50$ , $P = 0.0668$                                                                  |                                   |                          |
| clt impurity                         | $Z_1 -2.36$ , $Z_2 -0.94$ . Rounded table areas give 0.1645;                               |                                   |                          |
| unrounded z values give about 0.1637 | (errata 6)                                                                                 |                                   |                          |
| clt LED                              | $Z = -2.00$ , $P = 0.0228$                                                                 |                                   |                          |
| clt machine $n=9$                    | $Z_1 -1.80$ , $Z_2 0.60$ , answer 0.6898                                                   |                                   |                          |
| estimator set 1                      | $\lambda = 0$ , $\text{Var}(t_1) 0.2s^2$ , $t_2 1.5s^2$ , $t_3 0.5556s^2$ , best $t_1$     |                                   |                          |
| estimator set 2                      | $\text{Var}(T_1) 3s^2$ , $\text{Var}(T_2) 29s^2$ , $\text{Var}(T_3) s^2/3$ , best $T_3$    |                                   |                          |
| dice Chebyshev                       | $\sigma^2 35/6$ , $k=3$ , bound $35/54 = 0.6481$ , exact $1/3$                             |                                   |                          |
| 600 throws                           | $1 - 5/24 = 19/24 = 0.7917$ , exact binomial 0.9754                                        |                                   |                          |
| marks Chebyshev                      | $k=2$ , $P \geq 0.75$                                                                      |                                   |                          |
| tube (errata 16)                     | $P(X < 150) = 1/3$ , $P(X > 150) = 2/3$ , all three $1/27$ , none $8/27$ (key $2/3$ WRONG) |                                   |                          |
| hypergeometric lot                   | mean $nK/N = 4 \times 5/25 = 0.80$ exactly, same as binomial mean                          |                                   |                          |
| battery sd                           | $\text{Var} 0.9475$ , exact sd 0.9734 $\rightarrow 0.97$ (key's 0.975 wrong, errata 12)    |                                   |                          |

## 6. Identities and algebra shortcuts

$$e^0 = 1$$

$$e^{-1} = 0.3679 \quad e^{-2} = 0.1353 \quad e^{-4} = 0.01832 \quad e^{-5} = 0.006738 \quad e^{-10} = 4.54e-5$$

$$C(n,r) = C(n,n-r)$$

sum  $p(x) = 1$  (algebraic check, use it on find-k questions)

35/12 and 35/6 (one die, two dice variance)

$(b-a)^2/12$  uniform variance, 12 is fixed

$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$  when independent, the minus stays plus

$$\Phi(-z) = 1 - \Phi(z)$$

$$\text{relative efficiency} = \text{Var}(T_2)/\text{Var}(T_1)$$

$$\text{MSE} = \text{Var} + \text{Bias}^2$$

## 7. The exact-vs-rounded values (write the exact one)

| question                | write                                  | not                                        |
|-------------------------|----------------------------------------|--------------------------------------------|
| insurance $P(X=4)$      | 0.1755                                 | 0.1745 (slide)                             |
| irregular die count     | 0.5499 $\rightarrow$ .55 0.549 (slide) |                                            |
| assignment Q2 $P(X>12)$ | 0.0916                                 | 0.0915 (key truncates)                     |
| battery sd              | 0.97                                   | 0.975 or 0.98 (key rounds twice)           |
| A2 C3 $P(X>2)$          | 0.8753                                 | 0.8754 (key sums rounded terms)            |
| exponential conditional | 0.6225                                 | 0.5679 (key wrong)                         |
| MTE 2024 CDF            | option D                               | option B (scheme wrong)                    |
| MTE 2025 Q8(ii)         | $E(t^2) \neq \theta^2$                 | $E(t^2) \neq 0$ (scheme drops $\theta^2$ ) |

## 8. Unit conversions that decide marks

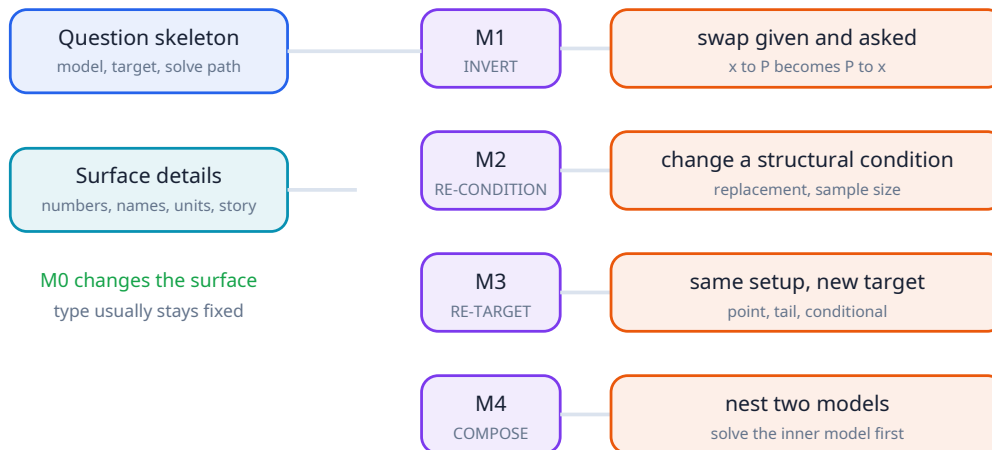
| 15 per hour                                                               | $\rightarrow$                       | 3 minutes = 0.05 hour | (rate in hours, time in hours) |
|---------------------------------------------------------------------------|-------------------------------------|-----------------------|--------------------------------|
| mean 2 (hours) $\rightarrow$ $\lambda = 1/2$ per hour                     |                                     |                       |                                |
| 4 minutes mean gap at 15/hour:                                            | $1/\lambda = 1/15$ hour = 4 minutes |                       |                                |
| days and hours: convert to ONE unit before substituting                   |                                     |                       |                                |
| RULE: the rate and the time MUST share a unit. Convert the odd one first. |                                     |                       |                                |



# 07 PATTERNS: how the setters mutate questions

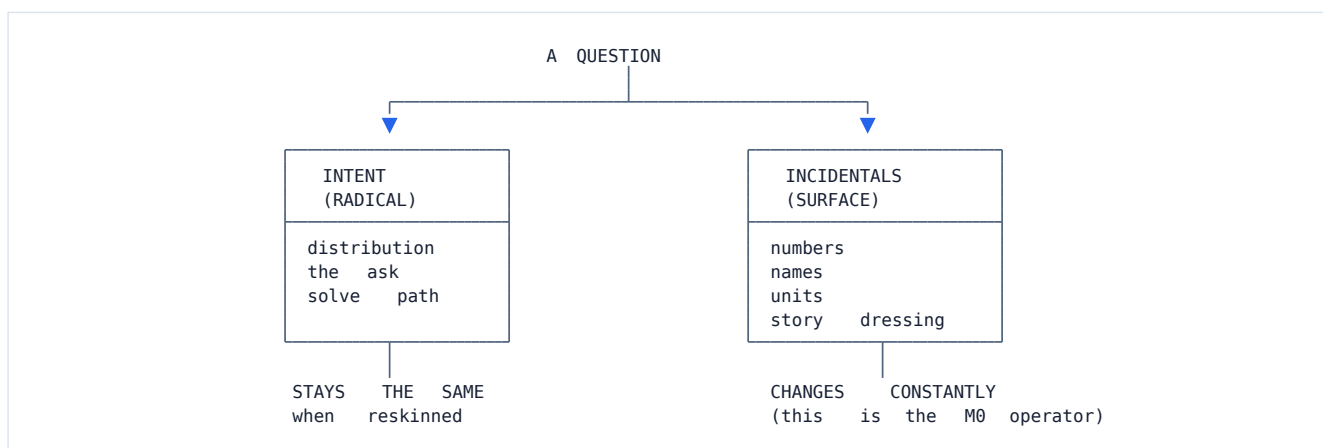
## THE MUTATION TREE (intent vs incidentals)

### Question mutation map



source match confidence belongs to the evidence ledger, not to visual similarity alone

### Question mutation operators



## THE FIVE OPERATORS AS A TREE

M0 RE-SKIN | numbers/names/units/story

TYPE UNCHANGED | <- common in the reviewed papers; exact rate unmeasured

M1 INVERT | swap given <-> asked

NEW NODE | x->P becomes P->x

M2 RE-CONDITION | flip a structural condition

NEW NODE | replace/no-replace, n<30 vs n≥30

M3 RE-TARGET | same setup, different target

NEW NODE | P(=k)->P(≤k)->P(>k)->conditional

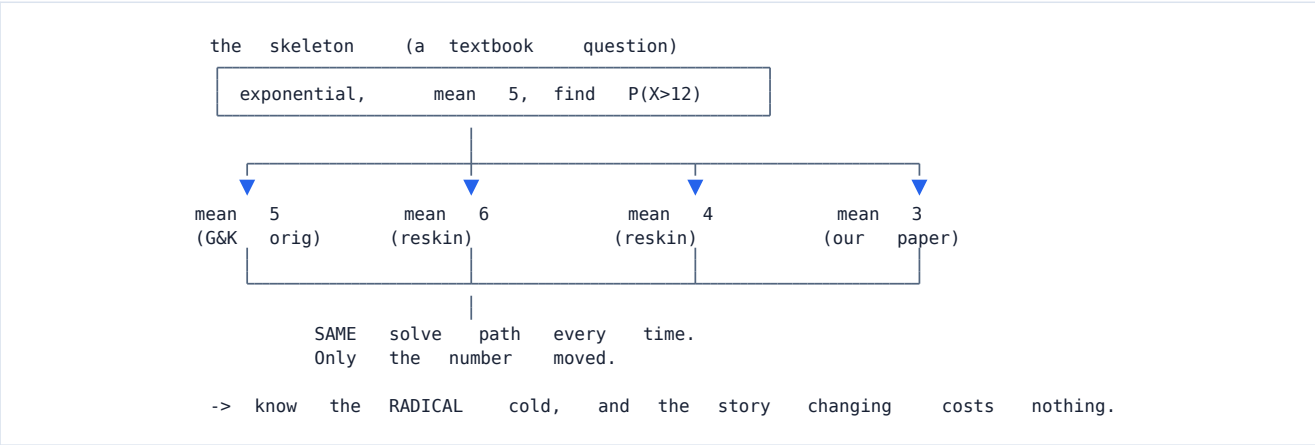
M4 COMPOSE | nest two models / chain

ALWAYS NEW | Poisson per min then binomial

## THE NINE SLOTS AS A 3x3 GRID

|                                                                                                                                   |                           |               |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|
| 1 POINT                                                                                                                           | 2 TAIL                    | 3 INTERVAL    |
| P(X=k)                                                                                                                            | P(X>k)                    | P(a≤X≤b)      |
| 4 MOMENTS                                                                                                                         | 5 PARAMS                  | 6 INVERSE     |
| E, Var                                                                                                                            | recover λ,p               | x for given P |
| 7 COUNT                                                                                                                           | 8 CONDITIONAL   9 COMPOSE |               |
| N x P                                                                                                                             | P(A B)                    | nest two      |
| many distribution questions fit one cell. Foundations, Chebyshev, estimation, and mixed questions need their own method families. |                           |               |

## THE VALUE-DELTA IDEA, DRAWN



Many confirmed questions reuse textbook or circulating skeletons, often with changed values or stories. That does not prove that every question is copied, or that future questions must reuse known structures. This file records confirmed value changes, candidate mutation patterns, and nearby siblings worth practising.

## 1. The two layers of a question

|                                                                                              |                                                                      |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| INTENT                                                                                       | the deep structure: which distribution, which ask, which solve path. |
| This is the RADICAL. It does not change when the story changes.                              |                                                                      |
| INCIDENTALS the numbers, names, units, and story dressing. These often leave the type fixed, |                                                                      |
| but unit or support changes can alter the mathematics, so recheck the model.                 |                                                                      |

That single split is the whole system. When a question looks new, find its radical: if the radical is old, it is an old question wearing a costume.

## 2. The mutation operators

|                                                                                            |                                                                 |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| M0 RE-SKIN                                                                                 | change numbers, names, units, story. TYPE UNCHANGED.            |
| pens become sensors become bulbs. Same math.                                               |                                                                 |
| examples: telephone mean 5 -> 6 -> 4 -> 3; subway 30min -> 4AM 15min                       |                                                                 |
| M1 INVERT                                                                                  | swap given and asked. NEW NODE.                                 |
| $x \rightarrow P(x)$ becomes $P \rightarrow x$ . moments -> parameters.                    |                                                                 |
| M2 RE-CONDITION flip a structural condition. NEW NODE when the formula changes.            |                                                                 |
| replacement on/off; $n < 30$ vs $n \geq 30$ ; discrete vs continuous.                      |                                                                 |
| M3 RE-TARGET                                                                               | same setup, different target. NEW NODE when the method differs. |
| $P(=k) \rightarrow P(\leq k) \rightarrow P(>k) \rightarrow$ conditional -> expected count. |                                                                 |
| example: ABES asks SD, our paper asks VARIANCE on the same density.                        |                                                                 |
| M4 COMPOSE                                                                                 | nest two models, or chain outputs. ALWAYS NEW NODE.             |
| Poisson per-minute then binomial over the minute.                                          |                                                                 |

### 3. The change table (does it create a new type or not)

| change                                           | new type? |
|--------------------------------------------------|-----------|
| numbers only                                     | NO        |
| names / units / story only                       | NO        |
| rounding / decimal places                        | NO        |
| same formula, different answer slot              | NO        |
| forward -> inverse                               | YES       |
| point -> tail or interval                        | YES       |
| unconditional -> conditional                     | YES       |
| single -> expected count                         | YES       |
| independence assumed -> not assumed              | YES       |
| with replacement -> without                      | YES       |
| n small -> n large (if the distribution changes) | MAYBE     |
| one model -> two models nested                   | YES       |
| E -> Var -> $E[h(X)]$                            | YES       |
| efficiency -> MSE comparison                     | YES       |

### 4. The value-delta log (mutations used on US, with the source)

|                         |                                         |                  |
|-------------------------|-----------------------------------------|------------------|
| telephone exponential   | mean 5 (G&K ch5 ex10) -> 6 -> 4 -> 3    | M0 x3            |
| uniform Chebyshev       | (-1,3) G&K ex15b -> (-1,1)              | M0               |
| subway/train wait       | 30-min subway G&K -> 15-min 4AM window  | M0 + M2          |
| Chebyshev inverse       | sigma 2, 21/25 G&K -> sigma 3, 24/25    | M0               |
| machine life            | "bread-making machine" -> "a machine"   | M0               |
| density $kx^3(4-x)^2$   | ABES sample asks SD, our paper asks VAR | M3               |
| pens defective          | G&K family of 18 -> 12 pens, 3 parts    | M0 (source OPEN) |
| Poisson $Y=2X$ variance | G&K answer 14 -> VERBATIM (uncostumed)  | M0               |
| $t^2$ biased estimator  | G&K estimation ex6 -> VERBATIM          | M0               |
| six-coins 6400          | G&K 7.26 family                         | M0               |
| Poisson 9P/90P          | G&K 7.30 family                         | M0               |
| soldiers 68.22          | G&K ch8 family                          | M0               |

Pattern in the currently confirmed source matches: M0 reskins dominate. M3 appears once in that confirmed set and M4 is less common. This is a study priority, not a claim about all 502 instances or a guarantee about a future paper.

## 5. The nine slots (the template every distribution shares)

|        |                             |                                                        |
|--------|-----------------------------|--------------------------------------------------------|
| slot 1 | point                       | $P(X = k)$                                             |
| slot 2 | tail                        | $P(X > k)$ , $P(X \geq k)$ , complement                |
| slot 3 | interval                    | $P(a \leq X \leq b)$                                   |
| slot 4 | moments                     | $E$ , $\text{Var}$ , $E(X^2)$                          |
| slot 5 | parameters                  | recover a parameter from a given probability or moment |
| slot 6 | inverse                     | quantile / cutoff / min n                              |
| slot 7 | expected count $N \times P$ |                                                        |
| slot 8 | conditional                 | $P(A   B)$ inside the model                            |
| slot 9 | composition                 | nest two models                                        |

These nine slots cover the main distribution questions. Probability foundations, Chebyshev, estimation, and mixed-model questions sit outside this grid. See 04-METHODS.md for those paths.

## 6. New type nodes the radical analysis generates (be ready for these)

|                                                                                        |                                    |                                                                       |              |
|----------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------|--------------|
| 7.a                                                                                    | DESIGN / MIN-SIZE<br>INVERSION     | "how large must n be so that ..."                                     | HIGH<br>RISK |
| binomial: solve n from the stated p and target probability                             |                                    |                                                                       |              |
| CLT: solve n from the stated sigma and SE target                                       |                                    |                                                                       |              |
| Chebyshev: k for a given guarantee                                                     |                                    |                                                                       |              |
| 7.b                                                                                    | PARAMETER<br>RECOVERY<br>(general) | solve for l, p, mu, sigma from a probability or moment                |              |
| examples seen: $p=5/8$ , $l=10$ , $l=2.3026$ , $\mu=87.18$ , $b=5.5$ , moment matching |                                    |                                                                       |              |
| 7.c                                                                                    | DERIVED-VARIABLE<br>ALGEBRA        | min/max/sum/difference of rvs                                         |              |
| two dice max $(2m-1)/36$ , Poisson sum additivity,                                     |                                    |                                                                       |              |
| independent $B(n_1,p)+B(n_2,p)=B(n_1+n_2,p)$                                           |                                    |                                                                       |              |
| 7.d                                                                                    | CONDITIONAL ON<br>AN INEQUALITY    | for $k \geq j$ ,                                                      |              |
| $P(X=k   X \geq j) = P(X=k)/P(X \geq j)$ . The pens sibling gives 0.3207.              |                                    |                                                                       |              |
| 7.e                                                                                    | DECISION /<br>OPTIMISATION         | choose the best option from an expectation                            |              |
| magazine 3 vs 4 copies, capacity sizing, which estimator to use                        |                                    |                                                                       |              |
| 7.f                                                                                    | APPROXIMATION<br>CHOICE            | Poisson for binomial (asked),<br>binomial $\rightarrow$ normal with a |              |
| continuity idea ( $z = 1.123$ )                                                        |                                    |                                                                       |              |
| 7.g                                                                                    | MSE /<br>BIAS-VARIANCE             | $MSE(T_4) = 25.75$ vs $Var(T_3) = 0.3333$ , the honest compare        |              |
| 7.h                                                                                    | SHEET GAPS                         | geometric E and Var, hypergeometric mean: needed, not on              |              |
| the provided sheet                                                                     |                                    |                                                                       |              |

## 7. Siblings one mutation away (not yet asked, risk-ranked)

| rank | sibling                                          | the mutation from something asked              |
|------|--------------------------------------------------|------------------------------------------------|
| 1    | normal one-unknown recovery                      | from the two-unknown system, drop one equation |
| 2    | binomial two-sided interval $P(2 \leq X \leq 4)$ | from the one-sided tails (0.3367)              |
| 3    | CLT for a SUM not a mean                         | sigma scales by $\sqrt{n}$                     |
| 4    | discrete conditional $P(X=k \mid X \geq j)$      | from the continuous conditional                |
| 5    | min-n design from stated $p$ , sigma, and target | from parameter recovery                        |
| 6    | geometric variance / tail                        | from the geometric mean (graded once)          |
| 7    | independence CHECK question                      | from the independence rules                    |
| 8    | MSE comparison                                   | from the efficiency question with bias allowed |
| 9    | Poisson between-values                           | from the tail family                           |
| 10   | valid-cdf check                                  | from the find-k normalisation                  |

These are the questions most likely to appear that we have NOT drilled. If time is short, do the top four.

## 8. The recognition routine (apply to any unseen question)

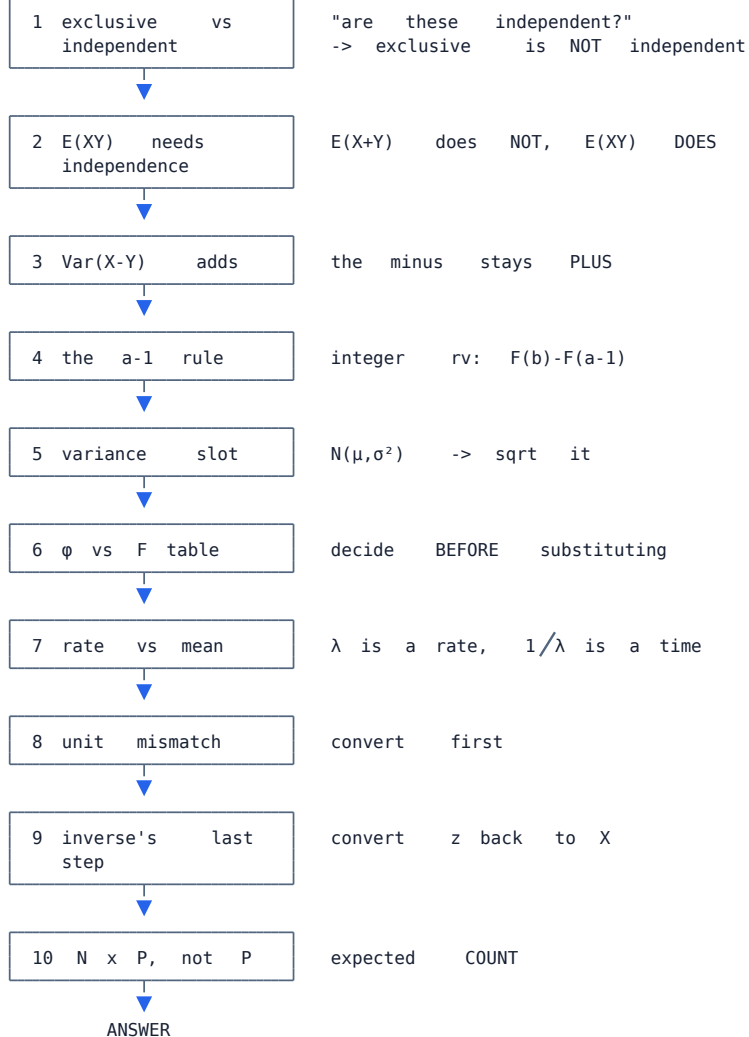
|                                                                          |  |
|--------------------------------------------------------------------------|--|
| 1 read the story, ignore the numbers                                     |  |
| 2 name the distribution (or "none", if it is event probability)          |  |
| 3 name the slot (1 to 9 above)                                           |  |
| 4 if a slot-5 or slot-6, it is an inversion: set up, solve, convert back |  |
| 5 if two distributions appear, it is slot 9: do the inner one first      |  |
| 6 check the boundary word (at least / more than) and the support         |  |
| 7 check the units                                                        |  |
| 8 then compute                                                           |  |

# 08 TRAPS: every trap and every source errata

## THE TRAPS AS A GAUNTLET

a question walks through and each trap tries to catch it:

START →





## THE ERRATA, BY WHERE THEY LIVE

| SOURCE                | ERRATA                                        |
|-----------------------|-----------------------------------------------|
| ppt3 (binomial/Poiss) | E1 insurance 0.1745->0.1755                   |
|                       | E2 die count 0.549->0.5499                    |
| ppt4 (continuous)     | E4 rate/mean wording p040                     |
|                       | E8 figure $\sigma=10$ -> use 5                |
| notes deck            | E13 bus integral bound 3 vs 2                 |
| clt deck              | E6 impurity Z2 -0.4 -> -0.94                  |
| assignments           | E3 0.0915->0.0916                             |
|                       | E5 second edition broken                      |
|                       | E9 0.5679->0.6225                             |
|                       | E10 mean 37.5->37.2                           |
|                       | E11 0.808->0.80                               |
|                       | E12 sd 0.975->0.97                            |
|                       | E14 0.8754->0.8753                            |
|                       | E16 tube 2/3 -> 8/27                          |
|                       | E17 row sums to 0.9                           |
| Chebyshev deck        | E15 Q3 slide mis-stated                       |
| MTE schemes           | E19 CDF option B -> D                         |
|                       | E20 infinite bounds -> 0..4                   |
|                       | E21 $E(t^2) \neq 0$ -> $E(t^2) \neq \theta^2$ |
| OUR OWN mock          | E18 "T4 smallest var" FALSE                   |

## THE 10-TRAP NIGHT LIST, AS A CHECKLIST

### NIGHT BEFORE THE PAPER:

- ☐ 1 exclusive vs independent can state the difference in one line
- ☐ 2  $E(XY)$  needs independence vs  $E(X+Y)$  always
- ☐ 3  $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$  the minus stays plus
- ☐ 4  $P(a \leq X \leq b) = F(b) - F(a-1)$  integer rv only
- ☐ 5  $N(\mu, \sigma^2)$  second slot is VAR sqrt before z
- ☐ 6  $\phi(z)$  area-from-0 vs  $F(z)$  cum decide first
- ☐ 7  $\lambda$  rate,  $1/\lambda$  mean exponential
- ☐ 8 convert units FIRST 15/hr with 3 min
- ☐ 9  $X = \mu + z\sigma$  the inverse's last step
- ☐ 10 expected count =  $N \times P$  not P

Two sections. First the concept traps (mistakes the QUESTIONS are built to catch). Then the source errata (numbers printed wrong in the slides and keys, with the correct value).

## Part 1: concept traps

|                                                                                                    |                                                                                                                    |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1                                                                                                  | EXCLUSIVE vs INDEPENDENT. Mutually exclusive means no shared outcome. Independent means                            |
| $P(A \cap B) = P(A)P(B)$ . Exclusive events with positive probability are NOT independent. This is |                                                                                                                    |
| the most common true/false the paper can carry.                                                    |                                                                                                                    |
| 2                                                                                                  | $E(XY) = E(X)E(Y)$ needs INDEPENDENCE. $E(X+Y) = E(X) + E(Y)$ does NOT. The paper probes exactly                   |
| this contrast.                                                                                     |                                                                                                                    |
| 3                                                                                                  | $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ when independent. The minus stays plus. Squaring kills the sign. |
| 4                                                                                                  | THE a-1 RULE. For an integer rv, $P(a \leq X \leq b) = F(b) - F(a-1)$ . Writing $F(a)$ is the slip. For            |
| a continuous rv it does not matter.                                                                |                                                                                                                    |
| 5                                                                                                  | THE VARIANCE SLOT. $N(\mu, \sigma^2)$ : the second number is the VARIANCE. Square-root it before                   |
| a z-score. "mean 70, variance 25" $\rightarrow \sigma = 5$ .                                       |                                                                                                                    |
| 6                                                                                                  | TWO TABLE CONVENTIONS. $\phi(z)$ is often the area from 0 to z; $F(z)$ is cumulative from $-\infty$ .              |
| MIXING them is the biggest normal error. Decide which the paper gives before substituting.         |                                                                                                                    |
| 7                                                                                                  | RATE vs MEAN. Exponential: $\lambda$ is a rate, $1/\lambda$ is the mean TIME. Getting the                          |
| reciprocal backwards changes the answer completely. (Slides get this wrong too, errata 4.)         |                                                                                                                    |
| 8                                                                                                  | UNIT MISMATCH. A rate per hour with a time in minutes. Convert to one unit first.                                  |
| 9                                                                                                  | THE INVERSE'S LAST STEP. Get z from the table, then convert $X = \mu + z \sigma$ . Forgetting                      |
| the conversion leaves the answer in z-space.                                                       |                                                                                                                    |
| 10                                                                                                 | EXPECTED COUNT vs PROBABILITY. "Expected number out of N" is $N \times P$ , not P. (0.5499 not                     |
| $5.5e-5$ .)                                                                                        |                                                                                                                    |
| 11                                                                                                 | CHEBYSHEV k IN SIGMA UNITS. $k = \text{eps}/\sigma$ . Raw units in, convert first. And the bound is                |
| an UPPER bound on the tail, so report the exact value too when the distribution is known.          |                                                                                                                    |
| 12                                                                                                 | CONSISTENCY vs SMALL VARIANCE. A fixed-n small variance is not consistency. Consistency is                         |
| about n growing. T1 with $\text{Var } 3\sigma^2$ at fixed $n=3$ is the trap.                       |                                                                                                                    |
| 13                                                                                                 | BIASED BEFORE VARIANCE. In "most efficient", a biased estimator is disqualified first,                             |
| even if its variance is smaller. Efficiency is among the UNBIASED only.                            |                                                                                                                    |
| 14                                                                                                 | EXACT vs ROUNDED. Carry exact values to the final line. Rounding mid-way and then rooting                          |
| or summing gives a slightly wrong number the key itself sometimes prints.                          |                                                                                                                    |

|                                                                                         |                                                                                                     |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1                                                                                       | EXCLUSIVE vs INDEPENDENT. Mutually exclusive means no shared outcome. Independent means             |
| 15                                                                                      | SUPPORT CHECK. A density is zero outside its support. Integrating over $(-\infty, \infty)$ when the |
| support is finite is a setup error (a scheme printed this, errata 20).                  |                                                                                                     |
| 16                                                                                      | VALID PDF CHECK. A pmf must sum to 1 over its support. A row that sums to 0.9 is not a pmf          |
| (errata 17).                                                                            |                                                                                                     |
| 17                                                                                      | "USE THE APPROXIMATION". When the paper says use the Poisson approximation to the                   |
| binomial, set $l = np$ , not the exact binomial.                                        |                                                                                                     |
| 18                                                                                      | LARGE SAMPLE LANGUAGE. "The mean of a sample of $n$ " divides by the STANDARD ERROR, not by         |
| sigma. A single observation uses sigma.                                                 |                                                                                                     |
| 19                                                                                      | CLT CONDITIONS. $n \geq 30$ is the course heuristic, not a universal theorem cutoff. For a          |
| small non-normal population, do not assume the normal approximation without checking    |                                                                                                     |
| shape and tail behaviour.                                                               |                                                                                                     |
| 20                                                                                      | NORMAL POPULATION FALLBACK. A normal population is exactly normal at any $n$ , so a small- $n$      |
| question is fine if the population is stated normal. No large- $n$ heuristic is needed. |                                                                                                     |

## Part 2: source errata (the slides and keys print these wrong)

Use the computed value. Each entry is verified in the repo (reports/09-ERRATA.md).

|                                                                                           |                                                                                         |              |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------|
| E1                                                                                        | insurance Poisson: slide 0.1745, correct 0.1755                                         | (ppt3 p027)  |
| E2                                                                                        | irregular die count: slide 0.549, correct 0.5499                                        | (ppt3 p018)  |
| E3                                                                                        | assignment Q2 $P(X > 12)$ : key 0.0915, correct 0.0916                                  | (truncation) |
| E4                                                                                        | exponential rate/mean meaning inverted on p040, read p041 instead                       |              |
| E5                                                                                        | assignment 1 long Q2 normalisation: the 12k version is a broken SECOND EDITION.         |              |
| batch-1 edition: $10k^2 + 9k = 1$ , $k = 1/10$ exact, key checks out.                     |                                                                                         |              |
| second edition: $12k + 10k^2 = 1$ , $k = 0.0782$ , but its own key prints 1/10 which does |                                                                                         |              |
| NOT satisfy its own table. Do not drill the second edition.                               |                                                                                         |              |
| E6                                                                                        | impurity Z: slide prints $Z^2 = -0.4$ , correct -0.94. Rounded table areas give         |              |
| 0.1645; unrounded $z$ values give about 0.1637. The slide's 0.1644 is also wrong.         |                                                                                         |              |
| (clt p014)                                                                                |                                                                                         |              |
| E7                                                                                        | $N(\mu, \sigma^2)$ : the prose says variance, the notation says squared. Second slot IS |              |
| variance.                                                                                 |                                                                                         |              |

|                                                                                                                           |                                                                                                         |                           |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------|
| E1                                                                                                                        | insurance Poisson: slide 0.1745, correct 0.1755                                                         | (ppt3 p027)               |
| E8                                                                                                                        | ppt4 p030 figure labels $\sigma = 10$ under $\mu = 8$ , but the arithmetic uses $\sigma = 5$ .          |                           |
| Use 5. (errata 8)                                                                                                         |                                                                                                         |                           |
| E9                                                                                                                        | assignment 2 B Q5: key prints 0.5679, correct 0.6225                                                    | (exponential conditional) |
| E10                                                                                                                       | assignment 2 C Q2: key prints mean 37.5, correct 37.2 (the key's own cutoff agrees                      |                           |
| with 37.2). Use $\mu = 37.2$ , $\sigma = 28.2$ , cutoff 30.4.                                                             |                                                                                                         |                           |
| E11                                                                                                                       | assignment 1 long Q3: key prints 0.808, both cases are exactly 0.80 (hypergeometric                     |                           |
| mean equals binomial mean here)                                                                                           |                                                                                                         |                           |
| E12                                                                                                                       | assignment 1 app Q3: key rounds twice, sd 0.975 $\rightarrow$ 0.98. Exact is 0.9734 $\rightarrow$ 0.97. |                           |
| E13                                                                                                                       | deck bus example p127/p129: the printed first integral bound (3) contradicts the stated                 |                           |
| event ( $Y < 2$ ). Final answer 2/5 is correct for " $Y < 2$ or $Y > 6$ ". Use $P(Y < 2) + P(Y > 6) = 0.08 + 0.32 = 0.40$ |                                                                                                         |                           |
| E14                                                                                                                       | assignment 2 C Q3: key 0.8754, exact 0.8753.                                                            |                           |
| E15                                                                                                                       | Chebyshev L10-11 Q3 slide: states one row, works another. It also writes a reciprocal                   |                           |
| template. USE Q1 ( $E = 3$ , $E(X^2) = 13$ , bound 21/25) AND Q2 ( $\mu = 10$ , $\text{var} = 4$ ) ONLY. Do not quote     |                                                                                                         |                           |
| Q3's row. (Full retraction detail in the repo errata, incl. two claims that were                                          |                                                                                                         |                           |
| themselves retracted after checking.)                                                                                     |                                                                                                         |                           |
| E16                                                                                                                       | tube question (2024 A1 Q12 and 2025-26 Q14): key part (ii) prints 2/3, correct is the                   |                           |
| probability that ALL THREE survive = $(2/3)^3 = 8/27$ . The key carried the single-tube                                   |                                                                                                         |                           |
| number into the three-tube part. Parts (i) 1/27, (iii) 0.25, (iv) 1.7 are correct.                                        |                                                                                                         |                           |
| E17                                                                                                                       | 2024-25 assignment 1 Q16: the probability row sums to 0.9, not 1. It is not a valid pmf.                |                           |
| State the renormalisation you choose (divide by 0.9) or ask for the correction.                                           |                                                                                                         |                           |
| E18                                                                                                                       | mock paper B4 (our own): the Note says T4 has the smallest variance. False, T4 is third                 |                           |
| of four (order T3, T2, T4, T1). T4 is biased, keep that part, drop the superlative.                                       |                                                                                                         |                           |
| E19                                                                                                                       | 2024 MTE QA2 CDF: scheme marks B, correct is option D (the CDF must accumulate the mass                 |                           |
| from 0 to 1). Option B gives $F(1) = 1.5$ , impossible.                                                                   |                                                                                                         |                           |
| E20                                                                                                                       | 2025 MTE scheme Q4: displays infinite integration bounds, but the support is $0 < x < 4$ and            |                           |
| the arithmetic uses 0 and 4. The constants are right, the printed bounds are the slip.                                    |                                                                                                         |                           |
| E21                                                                                                                       | 2025 MTE scheme Q8(ii): prints $E(t^2) \neq 0$ , correct is $E(t^2) = \theta^2$ .                       |                           |

## The 10-item trap list to read the night before

|    |                                               |
|----|-----------------------------------------------|
| 1  | exclusive vs independent                      |
| 2  | $E(XY)$ needs independence, $E(X+Y)$ does not |
| 3  | $\text{Var}(X-Y)$ adds                        |
| 4  | the $a-1$ in $F(b)-F(a-1)$                    |
| 5  | variance slot in $N(\mu, \sigma^2)$ , sqrt it |
| 6  | $\phi$ vs $F$ table convention                |
| 7  | rate vs mean in the exponential               |
| 8  | unit mismatch                                 |
| 9  | the inverse's last conversion step            |
| 10 | expected count = $N \times P$ , not $P$       |

# 09 QUESTIONS: the tagged MTE question inventory

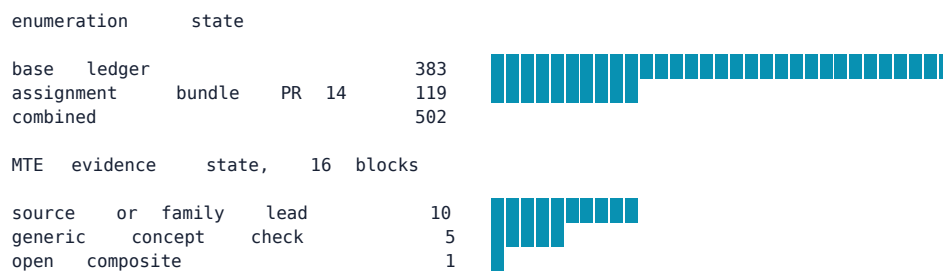
## THE QUESTION INVENTORY AS A GRID

|                                    |                                  |                  |        |               |            |
|------------------------------------|----------------------------------|------------------|--------|---------------|------------|
| the two MTE papers (16 blocks)     |                                  |                  |        |               |            |
| M24                                | A1 A2 A3 B1 B2 B3 B4<br>C1(OPEN) |                  |        |               |            |
| M25                                | Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8          |                  |        |               |            |
| 10 source/family leads             | 5 concept checks                 | 1 open composite |        |               |            |
| assignments (A1 + A2 = 52 blocks)  |                                  |                  |        |               |            |
| A1                                 | 10 MCQ                           | 6 short          | 4 long | 4 application |            |
| A2                                 | 12 A                             | 8 B              | 4 C    | 4 D           |            |
| the teaching decks (60 blocks)     |                                  |                  |        |               |            |
| notes                              | 30 blocks (D1-1 .. D1-30)        |                  |        |               |            |
| ppt3                               | 6                                | ppt4             | 7      | clt           | 5 ppt5   5 |
| lms-th                             | 7 (2 unique)                     |                  |        |               |            |
| re-teach decks (30 blocks, pass 2) |                                  |                  |        |               |            |
| L1-7                               | 13                               | L8-9             | 4      | L12-13   6    | L14-15   7 |

## THE SLOT COVERAGE MAP (what is asked vs what is one mutation away)

| SLOT:                                                   | 1       | 2 | 3 | 4 | 5 6 7 8 9   |
|---------------------------------------------------------|---------|---|---|---|-------------|
| pt tl iv mo pa in ct cd co                              |         |   |   |   |             |
| BINOMIAL                                                |         | X | X | S | X X S X S S |
| POISSON                                                 |         | X | X | . | X X . X . X |
| UNIFORM                                                 |         | X | X | X | X S S . . S |
| NORMAL                                                  |         | X | X | X | X X X X . X |
| EXPONENTIAL                                             | X       | X | X | X | X . . X X   |
| X = asked in our corpus                                 |         |   |   |   |             |
| S = a sibling, one mutation away, NOT yet asked         | <- risk |   |   |   |             |
| . = no evidence                                         |         |   |   |   |             |
| the S cells are where the next paper can surprise you.  |         |   |   |   |             |
| the four highest-risk siblings: normal one-unknown,     |         |   |   |   |             |
| binomial interval, CLT-for-a-sum, discrete conditional. |         |   |   |   |             |

## PROVENANCE TRUTH, VISUALISED



The 502 count measures enumerated question instances, not unique questions and not sourced questions. No content-family ids have been assigned yet, so corpus-wide deduplication and source attribution remain open work.

Every MTE-scope question block we hold, tagged by distribution, slot (the nine in 07-PATTERNS), and where it lives. This is the drill list. "In scope" means lectures 1-21.

The nine slots: 1 point, 2 tail, 3 interval, 4 moments, 5 parameter, 6 inverse, 7 count, 8 conditional, 9 compose.

## A. The two MTE papers (16 blocks, evidence states kept separate)

| id     | source page   | question                                               | dist         | slot             | status                |
|--------|---------------|--------------------------------------------------------|--------------|------------------|-----------------------|
| M24-A1 | mte 2024 p001 | Poisson transform concept                              | Poi          | 4                | generic concept       |
| M24-A2 | mte 2024 p001 | triangular density, mean, variance                     | contin       | 4+5              | traced G&K 8-1-5      |
| M24-A3 | mte 2024 p001 | Chebyshev forms, pick the false ones                   | Cheb         | -                | concept               |
| M24-B1 | mte 2024 p001 | $\text{Var}(X-2Y)$ , $\lambda^2$ , $\mu^3$ , answer 14 | Poisson      | 4                | VERBATIM G&K          |
| M24-B2 | mte 2024 p001 | $f(x)=Ae^{-x/5}$ telephone                             | Exp          | 4                | reskin G&K ch5 ex10   |
| M24-B3 | mte 2024 p001 | Chebyshev interval $(-1,1)$ vs $(-1,3)$                | Cheb         | 2                | reskin G&K ex15b      |
| M24-B4 | mte 2024 p001 | uniform wait, answer $1/3$                             | Unif         | 3                | G&K subway            |
| M24-C1 | mte 2024 p002 | rain $N(2.6,34.5)$ + $5/3$ binomial week               | Normal+Bin 9 | OPEN (composite) |                       |
| M25-Q1 | mte 2025 p001 | axiom of total mass                                    | prob         | -                | concept               |
| M25-Q2 | mte 2025 p001 | substitution drill                                     | prob         | -                | concept               |
| M25-Q3 | mte 2025 p001 | sufficient statistics MCQ                              | Est          | -                | concept cross-check   |
| M25-Q4 | mte 2025 p001 | density $kx^3(4-x)^2$ , find $k$ , mean, var           | contin       | 4+5              | ABES sample (asks SD) |
| M25-Q5 | mte 2025 p001 | Chebyshev find $c$ , $\mu$ 10, var 4, bound .04        | Cheb         | 6                | deck L10-11 Q2(iv)    |
| M25-Q6 | mte 2025 p001 | exponential reskin, mean 5 $\rightarrow$ 3             | Exp          | 4                | G&K ch5 ex10 reskin   |
| M25-Q7 | mte 2025 p001 | machine life sample mean, $n=9$                        | Normal       | 3                | Walpole P&S 8.25      |
| M25-Q8 | mte 2025 p002 | (i) Poisson notes (ii) estimation chapter              | Poi+Est      | 9                | mixed                 |



## B. Assignment 1 (24 blocks) and Assignment 2 (28 blocks)

| id        | question                                                  | dist                | slot       |
|-----------|-----------------------------------------------------------|---------------------|------------|
| A1 MCQ1   | 3 children, coin, pmf and cdf                             | discrete            | 1+3        |
| A1 MCQ5   | Chebyshev applicability                                   | Cheb                | 2          |
| A1 MCQ10  | Chebyshev applicability                                   | Cheb                | 2          |
| A1 short1 | pmf $1/8 \ 3/8 \ 3/8 \ 1/8$ + cdf                         | discrete            | 3          |
| A1 short2 | mortality pdf $P(X>12) = 0.0916$                          | contin              | 2          |
| A1 short3 | $f(x)=6x(1-x)$ , valid pdf, $b=1/2$                       | contin              | 5+1        |
| A1 short4 | geometric expectation, $E=2$                              | geom(gap)           | 4          |
| A1 short5 | dice Chebyshev 35/54 vs 1/3                               | Cheb                | 9          |
| A1 short6 | 600 throws, 19/24                                         | Bin+Cheb            | 9          |
| A1 long1  | mortality integral + conditional 0.4863                   | contin              | 3+8        |
| A1 long2  | pmf with unknown $k$ , $k=1/10$ , $P(X<6)=0.81$           | discrete            | 5+3        |
| A1 long3  | lot of 25, 5 defective, mean 0.8 both ways                | Hyper(no slide!) 7  |            |
| A1 long4  | four bad oranges 12/19, 32/95, 3/95                       | discrete            | 3          |
| A1 app1   | sensor batch, $E=1.2$ , $\text{Var}=0.86$                 | discrete            | 4          |
| A1 app2   | CDF 0.15,0.50,0.80,1.00                                   | discrete            | 4          |
| A1 app3   | battery $E=3.45$ , $\text{Var}=0.9475$ , $\text{sd}=0.97$ | discrete            | 4          |
| A1 app4   | marks Chebyshev, 75 percent                               | Cheb                | 9          |
| A2 A2     | binomial moments                                          | Bin                 | 4          |
| A2 A3     | Poisson                                                   | Poi                 | 1          |
| A2 A4     | uniform $E$ , $\text{Var}$                                | Unif                | 4          |
| A2 A7     | Poisson                                                   | Poi                 | see source |
| A2 A9     | binomial max successes                                    | Bin                 | 4          |
| A2 A10    | uniform parameters                                        | Unif                | 5          |
| A2 A11    | normal 68/95/99.7 landmarks                               | Normal(no slide!) 4 |            |
| A2 A12    | Poisson additivity                                        | Poi(no slide!) 9    |            |
| A2 B1     | Poisson rejected $P(X\geq 4)=0.2424$                      | Poi                 | 2          |
| A2 B2     | uniform                                                   | Unif                | see source |
| A2 B3     | normal interval 45-62                                     | Normal              | 3          |
| A2 B4     | 5000 batteries $N \times P$                               | Normal              | 7          |
| A2 B5     | exponential conditional 0.6225                            | Exp(no slide! 8)    | 8          |
| A2 B8     | uniform $ X $ intervals                                   | Unif                | 3          |
| A2 C1     | exponential memoryless                                    | Exp(no slide!) 8    |            |
| A2 C2     | two-unknown normal 37.2/28.2/30.4                         | Normal              | 5+6        |
| A2 C3     | Poisson $P(X>2)=0.8753$                                   | Poi                 | 2          |
| A2 C4     | uniform rounding                                          | Unif                | 1          |
| A2 D1     | 10000 bulbs $N \times P$                                  | Normal              | 7          |
| A2 D2     | exponential unit conversion                               | Exp                 | 5          |
| A2 D3     | binomial moments                                          | Bin                 | 4          |

| id       | question               | dist   | slot |
|----------|------------------------|--------|------|
| A2 MCQ5  | exponential memoryless | Exp    | 8    |
| A2 MCQ11 | normal landmarks       | Normal | 4    |

## C. The teaching decks' own questions (60 blocks, the practice pool)

|                                      |                                                             |
|--------------------------------------|-------------------------------------------------------------|
| notes deck (30 blocks, p041 to p147) | see 10-SLIDE-MAP.md for the page-by-page list               |
| ppt3 (6 blocks)                      | 5-coin, pens x3, irregular die x2                           |
| ppt4 (7 blocks)                      | uniform, normal chain, inverse, exponential                 |
| clt (5 blocks)                       | lightbulbs, ATM, impurity, LED, machine                     |
| ppt5 (5 blocks)                      | response time, packets, battery CI, comparison set 1, set 2 |
| lms-theory (7 total, 2 unique)       | sufficiency examples                                        |

## D. The re-teach deck questions (30 blocks, use as pass 2)

|                                                                                                  |                                                                                 |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| L1-7                                                                                             | 13 blocks: L17-01..L17-13 (faculty computers, basketball, camera, rare disease, |
| gas pumps, cylinder pmf [OPEN], county forms, batteries, boards, dice max, cdf proof,            |                                                                                 |
| E on -3/6/9)                                                                                     |                                                                                 |
| L8-9                                                                                             | 4 blocks: L89-01..L89-04 (bus pdf, CDF-to-density [OPEN], kidney E(Y), Pareto)  |
| L12-13 6 blocks: L1213-01..06 (5-coin [OPEN], pens, irregular die, Poisson x3 [one OPEN])        |                                                                                 |
| L14-15 7 blocks: L1415-01..07 (uniform, normal with mean 8 and sd 5, inverse cutoff, exponential |                                                                                 |
| McClave)                                                                                         |                                                                                 |

## E. The open items (no source found yet)

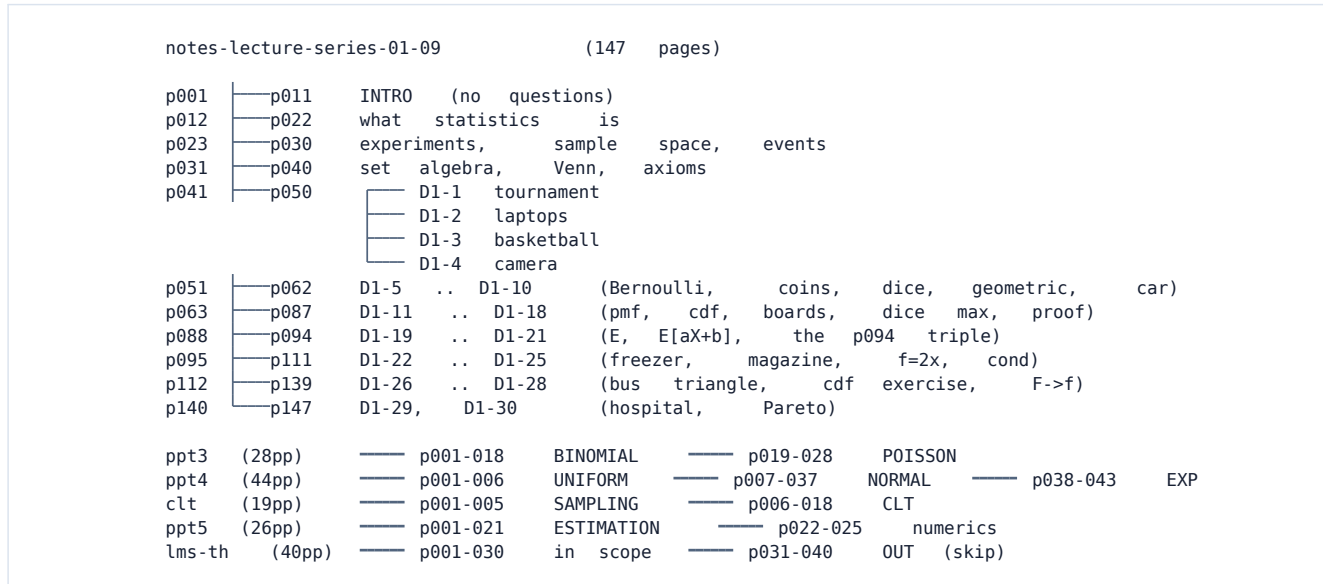
|                                                                                        |                                        |                     |
|----------------------------------------------------------------------------------------|----------------------------------------|---------------------|
| M24-C1                                                                                 | rain set + 5/3 binomial composite      | in scope, no source |
| L17-06                                                                                 | pmf table with k and CDF questions     | four-deck, open     |
| L17-13                                                                                 | pmf on -3/6/9, transformed expectation | four-deck, open     |
| L89-02                                                                                 | CDF to density                         | four-deck, open     |
| L1213-01 exactly 3 heads in 5 tosses                                                   | four-deck, open                        |                     |
| L1213-03 irregular die even outcomes                                                   | four-deck, open                        |                     |
| L1213-04 Poisson relation $P_1 = 0.2 P_2$                                              | four-deck, open                        |                     |
| L1213-05 Poisson calls exactly two each minute                                         | four-deck, open                        |                     |
| L1213-06 life-insurance Poisson approximation                                          | four-deck, open                        |                     |
| (the pens question circulates in several versions, but its originating source is OPEN) |                                        |                     |

## F. Provenance state (honest, as of 16 Sep)

|                                                                                       |  |
|---------------------------------------------------------------------------------------|--|
| Enumeration: 383 rows on this branch, plus 119 assignment-bundle rows on PR 14.       |  |
| Combined after merge: 502 gross instances.                                            |  |
| MTE evidence: 10 source or family leads, 5 generic concept checks, 1 open composite.  |  |
| A concept check confirms mathematical type, not copying or source provenance.         |  |
| Corpus-wide source matching is incomplete. No row carries a content family id yet, so |  |
| neither the unique-question count nor a corpus-wide sourced percentage is known.      |  |

# 10 SLIDE MAP: every page to topic to question

## THE SLIDE MAP AS A PICTURE



## WHICH SLIDE CARRIES WHICH EXAMINABLE ITEM

| the exam asks...    | the slide that teaches it  |
|---------------------|----------------------------|
| event probability   | notes p041-p050            |
| pmf / cdf build     | notes p063-p087            |
| E and Var           | notes p088-p111            |
| continuous pdf/cdf  | notes p112-p147            |
| binomial            | ppt3 p001-p018             |
| Poisson             | ppt3 p019-p028             |
| uniform             | ppt4 p001-p006             |
| normal              | ppt4 p007-p037             |
| exponential         | ppt4 p038-p043             |
| SE and CLT          | clt p001-p018              |
| estimation          | ppt5 p001-p021             |
| Chebyshev           | L10-11 deck (NOT notes!)   |
| everything in H1-H8 | NOTHING (the hidden layer) |

Page refs are the pipeline page numbers (md/<label>/pNNN.md), which match the PDF page order.

## notes-lecture-series-01-09 (147 pages) = lectures 1 to 9

| pages                                    | topic                                       | questions there        |
|------------------------------------------|---------------------------------------------|------------------------|
| p001-011                                 | course intro, syllabus, books               | none                   |
| p012-022                                 | what statistics is, importance, limitations | none examinable        |
| p023-030                                 | random experiment, sample space, events     | display examples only  |
| p031-040                                 | event algebra, Venn, axioms, rules          | none (setup for p041+) |
| p041-042                                 | tournament, 4 parts                         | D1-1                   |
| p043-045                                 | counting, permutations, combinations        | theory                 |
| p046-047                                 | laptops and desktops, 4 parts               | D1-2                   |
| p048-049                                 | basketball lineup, 2 parts                  | D1-3                   |
| p050                                     | camera conditional probability              | D1-4                   |
| p051-062                                 | random variables: Bernoulli, counting rv,   |                        |
| dice sum, two coins, geometric, car life | D1-5..D1-10                                 |                        |
| p063-087                                 | pmf, cdf: conditions, find k, build, graphs | D1-11..D1-18           |
| p088-094                                 | expectation: def, $E[h]$ , linearity        | D1-19..D1-21           |
| p095-111                                 | variance: def, shortcut, transforms         | D1-22..D1-25           |
| p112-139                                 | continuous: pdf, cdf, $F'=f$ , piecewise    | D1-26..D1-28           |
| p140-147                                 | continuous expectation and variance         | D1-29, D1-30           |

Detail on the pages that carry the graded skeletons:

|       |          |                                           |
|-------|----------|-------------------------------------------|
| D1-1  | p041-042 | tournament outcomes 16                    |
| D1-2  | p046-047 | laptops/desktops                          |
| D1-3  | p048-049 | basketball lineup                         |
| D1-4  | p050     | camera conditional                        |
| D1-5  | p052     | Bernoulli coin                            |
| D1-6  | p053     | two coin X                                |
| D1-7  | p054-055 | dice sum pmf                              |
| D1-8  | p056     | two coin Y                                |
| D1-9  | p057-059 | geometric N + verify sum = 1              |
| D1-10 | p060     | car lifetime                              |
| D1-11 | p065     | gas station pumps (NO solution)           |
| D1-12 | p066     | k probability function (key in A1 long 2) |
| D1-13 | p067-069 | tune-ups                                  |
| D1-14 | p070-071 | contractor forms                          |
| D1-15 | p072-073 | flashlight batteries                      |
| D1-16 | p080-081 | boards inspection                         |
| D1-17 | p082-083 | dice maximum M                            |
| D1-18 | p085     | cdf nondecreasing proof                   |
| D1-19 | p089     | die roll $E(X)$                           |
| D1-20 | p093     | prove $E[aX+b]$                           |
| D1-21 | p094     | $E(X)$ , $E(X^2)$ , $E(2X+1)^2$           |

|       |          |                                                |
|-------|----------|------------------------------------------------|
| D1-1  | p041-042 | tournament outcomes 16                         |
| D1-22 | p107-108 | freezer dealer 4 parts                         |
| D1-23 | p109-111 | magazine orders                                |
| D1-24 | p124-125 | $f(x)=2x$ verify + $P(X<1/2)$                  |
| D1-25 | p126     | continuous conditional 5/12                    |
| D1-26 | p127-129 | bus waiting 6 parts (errata 13)                |
| D1-27 | p138     | two cdfs exercise (NO solution)                |
| D1-28 | p139     | distribution function to density (NO solution) |
| D1-29 | p142     | hospitalization $E(Y)$                         |
| D1-30 | p146-147 | Pareto pdf 5 parts                             |

Pages with NO countable question (do not hunt): p012, p017, p021, p115-116, p133, p136, plus the boundary "display examples" on p023-027 (random experiment / sample space / mutually exclusive / independence illustrations).

## ppt3-discrete-prob-dist (28 pages) = lecture 12 binomial, 13 Poisson

|          |                                                                     |
|----------|---------------------------------------------------------------------|
| p001-018 | binomial: derivation, conditions, pmf, moments, pens, irregular die |
| p002-007 | 5-coin derivation, the $C(5,3)=10$                                  |
| p011     | recognition checklist                                               |
| p013-016 | pens $B(12,0.1)$ three parts                                        |
| p017-018 | irregular die, $p=5/8$ , expected count 0.5499                      |
| p019-028 | Poisson                                                             |
| p022     | Poisson as binomial limit                                           |
| p025     | ratio $P(X=1)=0.2P(X=2)$ , $l=10$                                   |
| p026     | telephone calls 2/min, THE NESTING 0.00145                          |
| p027     | insurance 5000 men, 0.1755 (errata 1)                               |
| p028     | tail $P(X>2)=0.8753$                                                |

## ppt4-continuous-prob-dist (44 pages) = lecture 14 uniform, 15 normal, 16 exponential

|          |                                                                            |
|----------|----------------------------------------------------------------------------|
| p001-006 | uniform: density, moments, length ratio, clipping, $ X $ intervals         |
| p007-037 | normal                                                                     |
| p010-011 | changing $\mu$ shifts, changing $\sigma$ widens                            |
| p017     | $Z=2.0$ worked                                                             |
| p027     | symmetry facts                                                             |
| p029-031 | $P(X < 8.6)$ $\mu=8$ $\sigma=5$ , $Z=0.12$ , 0.5478 (p030 figure errata 8) |
| p032-033 | $P(X > 8.6)=0.4522$                                                        |
| p034-037 | inverse 20 percent tail $\rightarrow 3.792$                                |
| p038-043 | exponential                                                                |
| p040     | rate/mean wording inverted (errata 4)                                      |
| p041     | correct $\mu = 1/\lambda$                                                  |
| p042-043 | arrivals 15/hour, $P(T < 3\text{min})=0.5276$                              |

## lms-standard-error-clt (19 pages) = lectures 17 to 18

|                                               |                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------|
| p001-005                                      | population vs sample, parameters vs statistics, $SE = \sigma/\sqrt{n}$ |
| p005                                          | lightbulb SE, $n=25 \rightarrow 20$ , $n=100 \rightarrow 10$           |
| p007-010                                      | the three rules, $n=30$ boundary                                       |
| p011-012                                      | ATM wait, $\mu=4$ , $\sigma=2$ , $n=36$ , $Z=1.50$ , 0.0668            |
| p013-014                                      | impurity, $\mu=4.0$ , $\sigma=1.5$ , $n=50$ , 0.1645 by rounded table, |
| about 0.1637 unrounded (errata 6, $Z=-0.94$ ) |                                                                        |
| p015-016                                      | LED, $\mu=50000$ , $\sigma=8000$ , $n=64$ , $Z=-2.00$ , 0.0228         |
| p017-018                                      | machine, normal population, $n=9$ exact, 0.6898                        |

## ppt5-estimation-summary (26 pages) = lectures 19 to 21

|          |                                                                      |
|----------|----------------------------------------------------------------------|
| p001-007 | parameter, statistic, estimator vs estimate, point vs interval       |
| p008-014 | unbiased, force-unbiased, consistency, efficiency, sufficiency       |
| p015-018 | comparison set 1 ( $t_1$ , $t_2$ , $t_3$ ; $\lambda=0$ ; $t_1$ wins) |
| p019-021 | comparison set 2 ( $T_1$ , $T_2$ , $T_3$ ; $T_3$ wins)               |
| p022-025 | response time 205 ms, packets 0.93, battery CI (boundary)            |

## lms-theory-of-estimation (40 pages)

|                         |                                                                              |
|-------------------------|------------------------------------------------------------------------------|
| p001-015                | estimation basics (same as ppt5 p001-007, the re-teach)                      |
| p016-030                | estimator properties (the 2 UNIQUE items: sufficiency examples, Poisson sum, |
| exponential $\bar{x}$ ) |                                                                              |
| p031-040                | confidence interval block, OUT OF MTE SCOPE, do not study                    |

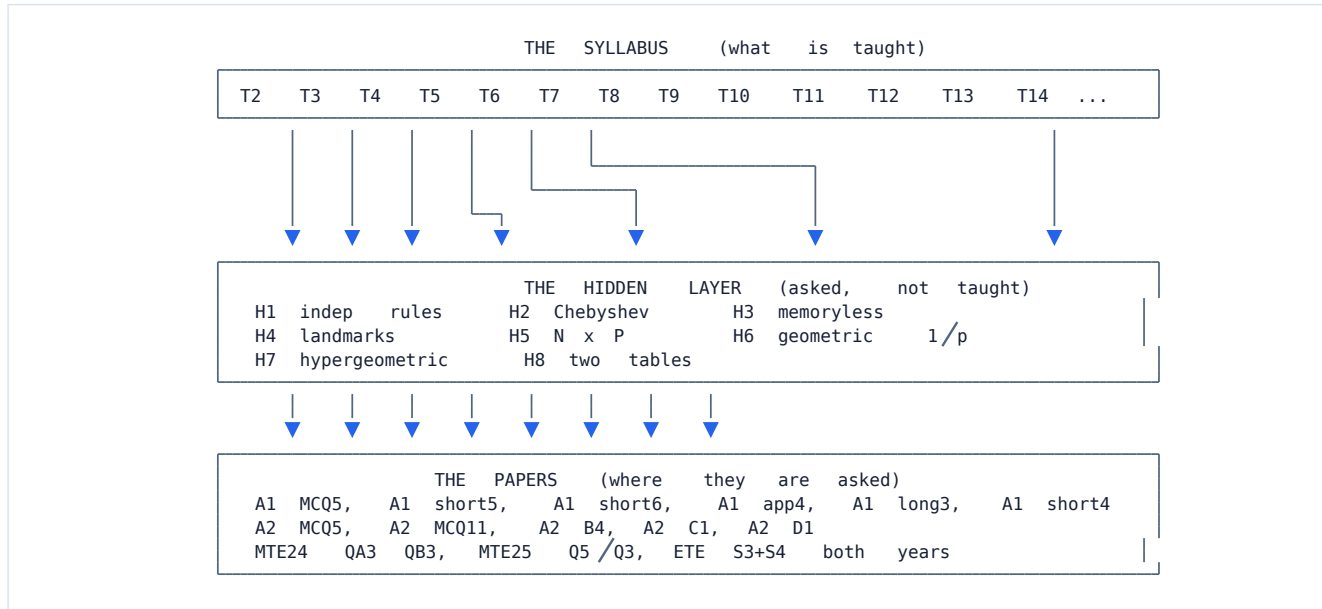
## The batch-2 S&P decks (not in md/, text in ~/mas2001-devore/corpus-text/)

|                |                                                                  |                                                                   |
|----------------|------------------------------------------------------------------|-------------------------------------------------------------------|
| L1-7           | (108 pp)                                                         | re-teach of the notes deck; 13 question blocks                    |
| L8-9           | (37 pp)                                                          | re-teach; 4 blocks (bus pdf, CDF-to-density, kidney E(Y), Pareto) |
| L10-11 (9 pp)  | CHEBYSHEV, the only new deck; Q1 and Q2 safe, Q3 mis-transcribed |                                                                   |
| L12-13 (28 pp) | re-teach of ppt3; 6 blocks, same pens numbers                    |                                                                   |
| L14-15 (44 pp) | re-teach of ppt4 + McClave; 7 blocks                             |                                                                   |

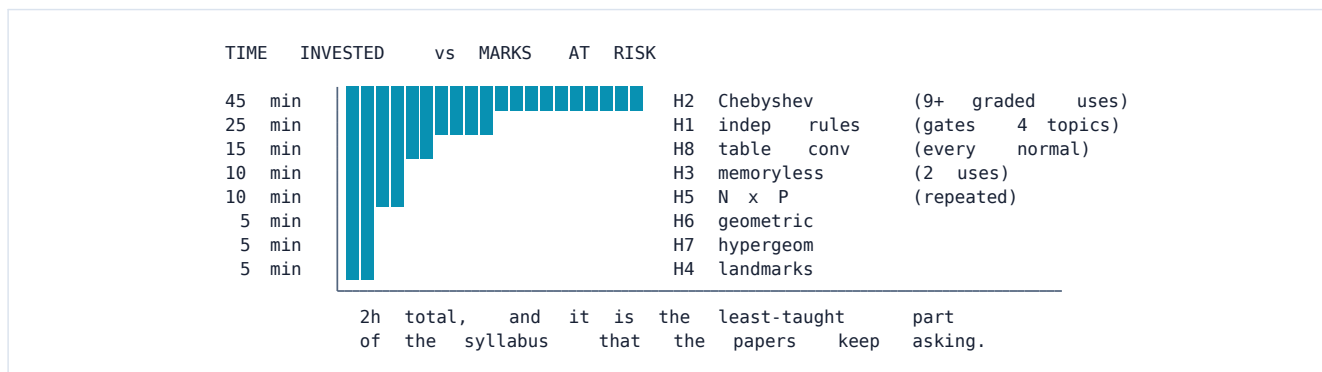


# 11 BEYOND SLIDES: what is asked but never taught

## THE HIDDEN LAYER, CONNECTED TO THE COURSE



## THE COST-BENEFIT OF DRILLING THE HIDDEN LAYER



## THE INDEPENDENCE RULES DRAWN (H1, the highest-value)

|                                                                                                         |                                   |                      |
|---------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------|
| INDEPENDENT X, Y:                                                                                       |                                   |                      |
| $E(XY)$                                                                                                 | $= E(X) \cdot E(Y)$               | needs INDEPENDENCE   |
| $\text{Var}(X+Y)$                                                                                       | $= \text{Var}(X) + \text{Var}(Y)$ | needs INDEPENDENCE   |
| $\text{Var}(X-Y)$                                                                                       | $= \text{Var}(X) + \text{Var}(Y)$ | the minus STAYS PLUS |
| $E(X+Y)$                                                                                                | $= E(X) + E(Y)$                   | ALWAYS, no condition |
| downstream uses:                                                                                        |                                   |                      |
| $\text{Var}(\text{sum of two dice}) = 35/12 + 35/12 = 35/6$                                             |                                   |                      |
| ↳ feeds the Chebyshev 35/54 bound                                                                       |                                   |                      |
| $\text{Var}(\bar{X}) = \sigma^2/n$                                                                      | $=$ this rule applied n times     |                      |
| ↳ feeds the CLT standard error                                                                          |                                   |                      |
| $\text{Var}(T)$ for estimator comparison                                                                |                                   |                      |
| ↳ the 6-8 mark estimation question                                                                      |                                   |                      |
| Poisson additivity: $\text{Poi}(\lambda_1) + \text{Poi}(\lambda_2) = \text{Poi}(\lambda_1 + \lambda_2)$ |                                   |                      |
| without H1, HALF the estimation block is unreachable.                                                   |                                   |                      |

These eight items are asked in the held graded material but are missing, lightly treated, or easy to overlook in the main converted teaching decks. They are high-value review items and several are prerequisites for later topics.

## H1. Independence rules of random variables

|                                                                                   |                                                                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| why hidden                                                                        | the notes deck teaches expectation to p094 and variance from p095, but the                                                  |
| product rule and the variance-of-a-sum rule sit on edges the conversion did not   |                                                                                                                             |
| capture as a named slide. Yet they gate four downstream topics.                   |                                                                                                                             |
| what it is                                                                        | $E(XY) = E(X) E(Y)$ when X, Y independent                                                                                   |
| $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ when independent              |                                                                                                                             |
| $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ when independent              | (the minus stays plus)                                                                                                      |
| contrast: $E(X + Y) = E(X) + E(Y)$ needs NO independence                          |                                                                                                                             |
| why it matters                                                                    | every dice-sum variance, every $\text{Var}(\bar{X})$ , every estimator variance, every                                      |
| Poisson additivity uses it. Without it, half the estimation block cannot be done. |                                                                                                                             |
| drill                                                                             | $\text{Var}(\text{one die}) = 35/12 \rightarrow \text{Var}(\text{sum of two}) = 35/6 \rightarrow$ the 35/54 Chebyshev bound |
| $\text{Var}(\bar{X}) = \sigma^2/n$ is this rule applied n times                   |                                                                                                                             |
| real ask                                                                          | R25S3-A1 asks $E[XY] = E[X]E[Y]$ directly (confirmed this is a graded question)                                             |

## H2. Chebyshev's inequality (whole topic)

|                                                                                             |                                                                                  |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| why hidden                                                                                  | batch 1 had no Chebyshev slide at all. The L10-11 deck arrived 15 Sep and closed |
| the gap, but it is the ONE deck that is genuinely new; everything else re-teaches.          |                                                                                  |
| what it is                                                                                  | $P( X-\mu  \geq k \sigma) \leq 1/k^2$ , complement form $\geq 1 - 1/k^2$         |
| k in sigma units. $k=2 \rightarrow 3/4$ , $k=3 \rightarrow 8/9$ , $k=4 \rightarrow 15/16$ . |                                                                                  |
| No distribution assumed. Always an upper bound on the tail.                                 |                                                                                  |
| why it matters                                                                              | graded 9+ times: A1 MCQ5, MCQ10, short5, short6, app4; MTE 2024 QA3, QB3;        |
| MTE 2025 Q5; ETE S3 and S4 both years; 2025-26 bundle Q5.                                   |                                                                                  |
| drill                                                                                       | dice 35/54 vs exact 1/3; 600 throws 19/24; marks 75 percent; inverse c from a    |
| stated bound (solve $1/k^2 =$ the given bound)                                              |                                                                                  |
| trap                                                                                        | Q3 of the L10-11 deck mis-states its own row (errata 15), use Q1 and Q2 only     |

## H3. The memoryless property

|                                                         |                                                                             |
|---------------------------------------------------------|-----------------------------------------------------------------------------|
| why hidden                                              | no slide states $P(T > s+t \mid T > s) = P(T > t)$ .                        |
| what it is                                              | the exponential does not age: surviving s more, given you already survived, |
| equals surviving s from scratch.                        |                                                                             |
| why it matters                                          | A2 MCQ Q5 and A2 section C Q1 both use it. It is one line to state and a    |
| full mark in a short answer.                            |                                                                             |
| drill                                                   | $P(T > 5 \mid T > 3) = P(T > 2) = e^{(-2\lambda)}$ .                        |
| This equals $e^{-1} = 0.3679$ only when $\lambda=0.5$ . |                                                                             |

## H4. The 68 / 95 / 99.7 landmarks

|                                 |                                                                       |
|---------------------------------|-----------------------------------------------------------------------|
| why hidden                      | no slide prints the empirical rule.                                   |
| what it is                      | for a normal distribution: 68.27 percent within 1 sigma,              |
| 95.45 within 2, 99.73 within 3. |                                                                       |
| why it matters                  | A2 MCQ Q11 uses it. Also the fast sanity check for any normal answer. |
| drill                           | "what fraction lies within 2 sigma" $\rightarrow$ 95.45.              |

## H5. Expected count, $N \times P$

|                                                                               |                                                                                                 |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| why hidden                                                                    | slides compute probabilities and expectations separately, never the "out of N                   |
| items, how many" phrasing as a named method.                                  |                                                                                                 |
| what it is                                                                    | expected number of items = $N \times P(\text{success})$ . It is expectation applied to a count. |
| why it matters                                                                | A1 app, A2 B4 (5000 batteries), A2 D1 (10000 bulbs), the irregular die count                    |
| (0.5499), and the new E25SUM-Q13 (800 families) and R25S4-B1 (heights > 6ft). |                                                                                                 |
| drill                                                                         | out of 10000 sets, expected = $10000 \times (3/8)^{10} = 0.5499$                                |
| trap                                                                          | report $N \times P$ , not $P$ . 0.5499, not $5.5e-5$ .                                          |

## H6. Geometric distribution

|                               |                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------|
| why hidden                    | the notes deck uses a geometric count (p057-059) but never names $E = 1/p$ as a        |
| result.                       |                                                                                        |
| what it is                    | $X$ = trials until the first success. $E(X) = 1/p$ , $\text{Var}(X) = q/p^2$ .         |
| why it matters                | A1 short Q4 asks the geometric expectation, $E = 2$ . Graded once in batch 1,          |
| so it is live.                |                                                                                        |
| drill                         | $E = 1/p$ . For $p = 1/2$ , $E = 2$ . The "number of failures before" convention gives |
| $1/p - 1$ , read the wording. |                                                                                        |

## H7. Hypergeometric mean

|                                             |                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------|
| why hidden                                  | no slide. It appears only in a question.                                                 |
| what it is                                  | drawing WITHOUT replacement from a finite lot: mean = $n \times K/N$ , the same value as |
| the binomial mean at the same $p$ .         |                                                                                          |
| why it matters                              | A1 long Q3: lot of 25, 5 defective, sample of 4, mean = $4 \times 5/25 = 0.80$           |
| exactly, both with and without replacement. |                                                                                          |
| drill                                       | $nK/N$ , and note it equals $np$ here. (errata 11: the key prints 0.808, wrong.)         |

## H8. The two normal table conventions

|                                                                                |                                                                                          |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| why hidden                                                                     | the deck uses a table but does not name $\phi(z)$ vs $F(z)$ as two different             |
| conventions.                                                                   |                                                                                          |
| what it is                                                                     | $\phi(z)$ is often tabulated as the area from 0 to $z$ ; $F(z)$ is the cumulative from   |
| $-\infty$ to $z$ . They differ by 0.5 for $z > 0$ .                            |                                                                                          |
| why it matters                                                                 | every normal and CLT numeric goes through a table. Mixing the conventions is             |
| the single biggest source of normal marks lost.                                |                                                                                          |
| drill                                                                          | $P(Z < 0.12)$ : from $\phi$ table, $0.5 + 0.0478 = 0.5478$ . From $F$ table, read 0.5478 |
| directly. Same answer, different route. Always decide which table the question |                                                                                          |
| gives BEFORE substituting.                                                     |                                                                                          |

## The one-block summary (all eight, one line each)

|    |                                                                                           |                     |
|----|-------------------------------------------------------------------------------------------|---------------------|
| H1 | $E(XY)=E(X)E(Y)$ , $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)$ , $\text{Var}(X-Y)$ adds | independence rules  |
| H2 | Chebyshev, $k$ in $\sigma$ , $1/k^2$ tail, complement $1-1/k^2$                           | the whole topic     |
| H3 | memoryless $P(T>s+t T>s)=P(T>t)$ , $e^{-1} = 0.3679$                                      | exponential         |
| H4 | 68.27 / 95.45 / 99.73                                                                     | normal landmarks    |
| H5 | expected count = $N \times P$                                                             | the count question  |
| H6 | geometric $E = 1/p$                                                                       | discrete lifeline   |
| H7 | hypergeometric mean = $nK/N$                                                              | without replacement |
| H8 | $\phi(z)$ area-from-0 vs $F(z)$ cumulative                                                | the table decision  |

Study order for these: H2 and H1 first (they gate the most), then H8 and H3, then H5, then H6, H7, H4. Total 2h, and it is the best-spent time before the paper.

# 12 DRILL: the practice set with answers

Solve, then check. Each item names its distribution and slot. Answers are exact where the source gives an exact value.

## The drill as a progress ladder

|      |   |              |             |         |                                    |
|------|---|--------------|-------------|---------|------------------------------------|
| RUNG | 1 | fundamentals | _____       | D1-D3   | probability, counting, complements |
| RUNG | 2 | discrete     | dists _____ | D4-D17  | binomial, Poisson                  |
| RUNG | 3 | continuous   | dists _____ | D18-D31 | uniform, normal, exponential       |
| RUNG | 4 | Chebyshev    | _____       | D32-D36 | bound, complement, inverse         |
| RUNG | 5 | sampling     | /CLT _____  | D37-D41 | SE, z, the CLT table               |
| RUNG | 6 | estimation   | _____       | D42-D46 | compare, point, interval           |
| RUNG | 7 | siblings     | _____       | D47-D50 | the mutation surface               |

|    |                             |
|----|-----------------------------|
| R7 | siblings (the risk surface) |
| R6 | estimation                  |
| R5 | sampling / CLT              |
| R4 | Chebyshev                   |
| R3 | continuous distributions    |
| R2 | discrete distributions      |
| R1 | fundamentals                |

climb in order. a rung is "done" when you can do it without notes.

## Warm-up: probability and counting

|     |                                                              |              |
|-----|--------------------------------------------------------------|--------------|
| D1  | Six faculty, choose 3 for laptops. How many ways?            |              |
| ans | $C(6,3) = 20$                                                |              |
| D2  | A camera chain: P(works each step) given. P(all three work)? |              |
| ans | product of the three, from the multiplication rule           |              |
| D3  | "At least one" heads in 3 fair coins.                        |              |
| ans | $1 - (1/2)^3 = 7/8$                                          | (complement) |

## Binomial

|     |                                                        |                                               |
|-----|--------------------------------------------------------|-----------------------------------------------|
| D4  | B(12, 0.1). $P(X=2)$ .                                 | ans $66 \times 0.01 \times 0.348678 = 0.2301$ |
| D5  | B(12, 0.1). $P(X \geq 2)$ .                            | ans $1 - 0.2824 - 0.3766 = 0.3410$            |
| D6  | B(12, 0.1). $P(X=0)$ .                                 | ans 0.2824                                    |
| D7  | Irregular die, $P(X=5)=2P(X=4)$ . Find p.              | ans 5/8                                       |
| D8  | Same die, 10 throws, P(no even).                       | ans $(3/8)^{10} = 5.499e-5$                   |
| D9  | Out of 10000 sets of 10, expected with no even number. | ans 0.5499                                    |
| D10 | B(600, 1/6). E and Var.                                | ans 100, 250/3                                |

## Poisson

|         |                                                                |                                         |
|---------|----------------------------------------------------------------|-----------------------------------------|
| D1<br>1 | $P(X=1)=0.2$ $P(X=2)$ . Find $\lambda$ . ans $\lambda = 10$    |                                         |
| D1<br>2 | Same, $P(X=0)$ .                                               | ans $e^{-10} = 4.54e-5$                 |
| D1<br>3 | Calls at 2/min. $P(\text{exactly } 2 \text{ in a minute})$ .   | ans $2e^{-2} = 0.2707$                  |
| D1<br>4 | $P(\text{exactly } 2 \text{ in each of } 5 \text{ minutes})$ . | ans $(2e^{-2})^5 = 32e^{-10} = 0.00145$ |
| D1<br>5 | 5000 men, death rate 0.001. $P(X=4)$ .                         | ans $e^{-5} 5^4/4! = 0.1755$            |
| D1<br>6 | $P(X>2)$ with $\lambda=5$ .                                    | ans 0.8753                              |
| D1<br>7 | "Rejected" $P(X\geq 4)$ with $\lambda=2.5$ . ans 0.2424        |                                         |

## Uniform

|     |                               |                       |
|-----|-------------------------------|-----------------------|
| D18 | $U(2,6)$ . f, mean, variance. | ans 0.25, 4, 1.3333   |
| D19 | $U(2,6)$ . $P(3<X<5)$ .       | ans $2/4 = 0.50$      |
| D20 | $U(2,6)$ . $P(X<1)$ .         | ans 0 (below support) |

## Normal

|     |                                                                                           |                           |
|-----|-------------------------------------------------------------------------------------------|---------------------------|
| D21 | $\mu=8$ , $sd=5$ . $P(X<8.6)$ .                                                           | ans $Z=0.12$ , 0.5478     |
| D22 | $\mu=8$ , $sd=5$ . $P(X>8.6)$ .                                                           | ans $1 - 0.5478 = 0.4522$ |
| D23 | $\mu=8$ , $sd=5$ . Find $x$ for $P(X<x)=0.20$ . ans $Z=-0.8416$ , $x = 8 - 4.208 = 3.792$ |                           |
| D24 | Marks: pass 40 = 46 percent, distinction 75 = 9 percent. Find $\mu$ , $\sigma$ .          |                           |
|     | ans $\sigma = 28.2$ , $\mu = 37.2$                                                        |                           |
| D25 | Same, re-exam cutoff at the 24 percent lower tail.                                        | ans 30.4                  |
| D26 | $N(100, 50^2)$ , $X=200$ . Find $Z$ .                                                     | ans 2.0                   |

## Exponential and related continuous models

|                                                                              |                                                                             |                                       |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------|
| D27                                                                          | 15/hour. $P(\text{gap} < 3 \text{ min})$ .                                  | ans 0.05 hr, $1 - e^{-0.75} = 0.5276$ |
| D28                                                                          | 15/hour. Mean gap.                                                          | ans 4 minutes                         |
| D29                                                                          | Mean 2. $P(X<1 \mid X<2)$ .                                                 | ans 0.6225                            |
| D30                                                                          | Gamma-shaped pdf $f(x)=x e^{-(x/3)}/9$ , $P(X>12)$ . ans $5e^{-4} = 0.0916$ |                                       |
| This is not an exponential density, even though its tail calculation uses e. |                                                                             |                                       |
| D31                                                                          | Memoryless: $P(T>8 \mid T>5)$ .                                             | ans $P(T>3) = e^{-3\lambda}$          |

## Chebyshev

|         |                                                                      |                                                              |
|---------|----------------------------------------------------------------------|--------------------------------------------------------------|
| D3<br>2 | Two dice, $P( X-7  \geq 3)$ bound.                                   | ans $\sigma^2=35/6$ , $k=3$ , $35/54 = 0.6481$ , exact $1/3$ |
| D3<br>3 | 600 throws, $P(80 \leq X \leq 120)$ lower bound.                     | ans $19/24 = 0.7917$                                         |
| D3<br>4 | Mean 70, var 25, $P(60 < X < 80)$ lower bound.                       | ans $k=2$ , 0.75                                             |
| D3<br>5 | $\mu=10$ , $\text{var}=4$ , $P(5 < X < 15)$ and $P( X-10  \geq 3)$ . | ans $21/25=0.84$ , and $4/9=0.4444$                          |
| D3<br>6 | Find $c$ so $P( X-10  \geq c) \leq 0.04$ , var 4.                    | ans $k=5$ , $c=10$                                           |

## Sampling, CLT

|     |                                                                                         |                                        |
|-----|-----------------------------------------------------------------------------------------|----------------------------------------|
| D37 | $\sigma=100$ , $n=25$ then $n=100$ . Find SE.                                           | ans 20 then 10                         |
| D38 | $\mu=4$ , $\sigma=2$ , $n=36$ . $P(\bar{X} > 4.5)$ .                                    | ans $Z=1.50$ , 0.0668                  |
| D39 | $\mu=4.0$ , $\sigma=1.5$ , $n=50$ .<br>$P(3.5 < \bar{X} < 3.8)$ .                       |                                        |
|     | ans $Z1=-2.36$ , $Z2=-0.94$ ; 0.1645 from rounded table areas, about 0.1637 unrounded   |                                        |
| D40 | $\mu=50000$ , $\sigma=8000$ , $n=64$ .<br>$P(\bar{X} < 48000)$ . ans $Z=-2.00$ , 0.0228 |                                        |
| D41 | $\mu=7$ , $\sigma=1$ , $n=9$ (normal pop).<br>$P(6.4 < \bar{X} < 7.2)$ .                | ans $Z1=-1.80$ ,<br>$Z2=0.60$ , 0.6898 |

## Estimation

|     |                                                                                                    |            |
|-----|----------------------------------------------------------------------------------------------------|------------|
| D42 | Sample of 5 normal; $t1=\text{mean}$ , $t2=(X1+X2)/2+X3$ ,<br>$t3=(2X1+X2+\lambda X3)/3$ .         |            |
|     | Find $\lambda$ , the unbiased ones, variances, the best.                                           |            |
|     | ans $\lambda=0$ ; $t1$ and $t3$ unbiased, $t2$ not; Var 0.2, 1.5, 0.5556<br>$\sigma^2$ ; best $t1$ |            |
| D43 | $T1=X1+X2-X3$ , $T2=2X1+3X3-4X2$ , $T3=(\lambda X1+X2+X3)/3$ .                                     |            |
|     | Find $\lambda$ , unbiased set, variances, best, and comment on consistency.                        |            |
|     | ans $\lambda=1$ ; all unbiased; Var 3, 29, $1/3 \sigma^2$ ; best $T3$ .<br>Fixed $n=3$ does not    |            |
|     | prove consistency; the sequence $\bar{X}_n$ is consistent under usual iid assumptions.             |            |
| D44 | OUT OF MTE SCOPE: $n=25$ , $\bar{X}=8.4$ , $\sigma=1.5$ , 95 percent interval.                     |            |
|     | ans $8.4 \pm 0.588 = (7.81, 8.99)$                                                                 |            |
| D45 | Response times sum 1640, $n=8$ . Point estimate.                                                   | ans 205 ms |
| D46 | 465 of 500 packets succeed. Point estimate of $p$ .                                                | ans 0.93   |



## The four high-risk siblings (from 07-PATTERNS, do these)

|                                                                               |                                                                                         |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| D47                                                                           | Normal one-unknown: 30 percent below 45 and 8 percent above 64, find $\mu$ , $\sigma$ . |
| D48                                                                           | Binomial two-sided interval: $B(12, 0.1)$ , $P(2 \leq X \leq 4)$ ?                      |
| D49                                                                           | CLT for a SUM: $n=40$ draws, $\mu=10$ , $\sigma=3$ . $P(\text{sum} > 440)$ ?            |
| ans sum $\sim N(400, 360)$ , $Z = (440-400)/\sqrt{360} = 2.108$ , tail 0.0175 |                                                                                         |
| D50                                                                           | Discrete conditional: $B(10, 0.5)$ , $P(X=5 \mid X \geq 3) = P(5)/(1-P(0)-P(1)-P(2))$   |

## The self-check rule

if you can do D4-D46 without notes:  
 Section B and C of the paper are MECHANICAL  
 if D47-D50 are also fine:  
 you are covered on the MUTATION SURFACE  
 if anything above is shaky:  
 go back to 04-METHODS.md for that SLOT before more drills

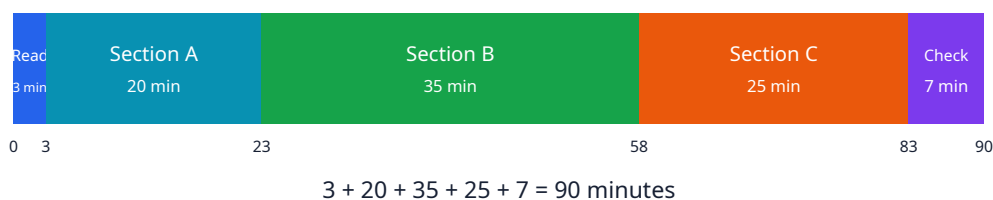
# 13 EXAM MECHANICS: format, marks, time, scoring

## 1. The format

MAS2001 Statistics and Probability, MID TERM  
 marks 30  
 scope lectures 1 to 21  
 sections A (MCQ) B (short) C (long)  
 ALL compulsory, no choice  
 CLOSED BOOK, calculator allowed

## 2. The time budget (90 minutes, 30 marks)

Ninety-minute practice budget



Section C is an 8-mark block  
 its topic may change

Leave seven minutes to check  
 boundaries, units, missed parts

Proportional ninety-minute practice budget

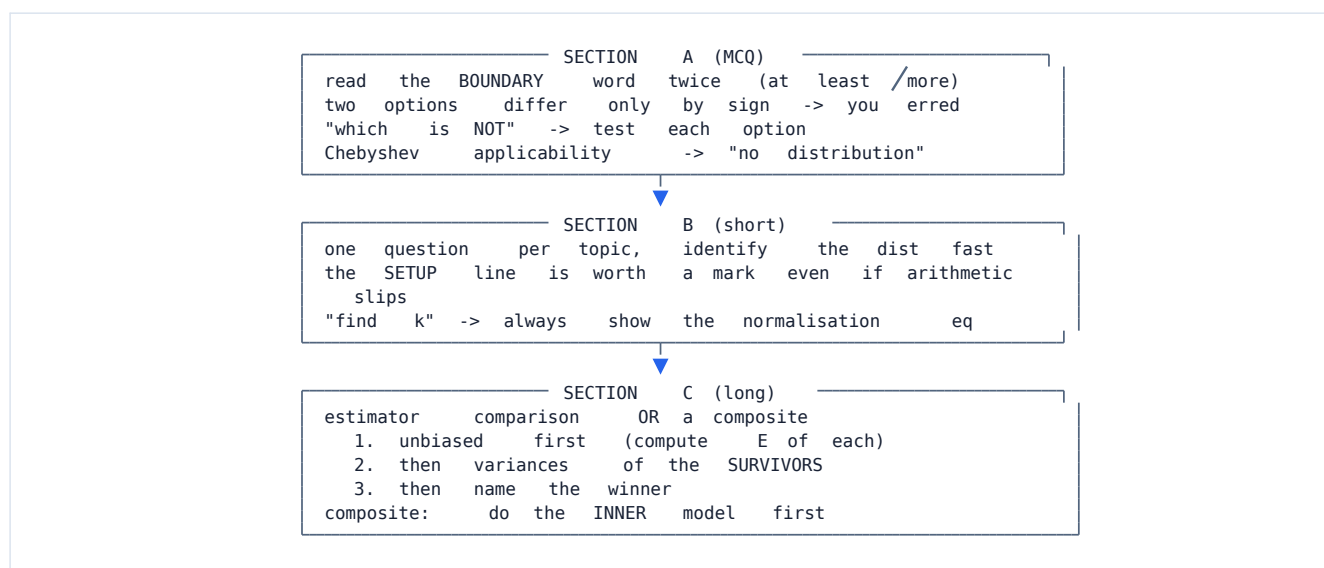
|        |  |                                   |        |
|--------|--|-----------------------------------|--------|
| 0 min  |  | read the whole paper first        | 3 min  |
| 3 min  |  | Section A (MCQs, ~1.5 min each)   | 20 min |
| 23 min |  | Section B (short, show the setup) | 35 min |
| 58 min |  | Section C (the estimator /comp)   | 25 min |
| 83 min |  | check and sweep for missed parts  | 7 min  |
| 90 min |  |                                   |        |

This split is a practice budget for the supplied 90-minute format.  
 Section C is an 8-mark combined block in both supplied papers, but its topic can change.

### 3. What earns marks (from the schemes)

| DO                                  | DO NOT                              |
|-------------------------------------|-------------------------------------|
| state " $X \sim B(12, 0.1)$ " first | drop a bare number                  |
| write the formula on its own line   | forget $z \rightarrow X$ conversion |
| keep exact values to the end        | round early and carry it            |
| for "show that", every algebra step | write 0.549 for 0.5499              |
| for comparisons, name the winner    | compare a BIASED estimator's        |
| AND the reason                      | variance against unbiased           |
| Chebyshev: bound AND exact value    | mix phi and F table conventions     |

### 4. Section tactics as a tree



### 5. The last-hour checklist

- ☐ 1. read 01-CHEATSHEET once, aloud, slowly
- ☐ 2. write the five distributions' mean/variance from memory
- ☐ 3. write the three Chebyshev thresholds: 3/4, 8/9, 15/16
- ☐ 4. write the hidden set: memoryless, landmarks,  $N \times P$ ,  $1/p$ ,  $nK/N$
- ☐ 5. read the 10-item trap list
- ☐ 6. no new material. nothing after lecture 21. no MLE, no MoM.

### 6. Confidence map (honest)

|            |                                                                                     |                                                                                                     |
|------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| STRONGEST  |  | five dists' moments, binomial /Poisson<br>point+tail, normal /CLT numerics,<br>estimator comparison |
| MEDIUM     |  | Chebyshev inverse, two-unknown normal,<br>exponential conditional, composition                      |
| WATCH      |  | the hidden set H1-H8 if not drilled,<br>min-n siblings, MSE comparison                              |
| SAFEST BET |  | the 16 MTE paper skeletons: all searched,<br>10 source /family leads, 5 concept checks, 1 open      |

## 7. Sources this deck draws on

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| reports/03-FORMULA-SHEET.md   | the verified closed forms                          |
| reports/04-QUESTION-BANK.md   | the worked answers                                 |
| reports/09-ERRATA.md          | all 21 source errata                               |
| reports/16,17,18,19,20        | provenance, intake, accounting, cross-check        |
| reports/11-QUESTION-ATLAS/    | topic universe, learn order, type space, mutations |
| md/                           | the converted slides and papers                    |
| ~/mas2001-devore/corpus-text/ | external reference text and OCR, outside the repo  |

## 8. The deck folder map, one more time

```
~/mas2001-mte-s2/deck/
README.md the folder map
00-INDEX.md the exam, the tree, the order
01-CHEATSHEET.md the whole paper in formulas
02-TERMS.md every term + its trap
03-LINGUISTICS.md wording -> ask -> move
04-METHODS.md method per question type
05-DISTRIBUTIONS.md 5 dists, 9 slots
06-NUMBERS.md every constant
07-PATTERNS.md the mutation system
08-TRAPS.md 20 traps + 21 errata
09-QUESTIONS.md the tagged inventory
10-SLIDE-MAP.md page -> topic -> question
11-BEYOND-SLIDES.md the 8 hidden items
12-DRILL.md 50 practice items
13-EXAM-MECHANICS.md this file
14-MASTER-STUDY-ORDER.md current order and gates
notes/ THE ACTUAL NOTES (11 files)
00-NOTES-INDEX the map
01..10 lecture-by-lecture teaching notes
```

# 14 MASTER STUDY ORDER: what to do, in what order

Built 16 Sep 2026. This is THE order document. It ties together everything we hold: the notes, the slide decks, the reference cards, the patterns, the drill, the papers. Follow it top to bottom.

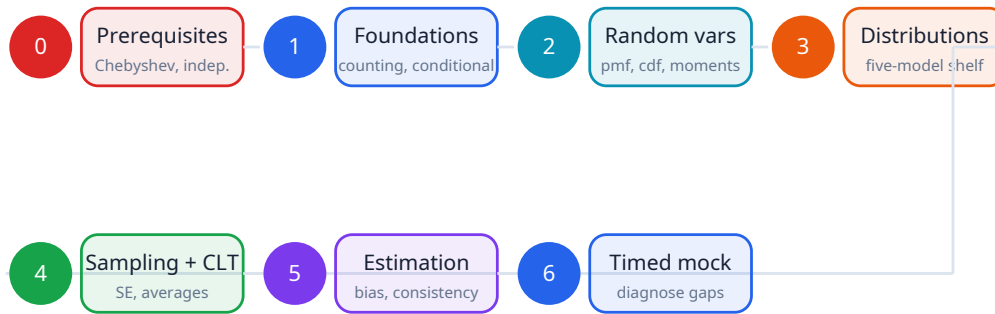
Supersedes [reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md](#) (that one was written before the notes folder existed; this one covers every artifact we now have).

## 0. THE ARTIFACT INVENTORY (what exists, where)

|                                                                  |                                               |            |
|------------------------------------------------------------------|-----------------------------------------------|------------|
| A. THE NOTES                                                     | deck/notes/                                   | (11 files) |
| the slides rewritten as teachable notes. READ THESE FIRST.       |                                               |            |
| you do NOT need to open a PDF to learn the theory.               |                                               |            |
| B. THE DECKS                                                     | the S&P batch-2 decks (text copies)           |            |
| L1-7, L8-9, L10-11 Cheb, L12-13, L14-15                          |                                               |            |
| these are PASS 2 / the visual check, NOT new teaching            |                                               |            |
| (except L10-11 Chebyshev, which IS new)                          |                                               |            |
| C. THE REFERENCE                                                 | deck/*.md                                     | (15 files) |
| cheatsheet, terms, linguistics, methods, distributions, numbers, |                                               |            |
| patterns, traps, questions, slide map, hidden layer, drill,      |                                               |            |
| mechanics. USE THESE WHILE SOLVING, not as reading material.     |                                               |            |
| D. THE PATTERNS                                                  | deck/07-PATTERNS.md                           |            |
| the mutation system. read AFTER two papers, then it pays off.    |                                               |            |
| E. THE PAPERS                                                    | md/paper-mte-*, paper-ete-*, resess-*, asgn-* |            |
| the drill material. papers FIRST, then assignments.              |                                               |            |
| F. THE ORIGINAL SLIDES the PDFs + md/ conversions                |                                               |            |
| ONLY open these for (a) Chebyshev deck, (b) L1-7/L8-9/L12-13/    |                                               |            |
| L14-15 question blocks. Everything else is in the notes already. |                                               |            |

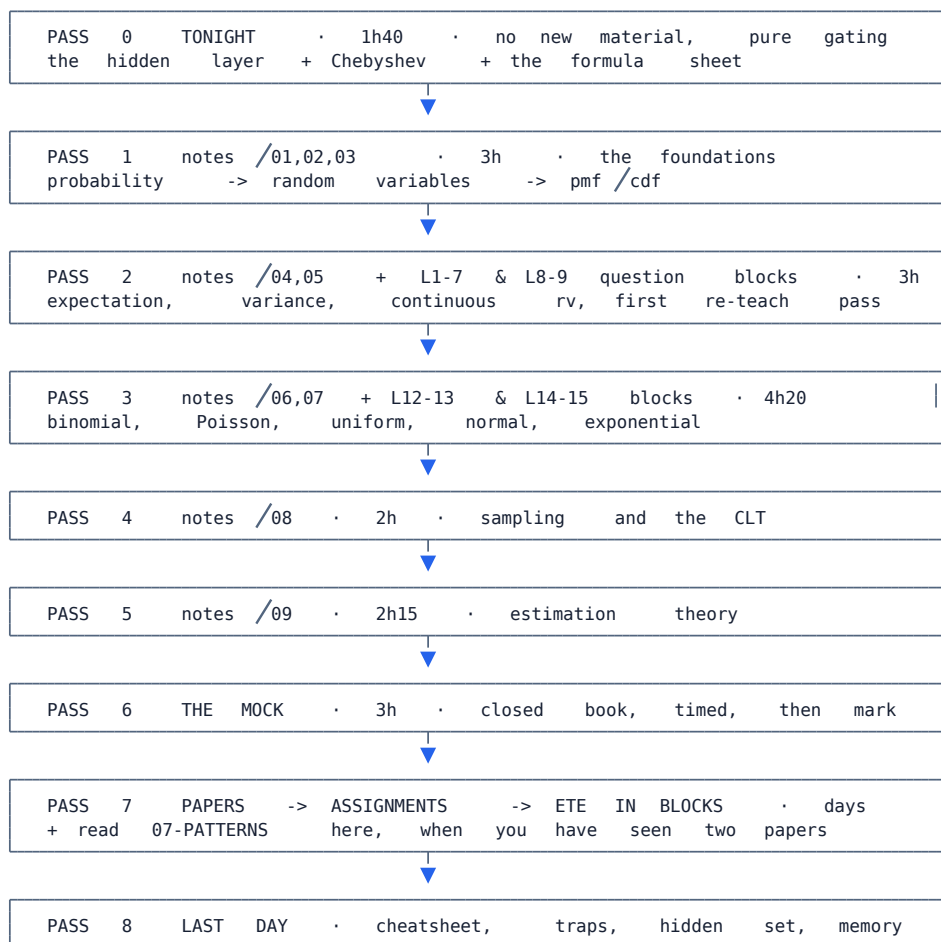
## 1. THE MASTER ORDER, ONE PICTURE

## Study order with dependency gates



advance only after the gate for the current pass is met

### Study order and dependency gates



## 2. PASS 0, TONIGHT, 1h40, THE GATING BLOCK

Do this FIRST. It gates four downstream topics and carries marks in every paper.

| #                                                      | WHAT                                                              | TIME |
|--------------------------------------------------------|-------------------------------------------------------------------|------|
| 0.1   notes/10 PART A: Chebyshev, all of it            | 30m                                                               |      |
|                                                        | (skip Q3, errata 15; do Q1 and Q2)                                |      |
| 0.2   notes/10 PART B: H1 independence rules           | 25m                                                               |      |
|                                                        | $E(XY)$ , $\text{Var}(X+Y)$ , $\text{Var}(X-Y)$ , then the drill: |      |
|                                                        | 35/12 $\rightarrow$ 35/6 $\rightarrow$ 35/54                      |      |
| 0.3   deck/11-BEYOND-SLIDES.md, H3 to H7               | 30m                                                               |      |
|                                                        | memoryless, landmarks, $N \times P$ , $1/p$ , $nK/N$              |      |
| 0.4   deck/01-CHEATSHEET.md, read aloud, twice         | 15m                                                               |      |
| GATE: write all of H1-H7 on blank paper, no notes.     |                                                                   |      |
| ALSO: the Chebyshev deck's own pages, 9 pages, text at |                                                                   |      |
| ~/mas2001-devore/corpus-text/L10-11-Cheb.txt           |                                                                   |      |

Why first: [07 - PATTERNS.md](#) shows Chebyshev is graded 9+ times; H1 gates the dice variance,  $\text{Var}(\bar{X})$ , every estimator variance and Poisson additivity. Doing this cold costs 1h40 now and saves rework in PASS 3, 4 and 5.

### 3. PASS 1, THE FOUNDATIONS, 3h

|     |                                               |     |
|-----|-----------------------------------------------|-----|
| 1.1 | notes/01-probability-foundations.md           | 55m |
|     | read fully, then DO the 3 worked examples     |     |
|     | (tournament, laptops, basketball) on paper    |     |
| 1.2 | notes/02-random-variables.md                  | 45m |
|     | the 36-cell dice grid, draw it yourself       |     |
|     | + deck/05-DISTRIBUTIONS.md first table        |     |
| 1.3 | notes/03-pmf-and-cdf.md                       | 60m |
|     | DO: gas pumps, find-k (the $10k^2+9k=1$ one), |     |
|     | boards inspection, dice maximum M, the proof  |     |
| 1.4 | deck/04-METHODS.md sections 1-5               | 20m |
|     | the solving paths for what you just read      |     |

## 4. PASS 2, EXPECTATION + VARIANCE + FIRST RE-TEACH, 3h15

|                                                               |                                                 |  |
|---------------------------------------------------------------|-------------------------------------------------|--|
| 2.1   notes/04-expectation-and-variance.md                    | 55m                                             |  |
|                                                               | DO: the die (3.5), the p094 triple (209),       |  |
|                                                               | the freezer 6-part chain, the magazine 3-vs-4   |  |
| 2.2   notes/05-continuous-rv.md                               | 65m                                             |  |
|                                                               | DO: $f=2x$ , the conditional 5/12, the bus      |  |
|                                                               | triangle 6 parts, cdf->pdf, the hospital Y      |  |
| 2.3   S&P L1-7 QUESTIONS ONLY (13 blocks, pass 2)             | 45m                                             |  |
|                                                               | text: ~/mas2001-devore/corpus-text/L1-7.txt     |  |
|                                                               | these are the SAME skeletons as notes/01-03     |  |
| 2.4   S&P L8-9 QUESTIONS ONLY (4 blocks)                      | 30m                                             |  |
|                                                               | bus pdf, CDF-to-density, kidney $E(Y)$ , Pareto |  |
| GATE: the freezer chain end to end. The L1-7 questions solved |                                                 |  |
| WITHOUT looking at the notes/01-03 answers.                   |                                                 |  |
| NOTE: do NOT read L1-7/L8-9 as new teaching. 54-55% of their  |                                                 |  |
| 5-grams exist in the notes deck already. They are pass 2.     |                                                 |  |

## 5. PASS 3, THE DISTRIBUTION SHELF, 4h20

One template per distribution: setup, pmf/pdf, moments, the corpus question, the trap.

|     |                                               |     |
|-----|-----------------------------------------------|-----|
| 3.1 | notes/06-binomial-poisson.md PART A binomial  | 55m |
|     | DO: pens 3 parts, irregular die $p=5/8$ ,     |     |
|     | 5-coin derivation                             |     |
|     | then L12-13 binomial block (pass 2)           |     |
| 3.2 | notes/06 PART B Poisson                       | 50m |
|     | DO: ratio $\lambda=10$ , THE NESTING 0.00145, |     |
|     | insurance 0.1755, the tail 0.8753             |     |
| 3.3 | notes/07 PART A uniform                       | 25m |
|     | DO: $U(2,6)$ , the length ratio 0.50          |     |
| 3.4 | notes/07 PART B normal                        | 70m |
|     | DO: the 8.6 chain, the 45-62 interval, the    |     |
|     | 20% inverse (3.792), the two-unknown (errata  |     |
|     | 10: $\mu$ 37.2, $\sigma$ 28.2, cutoff 30.4)   |     |
|     | then L14-15 normals (pass 2)                  |     |
| 3.5 | notes/07 PART C exponential                   | 40m |
|     | DO: 15/hr under 3 min (0.5276), the           |     |
|     | conditional 0.6225, memoryless, unit conv     |     |
|     | then L14-15 exponential (pass 2)              |     |
| 3.6 | deck/05-DISTRIBUTIONS.md, the 9-slot grid     | 20m |
|     | mark which slots are ASKED vs a SIBLING       |     |



## 6. PASS 4, SAMPLING AND CLT, 1h40

|     |                                                |     |
|-----|------------------------------------------------|-----|
| 4.1 | notes/08-sampling-and-clt.md, read fully       | 60m |
|     | DO: lightbulbs (SE 20/10), ATM (0.0668),       |     |
|     | impurity ( $Z_2 = -0.94$ , 0.1645 table), LED  |     |
|     | machines $n=9$ (0.6898)                        |     |
| 4.2 | deck/04-METHODS.md sections 14-15              | 15m |
|     | the SE and CLT solving paths                   |     |
| 4.3 | redo the machines $n=9$ case BLIND, then check | 25m |
|     | (it is the template: normal pop, small $n$ )   |     |

## 7. PASS 5, ESTIMATION, 2h15

|     |                                                       |     |
|-----|-------------------------------------------------------|-----|
| 5.1 | notes/09-estimation.md, read fully                    | 70m |
|     | DO BOTH comparison sets on paper, all steps:          |     |
|     | set 1 -> $\lambda=0$ , $t_1/t_3$ unbiased, $t_1$ wins |     |
|     | set 2 -> $\lambda=1$ , all unbiased, $T_3$ wins       |     |
|     | + the consistency trap (fixed $n$ is NOT it)          |     |
| 5.2 | the 3 real-life numerics                              | 25m |
|     | 205 ms (state estimator AND estimate),                |     |
|     | 0.93, the battery CI (7.81, 8.99) [boundary]          |     |
| 5.3 | lms-theory p024-p025 ONLY (sufficiency)               | 25m |
|     | SKIP p031-p040 entirely, out of MTE                   |     |
| 5.4 | deck/04-METHODS.md section 16                         | 15m |
|     | the estimator-comparison protocol                     |     |

## 8. PASS 6, THE MOCK, 3h

|                                                     |                                                     |  |
|-----------------------------------------------------|-----------------------------------------------------|--|
| 6.1   reports/07-MOCK-PAPER.md, CLOSED BOOK, TIMED  | 90m                                                 |  |
| 6.2   mark it against reports/08-MOCK-SOLUTIONS.md  | 30m                                                 |  |
|                                                     | (errata 18: the B4 note is wrong, T4 is             |  |
|                                                     | THIRD of four, not smallest)                        |  |
| 6.3   repair ONLY the misses, then redo those parts | 40m                                                 |  |
| 6.4   deck/08-TRAPS.md, the 10-item list, aloud     | 20m                                                 |  |
| GATE:                                               |                                                     |  |
| 24+/30                                              | -> proceed to PASS 7                                |  |
| 18-23                                               | -> the next morning goes to the WEAKEST pass only   |  |
| <18                                                 | -> the next morning re-reads notes/01-05 (the base) |  |

## 9. PASS 7, PAPERS AND ASSIGNMENTS, several days

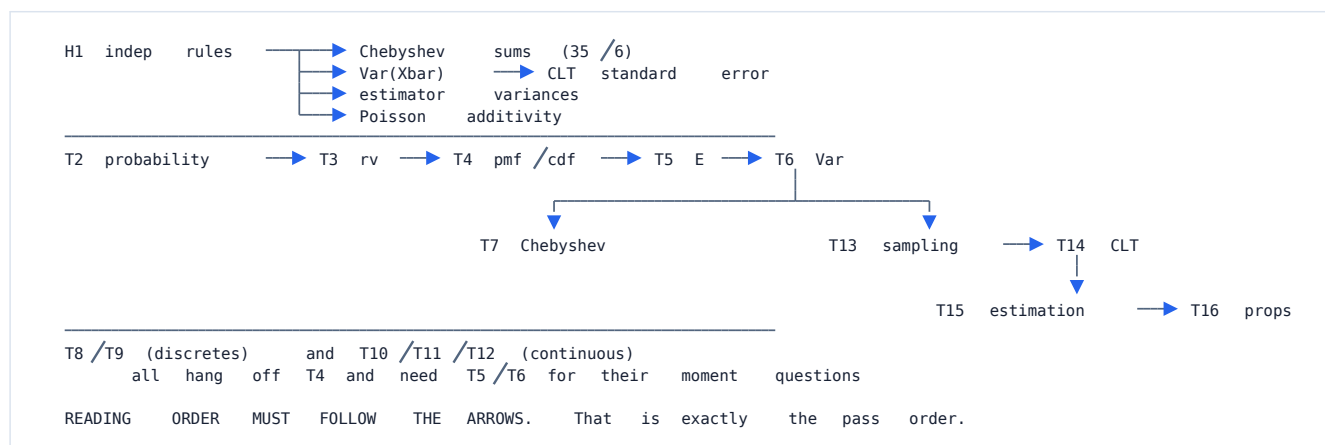
Order matters: papers reveal the SHAPE, assignments give the VOLUME.

|      |                                                                                                                                                                        |      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 7.1  | MTE 2025-26 paper, closed book, then mark<br>md/paper-mte-2025-26.md + the scheme                                                                                      | 60m  |
| 7.2  | MTE 2024-25 paper, same                                                                                                                                                | 60m  |
| 7.3  | READ deck/07-PATTERNS.md NOW<br>you have seen two papers, so the mutation<br>system will click. Then re-scan both papers<br>and label every question with its RADICAL. | 30m  |
| 7.4  | deck/09-QUESTIONS.md, the tagged inventory<br>confirm your labels against it                                                                                           | 20m  |
| 7.5  | 2025-26 assignment bundle, assignments 1 and 2<br>(the in-scope part). The full bundle has 119<br>rows; assignments 3-5 get a separate light pass.                     | 2h   |
| 7.6  | 2024-25 assignments 1 and 2 (in MTE scope)                                                                                                                             | 1h30 |
| 7.7  | 2024-25 assignments 3, 3-ep2, 4, 5 (LIGHT)                                                                                                                             | 45m  |
| 7.8  | ETE S3 and S4, IN blocks only (56 of 97)<br>where the same skeletons reappear under<br>exam wording                                                                    | 2h   |
| 7.9  | re-sess papers, IN blocks only                                                                                                                                         | 40m  |
| 7.10 | asgn-faculty-variant: DEDUP ONLY, do not drill<br>its key ( $k=1/10$ ) contradicts its row (0.0782)                                                                    | 10m  |

## 10. PASS 8, THE LAST DAY

|     |                                                                                         |     |
|-----|-----------------------------------------------------------------------------------------|-----|
| 8.1 | deck/01-CHEATSHEET.md, twice, aloud                                                     | 40m |
| 8.2 | every mean/variance pair, from memory<br>+ the three Chebyshev values (3/4, 8/9, 15/16) | 15m |
| 8.3 | the hidden set H1-H7, from memory                                                       | 15m |
| 8.4 | deck/08-TRAPS.md, the 10-item list                                                      | 10m |
| 8.5 | deck/13-EXAM-MECHANICS.md, the time budget                                              | 10m |

## 11. THE DEPENDENCY GRAPH (why this order)



## 12. THE ARTIFACT-TO-PASS TABLE (one lookup)

| ARTIFACT                      | PASS  | ROLE                  |
|-------------------------------|-------|-----------------------|
| notes/10 (Chebyshev + hidden) | 0     | GATING, read first    |
| deck/11-BEYOND-SLIDES.md      | 0     | GATING                |
| deck/01-CHEATSHEET.md         | 0,8   | read aloud            |
| notes/01 probability          | 1     | theory                |
| notes/02 random variables     | 1     | theory                |
| notes/03 pmf/cdf              | 1     | theory                |
| notes/04 expectation/variance | 2     | theory                |
| notes/05 continuous rv        | 2     | theory                |
| L1-7 questions                | 2     | pass 2                |
| L8-9 questions                | 2     | pass 2                |
| notes/06 binomial/Poisson     | 3     | theory                |
| notes/07 uniform/normal/exp   | 3     | theory                |
| L12-13, L14-15 questions      | 3     | pass 2                |
| notes/08 sampling/CLT         | 4     | theory                |
| notes/09 estimation           | 5     | theory                |
| lms-theory p024-025           | 5     | 2 unique items only   |
| mock + solutions              | 6     | integration gate      |
| MTE papers + schemes          | 7     | drill, shape first    |
| deck/07-PATTERNS.md           | 7     | after 2 papers        |
| deck/09-QUESTIONS.md          | 7     | label your radicals   |
| assignments (bundle, 2024-25) | 7     | drill, volume         |
| ETE + re-sess IN blocks       | 7     | drill under wording   |
| deck/12-DRILL.md              | any   | as a ladder, 7 rungs  |
| deck/04-METHODS.md            | while | lookup while solving  |
| deck/08-TRAPS.md              | while | lookup                |
| deck/02-TERMS.md              | while | lookup                |
| deck/03-LINGUISTICS.md        | while | lookup, before drills |
| deck/06-NUMBERS.md            | while | lookup                |
| deck/05-DISTRIBUTIONS.md      | while | lookup                |
| deck/10-SLIDE-MAP.md          | while | only if you open PDFs |
| deck/13-EXAM-MECHANICS.md     | 8     | night before          |
| lms-MLE, lms-MoM, handout     | NEVER | out of scope          |
| lms-theory p031-p040          | NEVER | out of scope          |

## 13. IF TIME IS SHORT (fallbacks, in priority order)

ONLY 1 DAY (7h20 core, plus 40m review reserve):  
 PASS 0 (1h40) + notes /01-03 skim (1h) + notes /06,07 skim (2h)  
 + mock (1h30) + cheatsheet (40m) + traps (30m)  
 -> the hidden layer + the shelf + the mock. Highest marks /hour.

ONLY 1 EVENING (3h40):  
 PASS 0 (1h40) + the two MTE papers, closed book, then mark (2h)  
 -> you will at least recognise every question shape.

ONLY 1 HOUR:  
 deck /01-CHEATSHEET.md read aloud, twice.  
 + the five distributions' mean /variance from memory.  
 -> the single highest-value hour available.

## 14. RULES THAT MAKE THIS ORDER WORK

|                                                                         |                                                                           |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| R1                                                                      | hidden prerequisites FIRST. they gate 4 downstream topics. 1h40.          |
| R2                                                                      | read a topic's NOTES before its S&P re-teach deck.                        |
| R3                                                                      | never read both of a pair cold:                                           |
| (notes deck, L1-7)                                                      | (notes deck, L8-9)                                                        |
| (ppt3, L12-13)                                                          | (ppt4, L14-15)                                                            |
| R4                                                                      | only L10-11 goes first, nothing else teaches Chebyshev.                   |
| R5                                                                      | every question is done WITH its deck or paper, never read in a batch.     |
| R6                                                                      | papers BEFORE assignments (shape first, volume second).                   |
| R7                                                                      | out-of-scope gets ONE light pass, never more.                             |
| R8                                                                      | read deck/08-TRAPS.md before opening raw source PDFs. The guided question |
| blocks in PASS 2 and PASS 3 may come earlier.                           |                                                                           |
| R9                                                                      | the mock is the only timed closed-book artifact; do not re-run it         |
| for comfort, only the failed parts.                                     |                                                                           |
| R10 the notes replace the PDFs for learning. Open the PDFs only for the |                                                                           |
| Chebyshev deck and the four S&P question blocks.                        |                                                                           |

## 15. THE DAILY SHAPE (once you are in PASS 1+)

```

09:00 read one notes / chapter end to end, no solving
10:00 solve every worked example in it on PAPER
11:00 the pass-2 deck block (if the pass has one)
12:00 break
13:00 deck /12-DRILL.md, one rung
14:00 mistakes: go to deck /04-METHODS for the slot
15:00 break
16:00 the pass gate (blank paper, no notes)
17:00 done. No new material after this.
```

do NOT read two chapters without solving between them.  
 the solve step is where the marks come from.

Master study order v1, 16 Sep 2026. Time estimates computed from the page counts and the worked-example density in each notes file. The errors/lies fix and the deep audit still await the user's

go.

# MAS2001 MTE ultra deck: notes + reference

Two things live here:

|                                                            |                                                          |
|------------------------------------------------------------|----------------------------------------------------------|
| notes/                                                     | THE ACTUAL NOTES. every slide rewritten as something you |
| read and learn from. 11 files, lecture by lecture.         |                                                          |
| you do NOT need to open a PDF to study from this.          |                                                          |
| (this dir)                                                 | THE REFERENCE LAYER. cheatsheet, terms, wording decoder, |
| methods, distributions, numbers, mutations, traps,         |                                                          |
| questions, slide map, hidden layer, drill, exam mechanics. |                                                          |
| these are the lookup cards you use WHILE solving.          |                                                          |

## Read order

|                                |                                    |
|--------------------------------|------------------------------------|
| 1. 14-MASTER-STUDY-ORDER.md    | the current order and checkpoints  |
| 2. notes /00-NOTES-INDEX.md    | the map of the notes               |
| 3. notes /01 .. notes /10      | read in order, lecture by lecture  |
| 4. this dir, as you solve      | cheatsheet, methods, traps on hand |
| 5. 12-DRILL.md                 | the practice set                   |
| 6. 13-EXAM-MECHANICS.md        | format, marks, last-hour list      |
| older measured schedule:       |                                    |
| .. /reports /11-QUESTION-ATLAS | /00-LEARN-ORDER.md                 |

## Everything is visual

Every file carries ASCII diagrams: trees, flowcharts, density-curve sketches, step plots, Venn diagrams, decision trees, tables. Plain text so it renders anywhere and never breaks.

## The one-line summary of each file

|                                                                            |                                                            |
|----------------------------------------------------------------------------|------------------------------------------------------------|
| reference (this dir)                                                       |                                                            |
| README.md                                                                  | this map                                                   |
| 00-INDEX.md                                                                | the exam, the 16 syllabus lines, the topic tree, the order |
| 01-CHEATSHEET.md                                                           | the whole paper in formulas + the master flowchart         |
| 02-TERMS.md                                                                | every term, exact meaning, its trap + the term map         |
| 03-LINGUISTICS.md                                                          | wording -> ask -> method + the decoder flowchart           |
| 04-METHODS.md                                                              | solving path per question type + the master decision tree  |
| 05-DISTRIBUTIONS.md                                                        | the 5 distributions, all 9 slots, all values               |
| 06-NUMBERS.md                                                              | every constant, z-values, landmarks, exact-vs-rounded      |
| 07-PATTERNS.md                                                             | the mutation system + the value-delta log + siblings       |
| 08-TRAPS.md                                                                | 20 concept traps + all 21 source errata                    |
| 09-QUESTIONS.md                                                            | the tagged question inventory + provenance truth           |
| 10-SLIDE-MAP.md                                                            | every slide page -> topic -> question                      |
| 11-BEYOND-SLIDES.md                                                        | the 8 hidden items (asked but never taught)                |
| 12-DRILL.md                                                                | 50 practice items with answers, as a 7-rung ladder         |
| 13-EXAM-MECHANICS.md format, time budget, what earns marks, last-hour list |                                                            |
| 14-MASTER-STUDY-ORDER.md the current study order and gates                 |                                                            |
| notes/                                                                     |                                                            |
| 00-NOTES-INDEX.md                                                          | the course map                                             |
| 01-probability-foundations                                                 | lecture 2, notes p012-p050                                 |
| 02-random-variables                                                        | notes p051-p062                                            |
| 03-pmf-and-cdf                                                             | notes p063-p087                                            |
| 04-expectation-and-variance                                                | notes p088-p111                                            |
| 05-continuous-rv                                                           | notes p112-p147                                            |
| 06-binomial-poisson                                                        | ppt3, lectures 12-13                                       |
| 07-uniform-normal-exponential ppt4, lectures 14-16                         |                                                            |
| 08-sampling-and-clt                                                        | lms-standard-error-clt, lectures 17-18                     |
| 09-estimation                                                              | ppt5 + lms-theory, lectures 19-21                          |
| 10-chebyshev-and-hidden                                                    | Chebyshev deck + the hidden layer                          |

## Grounding

Numerical answers are either copied from a named source page or recomputed in the audit. Known slide and key errors are marked inline. Provenance is a separate question: a correct answer does not prove where a question came from. See [09-QUESTIONS.md](#) and reports 16, 19, and 20 for the evidence states and open source searches.